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Task-Relevance of Topological Features and the Level Sets of Hidden Neurons

Course Paper

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Abstract

Interpreting a trained neural network includes a question about its individual units: what does a single hidden neuron respond to? A natural place to look for an answer is the data the network was trained on. Such a dataset may contain topological features, and one may ask whether echoes of those features can be found in the network itself afterwards.

The question is made computable by fixing the task to binary classification. In that case relevance splits into two notions. In the first, a feature is relevant to the task itself when the two classes are interleaved so that separating them becomes a question about an inside and an outside, a property of the labelled sample settled before any network is trained. In the second, it is relevant to the way a given network solved that task when one of its hidden neurons, read as a scalar function of the input, singles it out in its level sets. This narrower notion is the one measured here, by structural and homological scores and a descriptor of the level curve's shape.

Three properties of the data act on those level sets at once: the topological, geometric and density effects. Controlled settings separate them. Neither the density of the sample nor architectural capacity beyond what the task requires manufactures a witness for a task-irrelevant feature. What decides is relevance in the first sense: the same feature, point for point, is singled out by nearly every network where it is task-relevant and by none where it is not. The notion therefore has content. What the level curves follow, though, is the pair of objects, a feature together with the class inside it, and not the feature alone.

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1 Introduction and Motivation

Recently, neural networks have reached a level at which they can assist people in everyday life and in work-related tasks. To achieve such results many years of research were needed, and they led to the emergence of very complex systems that are highly capable but whose exact inner workings also evolved to be opaque.

The problem of neural network interpretability has become increasingly important for two reasons.

Firstly, from a practical point of view, the development of safe neural network models requires much deeper understanding than just general intuition. There are areas that do not tolerate unpredictable behavior, where the cost of error is too high, such as medicine, the financial industry, etc.

Secondly, the interpretation task appears to be complicated and highly interesting from the scientific point of view. The subject allows the use of various instruments that originate from different fields: physics, computer science and, of course, mathematics.

Speaking about the latter, this work focuses on the use of topological and geometrical instruments and aims to formulate rules that connect the topological structure of the training dataset with the structure that may be inferred from the neurons of the already trained neural network.

The question

The observation this paper starts from is easy to reproduce. Train a small multilayer perceptron to separate two classes of points in the plane, where one class traces a closed curve and the other fills the region that curve encloses. Take any hidden neuron of the trained network, read it as a scalar function of the network's input, and draw its level sets. In a large fraction of the networks trained this way, one of those level sets follows the curve in the data closely enough that anyone shown the picture would say that the neuron has found it.

It is tempting to conclude that the network has built the curve into that neuron. A single picture does not support such a conclusion, because the shape and the position of a neuron's level sets are influenced at once by several different properties of the dataset, and in one picture the effects of those properties are inseparable:

- **The topological effect.** The topological structure of the dataset may be essential for solving the task. A closed curve separates the plane into an inner and an outer region; here the points filling the inner region carry one label and the points of the curve itself carry the other, so the task is precisely the requirement to tell one region from the other. No network can solve it without accounting for that, and the requirement is purely topological: it is unaffected by any deformation that keeps the curve closed and keeps the opposite class inside it.
- **The geometric effect.** The same dataset also has geometry that the topological requirement leaves free: the forms of the two objects, their relative size, and where they stand with respect to each other. Two datasets may be indistinguishable topologically and differ in all of this, and when that geometry is changed the level sets change with it.
- **The density effect.** The same shapes can be sampled in different ways — more points on one object and fewer on the other, evenly spread or not. Level sets may cling to the denser parts of a sample, so that the picture reflects how the data were drawn rather than what they are.

Each of the three is a property of the data, and all three are observed through the same thing: the behavior of the level sets of the trained network's neurons. Their contributions cannot be told apart by looking at the pictures more carefully. They can be told apart by experiment — by varying one of them while the other two are held fixed, and by reading the outcome against a control that shares everything with the target except the one property under test. This paper reports four such experiments, together with the instruments built to make their outcome readable.

Two remarks about those instruments are needed before any of their numbers can mean something. First, they are not measurements borrowed from a mature literature: the scores used below were designed for this project and contain parameters chosen in this project, so their own behavior — where they respond, where they stay silent, how wide their baseline is — is itself part of what has to be established, and Section 6 is devoted to that. Second, the informal notion they make computable is broader than any one of them. I say that a neuron *witnesses* a feature when the feature is visible in that neuron, and this admits several formalisations. Two are used here: a *structural score*, which asks how closely the best component of a thickened level set follows the feature in shape and in position, and a *homological score*, which asks whether the topology of the feature is present in the superlevel filtration of the neuron without requiring any geometric agreement. A third could be built on the shape descriptor of Section 5.5. None of them is the definition of witnessing; each makes it measurable in one particular way, with its own blind spots, and the places where they disagree are among the more informative things we observe.

What this paper establishes

The four experiments are arranged so that each of them answers a question left open by the one before it. Every statement below is a statement about ensembles of independently initialised networks rather than about a single trained model: behind them stand 140 configurations of a dataset and an architecture with 50 networks trained in each, 7 000 networks in all. The numbers are given in Sections 7–10.

1. **The instruments respond where they should, and their baseline is known.** Before anything is said about a trained network, the scores are measured in two situations whose answer is fixed in advance: on a neuron built so that the feature lies in its level set by construction, and on congruent copies of the feature moved to places the task never involved. The first shows what a genuine agreement produces; the second shows what the absence of any agreement produces, and it turns out to be a distribution rather than a number — it depends on where the copy is placed, and it is wide enough that no verdict can be read without it.
2. **Whether a feature is needed for the task can be seen in individual neurons.** Two datasets carry the same curve, in the same place, point for point, and differ only in whether solving the task requires separating its inside from its outside. In nearly every network trained on the first there are neurons whose level sets follow the curve; in the networks trained on the second there are none. Since the curve is shared exactly, no property of the curve itself can account for the difference. Where the effect appears, it is not confined to one distinguished neuron: a large share of the network carries it.
3. **Neither the size of the network nor the density of the sample produces the effect.** Making the network wider or deeper than the task requires does not make a curve appear in the level sets where the task does not need it, and neither does training far past the point at which the task is solved. Redistributing the points between the objects of a dataset, at a fixed total count and with the shapes left where they are, moves neither configuration across the line: what was task-irrelevant stays unwitnessed and what was task-relevant stays witnessed, however the points are packed. Inside the task-relevant configuration the structural score does move, and the two things that move it are the number of points each object is given and the gap left between the two objects.
4. **The level sets do not follow the curve alone.** The first experiments measure in a configuration in which the curve and the region it encloses are concentric and both round, so that "the level set follows the curve" and "the level set follows the enclosed region" describe the same picture. Separating the two — by elongating one of them, by turning one across the other, by moving one off center — shows that these

are different statements, and that the shape, the orientation and the position of the separating level curve are determined mainly by the enclosed region, within limits that the curve sets. The influence of the curve is present and is weaker; it is seen where the enclosed region is round and does not compete with it. It follows that the effect can be removed without touching the curve at all, and that what the level sets respond to is the region enclosed between the two objects rather than the curve that bounds it. Which properties of that region they respond to is understood only in part: the shapes of its two boundaries, their relative orientation and their relative position all leave a trace, and no rule has been found that predicts the outcome from them.

5. **Scope.** All four experiments intervene in the data and never in the network, so every statement here is about agreement between measured quantities, and none of them says that a neuron encodes anything (Section 11.4).

Roadmap

Section 2 places the question among the works it grew out of. Section 3 fixes the objects: multilayer perceptrons, neurons as scalar fields, and the pair formed by a closed curve and the region it encloses. Section 4 separates three different meanings of "task-relevant", of which only the last one is about networks. Section 5 defines the instruments and Section 6 the protocol that turns their output into a result. Sections 7–10 are the four experiments, in the order in which each was made necessary by the previous one. Section 11 assembles them into a single statement, and Section 12 is the conclusion.

2 Background and Related Work

2.1 Topology of representations in neural networks

It was studied in [1] how the neural network influences the topology of the data as it passes through its layers. I consider this work to be the main inspiration for the current study. The authors consider the task of classification for points sampled from copies of interlinked tori in $\mathbb{R}^3$. They train a model to assign label 0 to the points from one of the interlinked tori and label 1 to the points from the other.

In a binary classification task the neural network is trained to distinguish between two classes. The output of the model is interpreted as a number from the interval $[0, 1]$ and may be treated as the probability that the point belongs to class 1. Points with values close to 0 are classified as class 0, and points with values close to 1 are classified as class 1.

From the topological point of view, this means that the final output of a well-trained classifier is very simple. The original dataset may have non-trivial geometry or topology, but after passing through all of the layers it is mapped to two separated regions near the endpoints 0 and 1. If one applies the final threshold, the output is reduced even further to the two-point label space $\{0, 1\}$. Therefore, successful classification necessarily simplifies the topology of the data at the level of the final output.

However, this does not explain how this simplification happens inside the network. The simplification of the topology in general may be achieved quite easily — just by mapping any topological space to one point. But in our case we obtain such a map for two disjoint subsets in the same ambient space $\mathbb{R}^n$. The neural network function is required to be defined in any point of the object space, so it performs some continuous transformation of $\mathbb{R}^n$ that still separates two subsets and maps them into two non-intersecting small neighborhoods on the interval $[0, 1]$.

Such expected behavior inevitably imposes very specific restrictions on the neural network. So, in [1] the authors implicitly make the following assumption: if a neural network was trained to an almost perfect accuracy, then it understood the data well, and with the data it should also understand its topological structure.

They do not clearly specify what exactly it means for a neural network to understand the topology of the training data; nevertheless, the observed behavior of topology simplification along the layers of the neural network may be viewed as one manifestation of such understanding.

In this paper I want to explore this assumption and make the term "neural network understands the topological structure of the data" more precise. In particular, I want to validate whether it is true beyond the basic intuition and in what sense it may be true.

2.2 From representations to neuron level sets

The neural network may be represented by the architecture and the weights that were learned during the training. From the point of view of topology, one may study the topology of the data as it passes through the layers of the neural network. But this gives us an understanding not of the neural network itself, but rather of how it acts on the training data.

A more direct way to study the neural network is to look at the learned neuron functions themselves. In this formulation the task becomes the task of inferring topological structure from scalar- or vector-valued functions defined by neurons.

This formulation appears to be more practice-oriented for the research of complex networks, whose training datasets may contain billions of objects and be extremely large and weakly structured. But at the same time taking this approach leads to a number of problems, the main one being the lack of information about which region of the data space to explore. The methods should remain computationally doable, so we cannot observe the whole $\mathbb{R}^n$ data space to study the values of the neuron function at each point.

One of the solutions for this obstruction was considered in [2], where the authors focused on local extrema of neurons, which are found in real neural networks by means of gradient ascent or descent. After reaching a close vicinity of local extrema they considered level sets of neurons and studied their topological and geometrical properties.

This supports the relevance of the neuron level-set approach. So, in this work I study level sets and the correspondence of their topological properties to those of the initial data. I want to understand under which circumstances the topological features of the data may also be found in the neurons, and how this correspondence depends on the chosen task. This work aims to bring more understanding of what the topological features found in the level sets of neurons mean in terms of the chosen task: whether they correspond to task-relevant information or arise for other reasons.

The order of these questions is the main reason the present work exists, and it is worth stating plainly. The long-term goal is the opposite one: to take a trained network, look into its neurons, and extract the structure it has learned without being told in advance what to look for. A structure found that way is worth something only if its presence means something. A level set may resemble a circle while the network has no use for that circle whatsoever — the finding would then be formally correct and completely empty, and nothing in the picture would say which of the two cases one is in. So the general problem cannot be attacked first. What has to come first is the controlled case: data whose topological structure we choose ourselves, where it can be asked whether that structure appears in the neurons at all, and whether its appearance depends on the task requiring it. The present work is that first step, and it is meant as the foundation of the second.

One more remark fixes the setting. At a regular value, a level set of a smooth function is a hypersurface, so in the plane it is a curve — an object of the same kind as the features our datasets carry. All the experiments below therefore take place in the plane, with closed curves as features.

2.3 Related work

The question of which neuron carries which feature has been asked in other languages as well, and it is worth saying how those differ from the question used here.

Linear probing. Gurnee et al. [4] locate neurons carrying a given feature by fitting k-sparse linear probes on activations. What is measured is the decodability of the feature from a representation of the data: a statement about a distribution of activations on the sample, which may succeed even when no individual neuron is geometrically organised around anything.

Superposition and polysemanticity. Elhage et al. [5] show that a network can represent more features than it has neurons, storing them along non-orthogonal directions, so that features need not align with coordinates at all; Lecomte et al. [6] add that polysemanticity may arise incidentally, without any pressure from limited capacity. This has a direct methodological consequence for the present work: a clean one-neuron-per-feature picture should not be expected, and a single striking example proves nothing. Every claim below is therefore a rate over an ensemble of independently initialised networks, with the network — not the neuron — as the unit of resampling.

Topological methods applied to networks. Rieck et al. [8] apply persistent homology to the weighted graph of a trained network and obtain a complexity measure of the model itself; Ramamurthy, Varshney and Mody [9] study the topology of the decision boundary for the purpose of model selection. Both speak about a whole model, and neither of them localises structure to a neuron.

These lines of work ask what a trained network contains, and answer in the language of linear decodability or of whole-model complexity. The question here is of a different kind: whether a structure that is present in the data can be located in the geometry of a single neuron, and whether its being located there depends on the task needing it. As long as that question is open, any structure extracted later from the neurons of a network may be a coincidence with no bearing on what the network does, and there is no way to tell one case from the other. This is why the datasets used below are ones whose topology is built deliberately, and why the same feature is measured in two tasks that differ only in whether they need it.

3 Setting and Definitions

3.1 Multilayer perceptrons

The neural network may be viewed as a complex function that is built by taking the composition of a number of much simpler building blocks, i.e. affine transformations and some non-linearity such as tanh, sigmoid or ReLU functions.

In this work I restrict attention to multilayer perceptrons. An MLP is a neural network in which each layer is given by an affine transformation followed by a coordinate-wise non-linearity. More precisely, for an input space $\mathbb{R}^n$, the hidden layers may be written as

$$f_i(x) = \sigma_i(W_i x + b_i),$$

where W_i is a matrix of weights, b_i is a bias vector, and σ_i is applied coordinate-wise. In the experiments below I use standard activation functions such as tanh, ReLU and sigmoid, with tanh being the main activation function.

Thus the whole neural network may be represented as a composition

$$F = f_{L+1} \circ f_L \circ \ldots \circ f_1.$$

Here $f_1, \ldots, f_L$ are hidden layers, while f_{L+1} is the output layer. In this work the output layer always ends with a sigmoid activation, so $F(x) \in [0, 1]$ and may be interpreted as the probability of class 1. For every hidden layer i, I denote by

$$h_i = f_i \circ f_{i-1} \circ \ldots \circ f_1$$

the representation of the input after the first i layers.

The choice of tanh as the main activation is made for the sake of the objects this work measures. A composition of affine maps with a smooth σ is smooth, so at a regular value its level sets are smooth curves, and quantities such as their shape and their length are defined in the ordinary sense. With ReLU the composition is only piecewise linear: level sets are polygonal, and the gradient vanishes identically on whole regions, so a neuron may be constant on the part of the plane we are asking about. This is a difference in the objects, not merely in the training, which is why the activation is treated as something to be varied and checked rather than assumed away; Section 8 reports what happens to the verdict under ReLU and sigmoid.

3.2 Hidden neurons as scalar fields

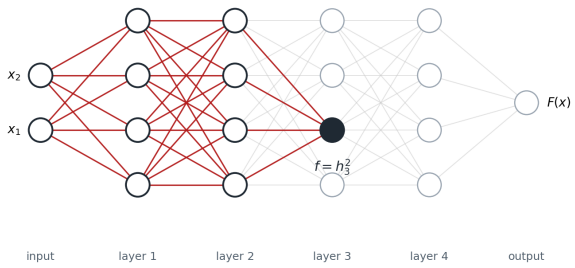


Figure 1: A neuron as a function of the input. The marked unit is the second neuron of the third hidden layer; the red edges are everything its value depends on, which is the whole network up to and including the layer before it. What comes after — drawn in grey — takes part in the network's answer $F(x)$ but not in the field f. Reading a neuron as a scalar field therefore means reading this sub-network, with all of its weights frozen, as a map $\mathbb{R}^2 \to \mathbb{R}$.

The main objects of this work are not the whole vector-valued maps h_i, but their scalar coordinate functions

$$h_i^j = \left(f_i \circ f_{i-1} \circ \ldots \circ f_1\right)^j,$$

where j is the index of a hidden neuron in the i-th layer. Each such coordinate is a scalar function on the input space, and therefore it is meaningful to study its level sets. In this sense, every hidden neuron is viewed as a scalar field on the original object space. Throughout the paper f denotes such a field: one neuron of one trained network, with every other weight of the network frozen, read as a function $f\colon \mathbb{R}^n \to \mathbb{R}$. Figure 1 shows which part of the network that function is: everything up to and including the layer before the neuron takes part in it, and everything after does not.

Each coordinate can be read before or after the non-linearity of its own layer. The two differ by σ_i. Where it is strictly monotone, as tanh and the sigmoid are, it sends level sets to level sets: $\{x : \sigma(g(x)) = \sigma(c)\} = \{x : g(x) = c\}$, so the family of level sets of a neuron is the same in both readings and only its labelling by c changes. ReLU is the exception, being constant on the negative half-line: read after it, every level there becomes one. All fields below are read after the activation.

3.3 The datasets

All datasets in this work are built from two objects in the plane. The *feature* is a closed curve — a circle or an ellipse — sampled evenly along its length and labelled class 0. The *interior* is a region filled with points of class 1. When the interior lies inside the feature, the two are separated by the region between them, which we will call the *gap*: the annulus bounded by the feature on the outside and by the interior on the inside. These three words are used with these meanings throughout.

The construction that all four experiments rest on is a *matched pair* of datasets, shown in Figure 2. Both contain the same circle of radius 1.5, sampled by the same 500 points, in the same place; both contain a disk of radius 1 carrying 500 points of the opposite class. In one of them the disk is placed inside the circle,

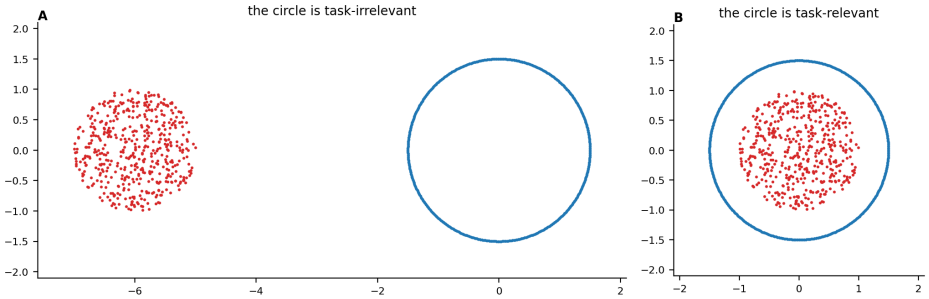


Figure 2: The matched pair. The blue circle is identical in the two datasets: the same 500 points, the same radius, the same position. The red class is placed outside it in (A), where a straight line solves the task, and inside it in (B), where the task is exactly the separation of the inside from the outside. Any difference measured between (A) and (B) is a difference in what the task requires, because there is no difference in the circle.

and the task cannot be solved without separating the inside of the circle from the outside. In the other the disk is moved away, and a single straight line solves the task, so nothing about the circle — that it is closed, that it is round, that it has an inside — is of any use.

The point of the construction is that it removes the explanation before it can be offered. If the circle were witnessed more often in one dataset than in the other, and the circles were merely similar, one could always suspect that the difference lay in their geometry. Here the two are the same set of points, so the suspicion has nothing to attach to.

The object the scores compare a neuron with is a finite point set A, the *target*. It is a sample of the region of the plane that we decide to interrogate, and it is chosen by us rather than read off the network or off the data. In the simplest case, and in Figure 2, the target is the feature itself; but it need not be, and in Section 9 the targets are circles that no point of the dataset lies on. Two obligations come with this freedom. The target must be sampled finely enough that the measurement is about the network and not about the sampling of A, which is a condition relating the spacing of A to the resolution of the grid (Section 5.2). And the control must be the same construction moved elsewhere, so that everything except position cancels in the comparison (Section 6.2).

The later experiments vary this pair in three ways, and it is convenient to have them together:

- **Architecture** (Section 8). The matched pair of Figure 2 is kept exactly as it is, and what changes is the network: widths from 1 to 40, depths from 1 to 10, three activations, and training continued far past the point at which the task is solved.
- **Density** (Section 9). The circle is replaced by a ring of a given width, sampled by the same number of points however wide it is, so that widening the ring thins it and brings its inner edge towards the centre; how the points are divided between the ring and the disk is a second axis of its own.
- **Geometry** (Section 10). The circle and the disk are replaced by ellipses, which can be elongated independently, turned across one another, and moved off center — so that "the level set follows the feature" and "the level set follows the interior" stop being the same statement.

3.4 Level sets

For a neuron f and a value $c \in \mathbb{R}$ the level set is

$$L_f(c) = \{x \in \mathbb{R}^n : f(x) = c\},$$

and we write $L_f(c) = \coprod_i L_f^i(c)$ for its decomposition into connected components. If c is a regular value of a smooth f, the level set is a smooth hypersurface; in the plane it is a disjoint union of curves, each of them either closed or leaving every bounded region.

Two things about this family are worth stating before any score is defined, because they decide what may be asked of it.

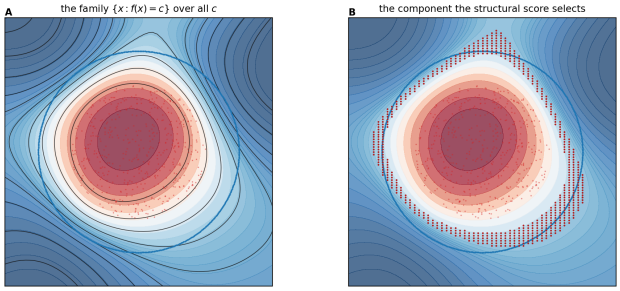


Figure 3: One neuron of the fourth layer of the network of Figure 4. (A) its level sets at thirteen values of c. (B) the single connected component picked out by the structural score of Section 5.3, which here follows the circle to within 4.1% of the circle's diameter. The component is drawn as the band of grid nodes it actually consists of: a level set is extracted at a finite resolution and with a finite tolerance, and the object that is scored is that band.

First, the object of a question is a *component*, not the whole level set. A neuron may have a closed curve around one part of the plane and an unbounded branch elsewhere at the same value, as several panels of Figure 4 show; asking whether $L_f(c)$ "is" a circle would be answered negatively by such a neuron for an irrelevant reason.

Second, the family is indexed by c, and nothing in the network distinguishes one value of c from another. Any question about a neuron and a feature is therefore a question about the whole family, and the instruments of Section 5 differ precisely in how they use it. The structural score *searches* the family: it looks for the level whose component comes closest to the feature and reports that best match, which is what Figure 3 shows. The homological score does not search but *filters*: it takes the nested family of superlevel sets $\{f \geq c\}$, follows how their topology changes as c decreases, and asks whether a cycle of the required kind persists over a long stretch of the sweep. The shape descriptor does neither: it fixes the level by a rule set outside the neuron and describes the curve that sits there.

That difference decides what each of them has to be controlled for. A search over levels is a genuine degree of freedom — the more levels are examined, the more likely it becomes that some component of some level happens to lie near the feature — so a structural score cannot be read without the negative control of Section 6.2. A filtration has no such freedom, and pays for it elsewhere: it establishes that a cycle is present without establishing where the cycle runs (Section 5.4).

Not every neuron can produce a closed curve at all, and for some of them this is a matter of arithmetic rather than of training.

Proposition 1. *Let F be an MLP on $\mathbb{R}^2$ whose activation σ is continuous and strictly monotone, and let $f = h_i^j$ be one of its neurons. Then no connected component of any level set of f is a closed curve, in each of the following two cases:*

(i) $i = 1$, that is, f belongs to the first hidden layer;

(ii) $w_1, \ldots, w_{i-1} \leq 2$, that is, no layer before the one f belongs to is wider than the input dimension.

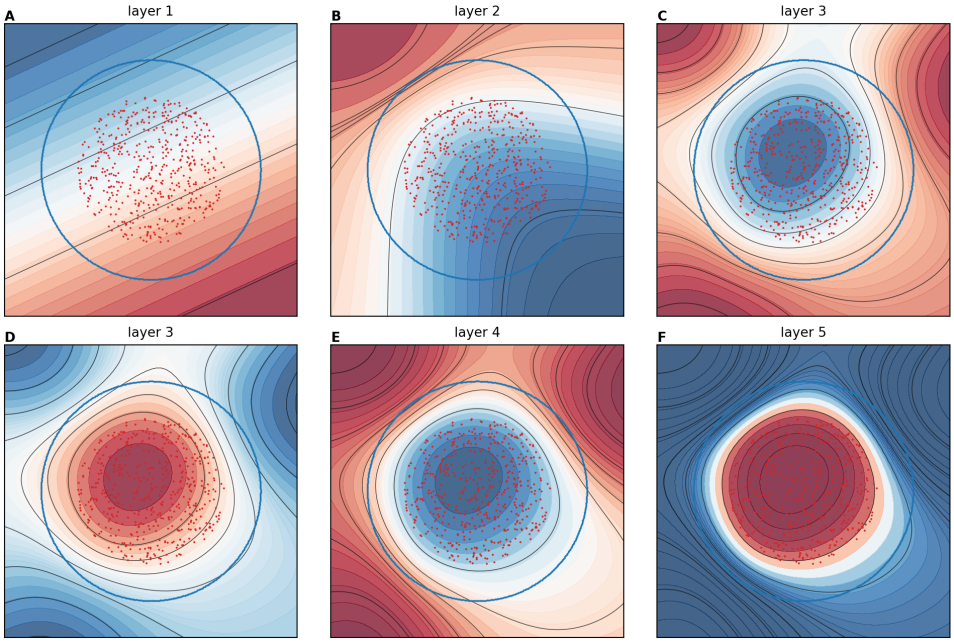


Figure 4: Six neurons of one trained network (tanh, five layers of width five, trained to 100% accuracy on the dataset of Figure 2B), each shown as a scalar field on the input plane in the order of the layers, (A) to (F); (C) and (D) are two neurons of the same third layer, and within a layer the neurons have no order, so only the layer is named. Colour is the value of the neuron, thin black lines are a few of its level sets; the blue curve and the red cloud are the two classes of the data. The first layer can only produce straight lines, and the second still produces open curves; closed curves appear from the third layer on, and by the last one they run close to the blue circle. That this ordering is systematic, and not a property of this particular network, is measured in Section 8.

Proof. In case (i) we have $f(x) = \sigma(w \cdot x + b)$ for a row w of W_1. If $w = 0$ the field is constant and its only non-empty level set is the whole plane, which is not a curve. Otherwise $f(x) = c$ holds exactly when $w \cdot x + b = \sigma^{-1}(c)$: a straight line if c lies in the range of σ, and the empty set if it does not.

For case (ii) write $f = \sigma(a \cdot h_{i-1} + \beta)$, where a is the j-th row of W_i. As σ is injective, the level sets of f are exactly the level sets of the scalar function $g = a \cdot h_{i-1} + \beta$, and if $a = 0$ then g is constant and there is nothing to prove.

Suppose first that every matrix $W_1, \dots, W_{i-1}$ is invertible; in particular every width before layer i equals 2. An affine invertible map of $\mathbb{R}^2$ is a homeomorphism, and σ applied coordinate-wise is a homeomorphism of $\mathbb{R}^2$ onto the open box $\text{range}(\sigma)^2$, so each layer, and hence h_{i-1} itself, is a homeomorphism of $\mathbb{R}^2$ onto an open set $\Omega \subseteq \mathbb{R}^2$. A level set of g is then $h_{i-1}^{-1}(L \cap \Omega)$ for a line L. Every connected component of $L \cap \Omega$ is a connected subset of a line, hence an interval, and a homeomorphism carries it to a set homeomorphic to that interval. No interval is homeomorphic to a circle.

Suppose instead that some layer before i has width 1 or a matrix of rank 1, and let k be the first such layer. By the previous paragraph h_{k-1} is a homeomorphism of $\mathbb{R}^2$ onto an open Ω, and every layer from k on depends on x only through the single number $\tau(x) = u \cdot h_{k-1}(x)$, where u spans the row space of W_k. Hence $f = \varphi \circ \tau$ for some $\varphi\colon \mathbb{R} \to \mathbb{R}$, and a level set of f is $h_{k-1}^{-1}(\{y \in \Omega : u \cdot y \in T\})$ for some $T \subseteq \mathbb{R}$. Let C be a connected component of that set. If $u \cdot y$ is constant on $h_{k-1}(C)$, then $h_{k-1}(C)$ lies in one line and C is homeomorphic to an interval, as before. If not, then by connectedness $u \cdot y$ takes every value of some interval on $h_{k-1}(C)$, so T contains an interval, the set $\{y : u \cdot y \in T\}$ contains an open strip, and C contains a non-empty open subset of $\mathbb{R}^2$. A subset of the plane homeomorphic to a circle has empty interior, so C is not a closed curve in either case. □

The same conclusion holds in $\mathbb{R}^n$ for networks all of whose hidden layers are at most n wide: Johnson [3] shows that such networks have only unbounded level sets, which is what makes them unable to approximate functions with compact level sets.

A third case is degenerate rather than geometric: a ReLU unit whose pre-activation is negative on a whole region is constant there, and the level set containing that region is two-dimensional. Such a neuron carries no information about anything inside the region, and in the ensembles of Section 8 about a quarter of the ReLU units are of this kind on the feature.

Proposition 1 is used twice in what follows, in opposite directions. It is a check on the instrument: a neuron of either class cannot be a witness whatever the network has learned, so a score that ever declares one of them a witness is broken, and for that reason such neurons are kept in the measurements rather than filtered out of them. And it is a statement about architectures: a network of depth one that solves a task requiring a closed curve has to solve it without any single neuron tracing that curve, so its encoding is necessarily distributed. Both readings are taken up in Section 8.

4 Task-relevance at three levels

The core of this work is the notion of task-relevance and task-irrelevance for topological features of the training data. Instead of giving a single formal definition I propose a three-level view of it, because the three levels are established by different means and only the last one is about neural networks at all.

4.1 Level 1: the informal notion

On the *intuitive* level we call a topological feature *task-relevant* if its topological structure is important for solving the task, and *task-irrelevant* if it is not.

In Figure 2A the circular shape of the blue class is irrelevant for the task: the data are linearly separable, and the form taken by either class on its own side of the separating line does not affect the accuracy of the separation. In Figure 2B the situation is the opposite. There the task is to separate the inner region from the curve that bounds it, and the very notions of inner region, outer region and boundary exist only because the blue class is a closed curve; in this case the circle is called task-relevant.

This level only relays the intuitive understanding of what it means for something to be relevant to a task, and it is a guide rather than a definition. “The task could not be solved without

it" quantifies over a hypothesis class, an accuracy, and a distribution, none of which the phrase names. For the two datasets of Figure 2 the informal argument can be made by hand and is sound, which is exactly why they are the datasets everything else is calibrated on; but no conclusion in this work rests on this level alone.

4.2 Level 2: relevance as a property of the data

The second level replaces the intuition by a computable quantity measured against a chosen threshold. It is still a statement about the data and the task, with no model involved — nothing has been trained at this point — but it is a number rather than an argument.

The intuitive level was stated in vague enough terms to apply to any task; to make it computable we fix the task to be binary classification. For another kind of task this level, and the one after it, would need different instruments. Under that restriction the relevant object is not the homology of one class, but the way in which the two classes are interleaved, and chromatic persistent homology (Section 5.6) computes exactly that. The part of it that matters is the kernel/relative class: a cycle of one class that is filled in by points of the other. A closed curve of one class whose inside is occupied by the other produces such a class; the same curve with the opposite class moved away does not.

This gives a definition of task-relevance which is a property of the point cloud, computable before any network exists, and it is the ground truth against which the informal level can be checked. On the pair of Figure 2 the two agree. What this level does not do is find anything inside a network: a dataset having a kernel class says that a classifier is forced to break that cycle, not that any particular neuron reflects it.

4.3 Level 3: relevance to the way a given network solved the task

Only at the third level does a trained network enter. Here we go through its neurons, read each of them as a scalar field, and ask whether the feature is singled out by any of them. There is more than one way in which a neuron can single a feature out — one of its level sets may follow the feature; it may have an extremum inside the region the feature bounds; a closed level curve of it may enclose that region without following the feature at all — and each of the instruments of Section 5 makes one of these precise.

A feature that is singled out in this way in the networks trained on one dataset and not in those trained on another is relevant to the way those networks solved the task. This is a narrower statement than either of the first two levels, and it is the one this work is about. It is also weaker than it may sound: that a feature is visible in a neuron does not mean the network would be any worse without it. A neuron may single out a feature and play no part in the answer, and separating the two requires removing the neuron and measuring what the network loses — which is not done here.

Two distinctions have to be kept at this level, and both are made precise later.

The first is between witnessing and its formalisations (Section 5.1). "The feature is visible in this neuron" is the notion; the structural and the homological scores are two ways of computing it, and neither of them is its definition. A verdict therefore always names the instrument that produced it.

The second is between witnessing and encoding. A witness is an agreement between a level set and a point set. That the network *uses* the feature is a stronger statement, and establishing it requires intervening in the network — removing the neuron and looking for a loss of accuracy that is concentrated near the feature. Nothing in this work does that, so all of its claims are of the first kind.

Relevance is thus either intuitive, or formal for the task itself with no model involved, or defined for a given model and the particular way it solves the task. The object of study is the last of the three; the first two serve as a sanity check and as the ground truth that it is measured against.

5 Instruments: how a neuron becomes a number

5.1 Witnessing, and why it has more than one formalisation

Informally, a neuron *witnesses* a feature when the feature is visible in that neuron. The word is doing a lot of work, and it is worth seeing how much before it is made computable, because the obvious readings of it are genuinely different statements:

- one of the neuron's level sets *follows* the feature — runs along it, at the right place and with the right shape;
- some level set of the neuron *encircles* the region the feature bounds, without necessarily running along the feature at all;
- the neuron has an extremum inside that region, so that the feature is where its values turn around.

None of the three implies another. A neuron with a wide basin over the interior encircles the feature at many levels and follows it at none; a neuron whose crest runs along the feature follows it without having an extremum inside it; and, as Section 10 shows, the first two come apart in practice often enough that the difference is the main finding of an experiment.

So a formalisation is a choice, not a discovery, and each one buys a definite question at the price of the others. This work uses two scores and one descriptor:

- the **structural score** (Section 5.3) measures the first reading: how close, in shape and position, the best component of a level set comes to the feature;
- the **homological score** (Section 5.4) measures the second: whether a cycle of the required kind is present in the neuron's superlevel structure near the feature;
- the **harmonic shape descriptor** (Section 5.5) measures nothing about the feature at all. It describes the level curve itself — where it is, how big it is, what shape it has — and returns no verdict, although a threshold on it would define a third notion of witnessing as legitimately as the other two.

The structural score carries most of the weight in Sections 7–9. That is a practical matter and not a claim of primacy: it is the one of the two scores that speaks about shape and position, and shape and position are what the manipulations of those experiments move. Where the two are read together they are read as two instruments, and their disagreement is treated as information (Section 5.8).

5.2 What every instrument reads

None of the three can be applied to a neuron directly: f is a function on the plane, and every one of them needs it as a finite object. That translation is made once and in the same way for all of them, and three of the choices made here are the ones the instruments inherit.

The evaluation box and the grid. A rectangle is fixed around the data — wide enough to hold the data and the controls of Section 6.2, and fixed *before* the run so that it cannot be adjusted once the scores are seen. Figure 5 shows it for the datasets used here, together with the controls it has to accommodate. The neuron is evaluated on a regular grid over that rectangle, giving a pair (grid points, values) that we call the *field sample*. The grid step h is the largest of the per-axis steps.

The sample is computed *once per neuron* and then shared: the structural score labels its nodes, the homological score builds a complex on them, the descriptor traces curves through them. This is partly economy — the experiments below score some

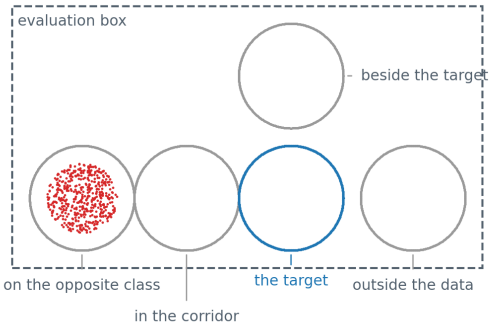


Figure 5: The negative control. The target is the circle at the origin; each grey copy is the same 500 points moved by a rigid motion, so it agrees with the target in shape, size, point count and topology, and differs from it in position alone. The four placements are assembled to cover the problem: on the opposite class, in the empty corridor between the two, beside the target, and outside the data entirely. The dashed rectangle is the evaluation box of Section 5.2: it has to hold all of them, which is why it is much larger than the data.

10^6 neuron-feature pairs, and evaluating the same field three times would be the dominant cost — and partly a matter of validity, since three instruments that disagreed might otherwise be disagreeing about which sample of the neuron they had seen.

How fine the grid must be. The resolution is not a free parameter and it is not transferable as a number. What transfers is the dimensionless ratio

$$N_A = \frac{\operatorname{diam}(A)}{h},$$

the number of grid steps across the object being looked for. Experiment 1 measures its scale directly, by re-running a probe whose answer is known — a neuron built so that its level set *is* the target, with a true geometric error of 0.0016 — at five resolutions, 25, 50, 100, 200 and 400 nodes per axis, which on its box are $N_A = 5.5$, 11, 22, 44 and 89. The probe scores 0.128, 0.095, 0.074, 0.034 and 0.021 respectively — these are the probe's numbers and not a bound on the instrument, which trained neurons beat on the same grid (Section 6.3). What the scan fixes is where the answers stop depending on the grid: at $N_A = 5.5$ the probe fails its own threshold outright, so a negative answer there is a statement about the grid and not about the network; by $N_A = 22$ the ordering of the answers is stable, and by $N_A = 44$ so are the counts. All runs below are set up with $N_A \geq 40$, which for the datasets of Figure 2 means 200 nodes per axis over a box of about thirteen units, and a grid step of 0.068.

The target, and how finely it is sampled. The object a score compares the neuron with is a finite point set A, a sample of the region we have decided to interrogate (Section 3.3). Its own spacing is part of the instrument. If $\delta(A)$ is the typical distance between neighbouring points of A, then a component lying exactly on the region still finds its nearest *sample point* about $\delta(A)/2$ away, and it cannot be localised better than about $h/2$ because it is read off the grid. The two are independent, so the best distance attainable by anything at all is about

$$\sqrt{(h/2)^2 + (\delta(A)/2)^2},$$

and the sampling of A is negligible exactly when $\delta(A) < h$. Above that the ruler is coarser than what it measures. Two rules follow and are used throughout: A is sampled uniformly and finely enough for the condition to hold, and A is never taken to be the dataset's own points when the number of those points is a manipulated variable — otherwise sparse configurations would be measured with a coarser ruler than dense ones, and the ruler would move with the very quantity under study.

In higher dimensions. A grid costs $\mathcal{O}(N^d)$ and is dead beyond three dimensions. The same field sample is then drawn as a uniform random cloud in the box instead, and the instruments that need only pairwise distances — the homological score above all — carry over unchanged, while those that need a grid do not. All experiments in this work are planar and use the grid.

5.3 The structural score

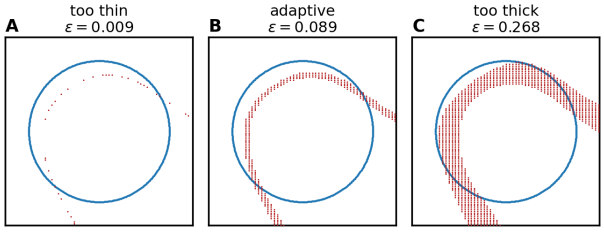


Figure 6: One level of one neuron, extracted with three tolerances. (A) a tenth of the adaptive value: thinner than the steps the field takes between neighbouring nodes, so the set breaks into fragments and the component that the score is about does not exist. (B) the rule used here, which makes the band about one grid step wide whatever the steepness of the neuron. (C) three times the adaptive value: a region whose shape has little to do with the level set.

Let f be a neuron and A the target. Because level sets are read off a grid, the object actually available is not $L_f(c)$ but its thickened version

$$L_f(c, \varepsilon) = \{x : |f(x) - c| \leq \varepsilon\},$$

whose connected components are found by labelling the grid nodes that satisfy the inequality. Distance between two finite point sets is the max-symmetrised Chamfer distance

$$D(K, A) = \max\left\{ \frac{1}{|K|} \sum_{k \in K} d(k, A),\ \frac{1}{|A|} \sum_{a \in A} d(a, K) \right\}.$$

One of the two averages alone would say only how well one set covers the other: a short arc lying exactly on A has a small average distance to A, and a large blob containing A has a small average distance from A. Taking the maximum of the two makes the quantity symmetric and forces agreement in both directions at once. It is still an average and not a supremum, which is a deliberate tolerance: a component may carry some spurious mass — a few nodes of the band running off elsewhere — without the score collapsing.

The *structural score* is the best such distance over the whole family,

$$\operatorname{Score}_{\mathrm{str}}(f, A) = \min_{c,\, i} D\big(L_f^i(c, \varepsilon), A\big).$$

Every number quoted below is this distance divided by $\operatorname{diam}(A)$, so that it reads as a fraction of the size of the object being looked for and is comparable between targets of different size.

Computed, the procedure is: sweep 101 levels c across the range the neuron takes on the grid; at each level extract $L_f(c, \varepsilon)$ and split it into connected components; discard components of fewer than five nodes; score every remaining component against A; keep the smallest number seen anywhere in the sweep. Figure 3 is one instance of that search: the whole family in (A), and in (B) the single component it returns.

The tolerance ε. One parameter cannot be left to taste, and it is the thickness of the band. It is measured in the *values* of f while what it controls happens in the *domain*, and the two are related by the grid: between two neighbouring nodes the field changes by some amount, and if ε is smaller than that amount the level passes *between* the nodes and is never recorded. The set extracted then falls into fragments, as in panel A of Figure 6, and the connected components that the score is about stop existing. Too large, and the band spreads over a region wide enough that its shape is no longer the shape of a level set, as in panel C.

The rule used here reads the grid rather than the neuron: ε is taken to be the largest difference between the values of f at neighbouring nodes within a few grid steps of the target — the largest step the field takes between two adjacent samples where the question is being asked. On the neuron of Figure 6 that is 0.089, collected from 4245 neighbour pairs inside four grid steps of the circle, on a grid of step 0.068 and a field spanning 1.82. A steep neuron therefore receives a *larger* tolerance in value units than a flat one, and that is the point: the band it produces is about one grid step wide in the plane either way. The tolerance is set by the discretisation, not by how strong the neuron is.

Two guards complete the rule. The sweep is run at two quantiles of that distribution of steps, the better of the two results being kept, so the search is over roughly two hundred (level, tolerance) pairs. And ε is never allowed below half the spacing between consecutive levels of the sweep, since a band thinner than that could fall between two levels and be missed by both.

Three properties that are used later. They all follow from the definition.

It is an *order statistic* — the minimum over that search — and not a fixed property of the neuron. Two runs are comparable only if the number of levels, the resolution and the ε rule agree, which is why those are recorded with every result; and the search is a real degree of freedom, which is why the negative control of Section 6.2 is part of the measurement rather than an addition to it.

It is *not localised*. Every component anywhere in the evaluation box is scored and the smallest wins, so a neuron is never credited for being near the target by being allowed to look only there — in contrast to the homological score, which does restrict, and pays for it in another currency.

It *mixes three things into one number*. Shape, size and position all enter $D(K, A)$, and only their sum is reported. A curve of the right shape in the wrong place and a curve of the wrong shape in the right place can score the same, which is the reason the descriptor of Section 5.5 exists.

Restricting the search. The minimum above runs over two independent coordinates: the *level*, which value of c and which tolerance to look at, and *space*, which connected component — that is, where in the box to look. Exhausting both is possible only because a grid hands over every component of every level at once. Where a grid is not available, one of the two has to be restricted, and each restriction defines a different instrument.

The ridge restriction keeps the whole domain and restricts the levels to the critical values $c^* = f(p^*)$ at the extrema of the neuron, which are found by gradient ascent and descent. It sees an encoding in which the extremum is attained along the feature itself — a crest running over the curve — since for such a neuron c^* is the level wanted.

The basin restriction does the opposite: it keeps the full sweep of levels and restricts the domain to a box of side $2\,\mathrm{diam}(A)$ around one extremum p^*. The side is the smallest that guarantees what is needed, because if p^* lies in the convex hull of A then every point of A is within $\mathrm{diam}(A)$ of it. This restriction sees the other arrangement, an extremum inside the region and the feature on a regular level around it — and the ridge restriction cannot: at a non-degenerate extremum the component at c^* is a single point, whose distance to a curve of radius r_A is about r_A, while the level that does follow the curve is regular and is excluded from the ridge candidates by construction.

The converse inclusion, however, mostly does hold, and it is worth stating because it is not symmetric. If the neuron has a crest along the feature then on both sides of the crest the field falls away; following it inwards from the crest it keeps falling, so inside the enclosed region it must turn around, and a non-degenerate ridge therefore comes with an interior extremum — which is exactly the anchor a basin search needs. Barring degeneracy, and provided the anchor found lies inside the hull of the feature so that the box around it contains the feature, a ridge configuration is also seen by the basin restriction. The two are not two halves of a partition: one is nearly contained in the other.

Two things are worth saying about any such restriction, because they are easy to get backwards. Both minimise the same quantity over a *subset* of the same candidates, so

$$\mathrm{Score}_{\mathrm{str}} \;\le\; \min\bigl(\mathrm{Score}_{\mathrm{ridge}},\, \mathrm{Score}_{\mathrm{basin}}\bigr),$$

and a restriction can therefore only miss a witness that the full score finds, never invent one. And a restriction does not remove coincidences by being stricter: it cuts the *search budget*, which lowers the chance of an accidental match — and lowers the floor with it, since the controls are scored by the same procedure. The value of a restriction is accordingly measured by what it does to the distance between the target and its own floor, never by what it does to the raw score. No experiment in this work uses either restriction: all of them are planar and run the full sweep. They are described because they are what the structural score becomes when the plane is left behind.

A third strategy does not restrict the search but changes the domain it runs on: fit a low-dimensional affine subspace to a meaningful set of points in the input space — the trajectory of the gradient ascent that reached p^*, say — and run the full two-dimensional procedure on the restriction of f to that plane. The order is not interchangeable, since projecting a level set and taking the level set of a projection are different operations: the subspace has to be fixed first and the level sets computed second.

5.4 The homological score

The structural score leaves one thing entirely open: it never asks the component it picks to be a closed curve, or to have the feature's topological type at all. The minimum runs over every component, the distance is an average in both directions, and an arc that hugs a large part of the feature is therefore a legitimate candidate that can score as well as a loop — a fact that is not a defect of the implementation but a consequence of the definition. So a low structural score says that something in the neuron lies where the feature lies; it does not say that the something is of the same kind as the feature.

The second score asks precisely that question: whether the neuron's structure near the feature contains a cycle of the required kind at all, regardless of where exactly it runs. It also has a property the first one does not: the construction is carried out on the sample itself rather than on a grid laid over the plane, so it is the one instrument here that survives the loss of the grid in higher dimensions.

The superlevel filtration. For a level c write

$$S(c) = \{x : f(x) \ge c\}.$$

As c decreases these sets only grow, $S(c') \subseteq S(c)$ whenever $c' > c$, so they form a nested family — a *filtration* — indexed by the level. Watching the topology of $S(c)$ as c sweeps downwards is watching the neuron being filled in from its maxima: components appear at the maxima, grow, and merge; holes open when a growing region closes a ring around something lower, and they disappear when the enclosed part is finally swallowed. Each such hole exists over an interval of levels, from the c at which it opens to the c at which it closes, and that interval is its *bar*; the length of the bar is the hole's *persistence*. The collection of bars of all dimensions is the persistence diagram of the filtration, and it is what the score reads. Long bars are structure of the neuron; short ones are the sampling.

Two examples make the whole construction concrete, and both can be computed by hand; Figure 7A draws them. Take the plane restricted to a disk D of radius 2.

A crest on the unit circle: $f(x) = -(\|x\| - 1)^2$. For $-1 < c < 0$ the superlevel set is the annulus $1 - \sqrt{-c} \le \|x\| \le 1 + \sqrt{-c}$, which has exactly one hole. At $c = 0$ the set is the unit circle itself, where the hole opens; at $c = -1$ the inner radius reaches

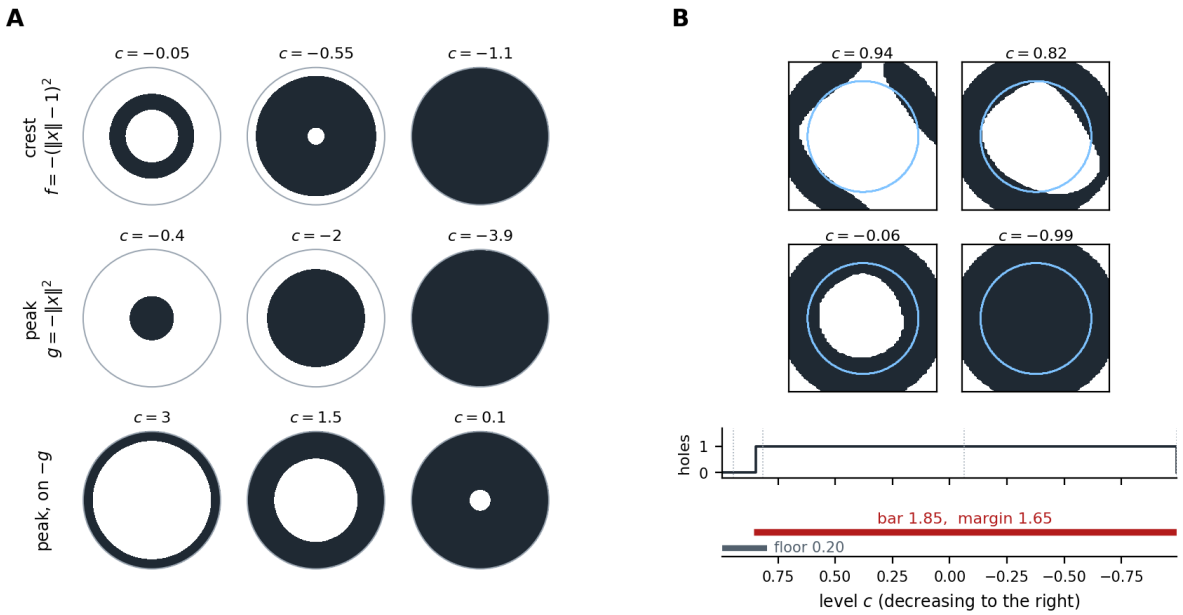


Figure 7: **(A)** The two fields computed by hand, on a disk of radius 2; dark is $\{f \geq c\}$. The crest on the unit circle gives an annulus for every c between 0 and -1, so its hole opens at 0 and closes at -1. The peak at the origin gives disks and no hole at all; read through $-g$ the same peak gives annuli and one hole. **(B)** The same construction on one neuron of the network of Figure 4, whose loop lives in the $-f$ orientation. The four pictures are the set at four decreasing levels, restricted to the neighbourhood the score uses, with the target circle drawn over it: before the hole opens, just after it opens, in the middle of its life, and after it has filled. Under them, the number of holes as c decreases, and under that the single bar of dimension one that this produces, with the absolute floor drawn as a length beside it. The interval in which the picture has a hole and the bar in the diagram are the same object seen twice.

the origin and the hole closes. The diagram therefore contains a single one-dimensional bar, from 0 down to -1, of persistence 1 — and the level at which it is born is exactly the crest.

A peak at the origin: $g(x) = -\|x\|^2$. Now every superlevel set is a disk about the origin, every one of them is contractible, and the diagram contains no one-dimensional bar at all. The structure is there — the level sets of g are circles around the origin, and they encircle anything placed near it — but the superlevel sets do not see it. They do see it for $-g$: the superlevel sets of $-g$ are the complements of disks, that is annuli in D, and one bar appears with persistence 4.

The second example is the reason the filtration is computed for both f and $-f$ and the better answer reported. Whether the opposite class sits at a maximum of the neuron or at a minimum is not something we get to assume, and sublevel sets of f are superlevel sets of $-f$.

From the continuous picture to a computation. The field is known at finitely many points, so the filtration has to be built on them. Three steps do it.

First, a complex is built on the sample, and it is built *once*: every point is a vertex, two vertices are joined by an edge when the distance between them is below a fixed r, and a triangle is filled in whenever its three edges are present. This r is a single length, not a parameter that sweeps: the filtration below is indexed by the value of f, and the complex it runs on does not change. Its value is set to twice the typical nearest-neighbour distance within the target, which is large enough to join the neighbours a grid produces and small enough not to join parts of the cloud that are genuinely apart.

Second, the function is carried onto that complex by the *lower-star* rule: a vertex v enters the filtration at $-f(v)$, and a simplex enters when its last vertex does, that is at the maximum of $-f$ over its vertices. Sweeping the parameter upwards then reproduces exactly the family $S(c)$ with c decreasing, and the persistence of this filtration is computed by the standard algorithm.

Third, the diagram is read for the dimension the feature requires — dimension one for a closed curve in the plane.

Which complex. The complex is fixed and the function sweeps, so every hole of the complex itself is a hole at every level: it is born when the filtration starts and never dies. Such a class is called essential, and since a diagram cannot carry an infinite death it is clipped at the end of the sweep, which gives it the largest persistence present. The decision rule of the next paragraph then measures the loop of interest against it.

This is what decides the choice between the two constructions, and Figure 8 shows the difference on a cloud with one empty region in it. A radius graph is a complex *inside* the sample: wherever the cloud has a gap wider than r, the graph has a hole, and that hole is a hole of the substrate — permanent. A Delaunay triangulation of the same points fills the convex hull instead, spanning the gap with a few long thin triangles. Its substrate is contractible, so it has no essential classes at all, and every bar in the diagram belongs to the function rather than to the sampling.

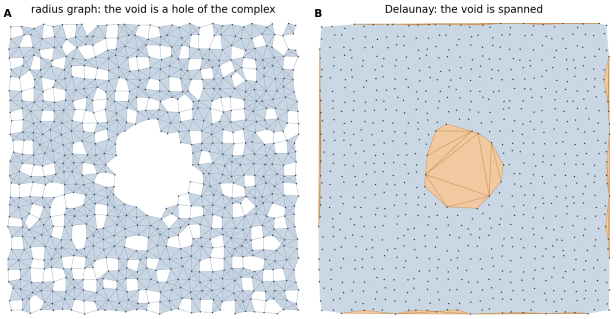


Figure 8: A cloud with an empty region, and the two substrates. (A) the radius graph, where the empty region is a hole of the complex and stays one at every level of the sweep. (B) the Delaunay triangulation of the same points, which spans the region with a few long triangles (in orange, together with the other simplices longer than the radius graph would join). The second substrate fills the convex hull, so it carries no holes of its own.

On a grid the two agree about where the points are, but not about the substrate, and the difference is large enough to change verdicts. Taking the 25 neurons of the trained network of Figure 4, the same field samples and the same decision rule, and

changing nothing but the complex: the radius graph returns a witness for 0 of them and the Delaunay complex for 16. All runs reported below use the second.

Locality. A persistent cycle somewhere else in the box says nothing about *this* feature, so before the filtration is built the sample is restricted to a ball around the feature's centroid, of one and a half times the feature's own radius. This is the price paid for not searching: the score only ever sees that neighbourhood, and the shape of what is inside it does not enter the answer. A fake target three times too large, or displaced by a full radius, returns the same verdict to three decimals; and when the feature is elongated the ball grows with its longest axis, which costs specificity in a way that Section 10 measures.

The verdict and its margin. A neuron counts as a witness when the longest bar of the required dimension satisfies two conditions at once: it is at least twice as long as the next longest bar of that dimension, and it exceeds an absolute floor set at a tenth of the range of values in the sample. Neither alone would do. Dominance alone lets a pair of tiny bars pass, since one of them may be twice the other by accident; the floor alone does not distinguish a single long loop from a diagram full of comparable ones.

Beside the verdict a continuous quantity is reported, the *margin*: the persistence of the longest bar minus whichever is larger, the runner-up or the floor. It is a useful number — it says how far from the decision the neuron is — but it is not equivalent to the verdict, and the two must not be confused: a bar can exceed the runner-up by a little, giving a small positive margin, while failing the dominance requirement, which is a ratio. An empty diagram is reported as undefined rather than as zero, since "no cycle was found" and "a cycle was found and lost" are different facts and a zero would place the first above every instance of the second.

The bar under Figure 7B is this arithmetic on the neuron shown there: one bar of dimension one, of length 1.85, against a floor of 0.20 and no runner-up at all, so the margin is 1.65 and the verdict is a witness.

The complement of the structural blind spot. What this score cannot say is *where* the cycle runs, and Section 10 is the experiment in which the two come apart: the elongation of a feature can take the structural verdict from one to almost zero while the homological verdict does not move at all, because the loop is still there — just not on the feature.

5.5 The harmonic shape descriptor

Both scores answer a question of the form "is the feature visible here". The experiments of Sections 9 and 10 need something else: given that the network has drawn a curve between the two objects — the feature and its interior — *whose shape is it*, *which way does it point*, and *where does it sit*. A number that compresses those three into one cannot answer, so the third instrument is not a score at all but a description.

A closed curve around a center is written in polar form and expanded,

$$r(\theta) = r_0\Big(1 + \sum_{k\geq 1} a_k \cos\big(k(\theta - \phi_k)\big)\Big),$$

which splits it into three parts that can be read separately: the *center* says where the curve is, the mean radius r_0 says how big it is, and the coefficients (a_k, ϕ_k) of the normalised profile say what shape it has. The harmonic $k = 2$ is elongation, ϕ_2 is the direction of the long axis, and $(1 + a_2)/(1 - a_2)$ is the familiar major-to-minor ratio of the ellipse that has the same second harmonic.

Figure 9 shows the three parts on a level curve taken from one of the runs of Section 10. Its mean radius is 1.52 and its second harmonic is 0.355, an aspect ratio of 2.10 — and that pair of numbers is not its shape. The first harmonic is 0.112 and the third 0.187, and an ellipse has no odd harmonics at all: they are the statement that the curve is not symmetric about its own centre, as the picture shows it is not. Reconstructing from the harmonics up to $k = 2$ therefore returns an ellipse with the right size, the right elongation and the right direction, which still misses the curve; adding the next four gives one that is hard to tell from the original. Shape here is a vector, and the comparison of two shapes has to be a comparison of vectors.

Two shapes are compared by the distance

$$d(A, B) = \sqrt{\tfrac{1}{2}\textstyle\sum_k \|u_k - v_k\|^2},$$

where $u_k = a_k(\cos k\phi_k, \sin k\phi_k)$ and likewise for v_k, the two coefficient vectors being compared. This is not an arbitrary choice of metric: by Parseval it is the L^2 distance between the normalised radial profiles, so $d = 0.10$ means a root-mean-square disagreement of ten per cent between the two radii, read directly as a fraction of the size. Keeping each harmonic as a vector rather than as a magnitude keeps the orientation: an ellipse along one axis and the same ellipse across it are at a distance from each other, which is what makes the experiment of Section 10 possible at all.

Summing over the harmonics is what makes the quantity a shape distance rather than a difference of elongations, and on a real pair the difference is not decorative. Between the curve of Figure 9 and the feature of its own cell the second harmonic carries 41% of d^2, the fourth 27% and the odd ones, which the feature does not have at all, 24%; the second harmonic alone would report 0.278 where the distance is 0.371.

What such a number is worth is fixed by the shapes the experiment actually puts in front of it. Figure 10 sets the four reference ellipses of Section 10 — aspects 1, 1.5, 2.5 and 4, all drawn at the same area — beside their harmonics and the distances between them. The smallest step the design asks the descriptor to resolve is a circle against a 1.5:1 ellipse, 0.144, and the whole span from a circle to 4:1 is 0.503. Against that, the error of the extraction itself, measured on a field whose level curves are known analytically and passed through the same code, is 0.010 once a curve is four grid steps across in mean radius — fourteen times smaller than the step it is asked to distinguish.

The expansion is truncated at $k = 6$, and the cut is measured rather than chosen: on a perfectly circular field sampled by the same grid every coefficient comes out below 0.004, so beyond that order the description is largely reporting the discretisation. What real structure is dropped with it is small: on the curve of Figure 9 the seventh and eighth harmonics carry 3.8% of the squared distance to its own feature, moving that distance from 0.371 to 0.379.

Which curve to describe. A score may take a minimum over the family of level sets; a description may not, since it would then be describing whichever curve happened to be closest to something, and the choice would be part of the answer. So the level has to be fixed by a rule, and the rule has to satisfy two conditions: it must be external to the neuron, and it must be symmetric in the two objects of the dataset — the interior and the feature. A rule that favoured either of them would prejudge exactly the question Section 10 asks.

Five curves are described rather than one, spread from the one object to the other; we call them the *rungs* of a ladder. Spreading them evenly is where a choice appears, because there are two spaces in which "evenly" can be meant, and they are the two sides of the map

$$c \longmapsto \text{ the curve of } \{f = c\} \text{ around the interior.}$$

The rungs may be spaced evenly in the *codomain* of that map — in the values of f — or evenly in its *domain* — in space. Both give a reference that is common to every neuron and every dataset; they differ in which space it is common in.

Evenly in the values. Let $\bar{f}_A$ be the mean value the field takes on the points of the feature and $\bar{f}_I$ the mean value it takes on the boundary of the interior — on the curve that bounds it, not

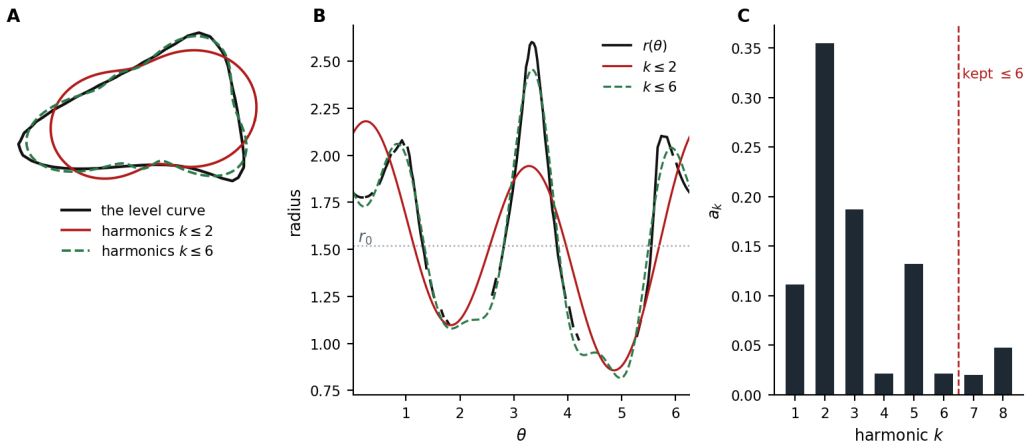


Figure 9: The descriptor on one level curve of one neuron — the middle rung of the ladder of Figure 11, on the median model of its cell. (A) the curve, and its reconstruction from the harmonics up to $k = 2$ and up to $k = 6$. (B) the same as a radial profile $r(\theta)$ about the curve's center, with the mean radius r_0 marked; the breaks in the profile are the angular sectors that received no point of the curve and are filled by interpolation. (C) the amplitudes a_k. The second dominates at 0.355, but the first, third and fifth are 0.112, 0.187 and 0.132 — an ellipse, which is what the $k \le 2$ reconstruction is, has none of them.

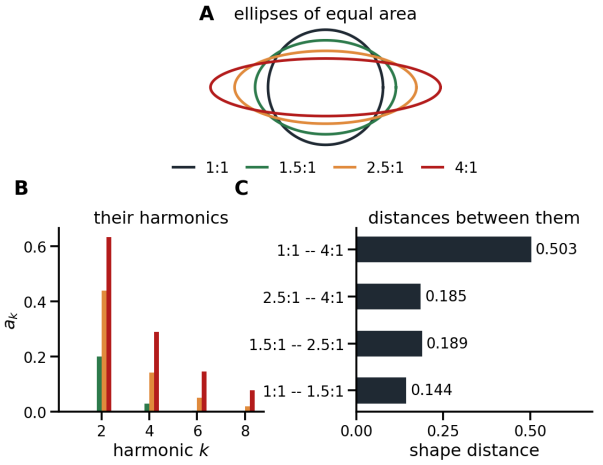


Figure 10: What the numbers mean. (A) ellipses of aspect 1, 1.5, 2.5 and 4, drawn at equal area — the four shapes the experiment of Section 10 builds its grid from. (B) their harmonics; an ellipse is almost purely a second harmonic, and elongating it raises a_2 and adds a little of the even harmonics above. (C) the shape distance between neighbouring pairs. The smallest difference the design has to resolve — a circle against a 1.5:1 ellipse — is 0.144, and the instrument's own error on a curve of this size is 0.010.

on the cloud of points that fills it. The field varies across the interior, so an average over the filled cloud is a different number, and it would move all five rungs at once. The rungs are the five levels

$$c_i = \bar{f}_I + \tfrac{i}{4}\left(\bar{f}_A - \bar{f}_I\right), \qquad i = 0, \dots, 4,$$

so that "rung 3" means the same fraction of the traversed range of values for every neuron.

Evenly in space. The other rule fixes five target radii, spaced evenly between the r_0 of that same boundary and the r_0 of the feature, and asks for the levels whose curves have those radii. A neuron cannot be asked that directly. What it offers is the map above, in the forward direction only — a level in, a curve out — so the rule has to be inverted, and the inversion is numerical.

Thirteen trial levels are placed across the same interval of values that the first ladder spans — the circles in panel C of Figure 11 — so that the two rules read one and the same stretch of the family rather than two different ones. At each of them the curve around the interior is extracted and its r_0 measured about *its own* centroid: what is being inverted has to be size alone, and a radius taken about a fixed point would report a curve that has moved as a curve that has grown. The table $c \mapsto r_0$ that this produces is then checked for monotonicity — a radius names a level only while the map is invertible — and each target is turned into a level by linear interpolation between the two trial levels that bracket it.

The targets themselves are not adjusted to what a given neuron happens to reach. A radius outside the range of its map comes back as a gap rather than as the nearest curve that does exist, because substituting the nearest curve would cost precisely what the construction is for: "rung 3" would then name a different place for every neuron, and two neurons would no longer be describing the same thing. Here "rung 3" means the same place in space for every neuron that has it at all.

Figure 11 puts both ladders on one neuron, and panel C says why they place their curves differently. The map $c \mapsto r_0$ of this neuron runs from 1.00 to 1.88 while the feature's own r_0 is 2.68, so three of the five target radii lie inside its range and the other two have no curve at all.

Which of the two is used. The experiments of Sections 9 and 10 read the ladder in values, for two reasons.

The first is about the neuron. Equal steps in value place the curves closer together *where the field is steeper.* A steep region is a fold of the field: the place where it swings from the values it takes on one class to the values it takes on the other, which is the part of the plane the network was forced to do something about — it had to change its output quickly over a small area there. A ladder in values concentrates the description on that fold by itself; a ladder in radius spreads the rungs evenly through space whether or not anything is happening at a given radius.

The second is availability. A level always exists, so a rung in values is defined for every neuron, while the inversion needs the map $c \mapsto r_0$ to be monotone, and over the neurons of Section 10 it is monotone for 52% of them. Counting the rungs that come back filled, from the inner one outwards, the ladder in values gives 0.68, 0.64, 0.59, 0.48 and 0.10 of the (cell, model) pairs, the ladder in radius 0.04, 0.64, 0.50, 0.19 and 0.05.

A missing rung means different things in the two. For the ladder in radius it means that the family contains no curve of that size. For the ladder in values the level exists by construction, and a miss means that at that level no curve closes around the interior: it breaks into pieces, or runs into the edge of the evaluation box.

From the grid to a profile. What the descriptor reads is a function $r(\theta)$; what the run holds is a field sampled at grid nodes. Two steps connect them, and each costs something that has to be reported with the number.

The first step is extraction. At a chosen level the curve is taken off the grid by marching squares: in every cell whose corners straddle the level the crossing is interpolated between the two

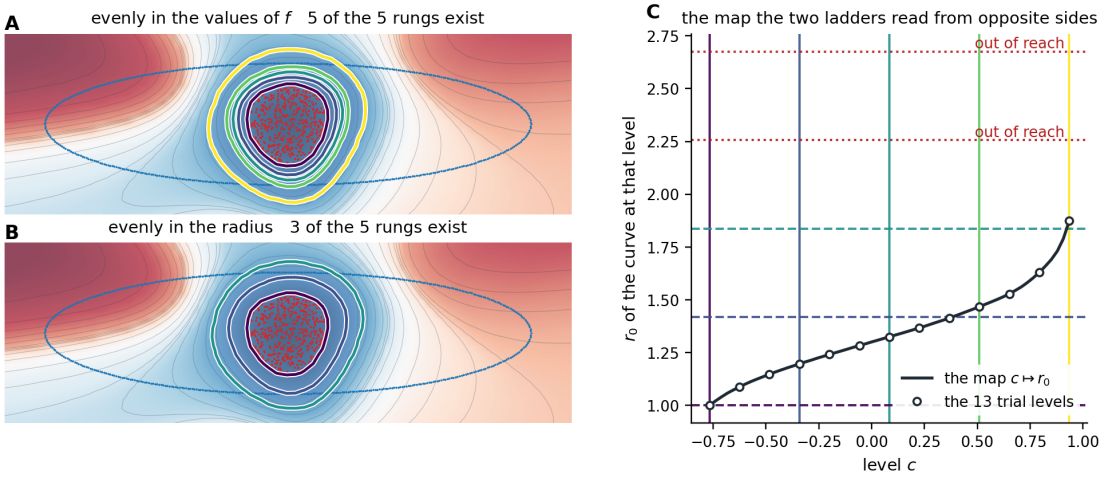


Figure 11: The two ladders on one neuron of one cell of Section 10 — a feature elongated 4:1 about a round interior, and the neuron of that cell whose two ladders come apart furthest. (A) the five rungs spaced evenly in the values of f. (B) the rungs spaced evenly in the radius; three of the five exist for this neuron. (C) the map $c \mapsto r_0$ that the two rules read from opposite sides, with the levels of the first ladder as vertical lines, the target radii of the second as horizontal ones, and the thirteen trial levels its inversion probes as circles on the map. The two radii above the range of the map are the two rungs that are missing from (B).

nodes, and the crossings are joined into polylines. The curve is then located to a fraction of a cell and carries no tolerance of its own — the band of Section 5.3, which the structural score reads off the same grid, is a different object with its own ε — and closedness becomes exact: a polyline is closed when it returns to its first vertex. Among the closed polylines at that level the one kept is the one that encloses the interior's centre, decided by counting how many times a ray from that centre crosses the polyline and calling the centre inside when the count is odd.

The second step is the profile, and it is needed because a polyline is a set of points in the plane and not yet a function of the angle. The points are binned by angle about the chosen centre into 180 equal sectors, and the radius of a sector is the mean radius of the points that fall into it. A curve drawn from a few hundred grid nodes does not reach all 180 of them, so the sectors that receive no point are filled by interpolation between their neighbours around the circle.

That filling is the cost, and it is measured rather than assumed: the fraction of sectors that were occupied is reported with every curve. On the curve of Figure 9 it is 0.76, and the gaps are visible in panel B as the breaks in the profile. It is a quantity about the instrument and not about the network — over the runs of Section 10 it correlates at 0.97 with r_0/h, the radius of the curve counted in grid steps, and at -0.01 with r_0 in units of length. A small curve is drawn from few nodes, is described largely by interpolation, and is flagged rather than trusted.

5.6 Chromatic TDA: relevance as a property of the data

The three instruments above all read a neuron. The fourth reads the data, and it is the one that makes the second level of Section 4.2 computable. It is also the only instrument here that this project did not build, so it is worth describing in its own terms before it is applied.

Persistence on a point cloud. The filtration of Section 5.4 was indexed by the value of a function. A point cloud carries a filtration of its own, indexed by scale. For a finite $X \subset \mathbb{R}^d$ and a scale α, the *alpha complex* C_α is the nerve of the balls of radius α about the points intersected with their Voronoi cells; by the nerve theorem it is homotopy equivalent to the union of those balls, and $C_\alpha \subseteq C_\beta$ whenever $\alpha \le \beta$. Homology with coefficients in a field turns this into vector spaces $H_p(C_\alpha)$ together with the linear maps $H_p(C_\alpha) \to H_p(C_\beta)$ induced by inclusion — a *persistence module*. A module of this kind decomposes into interval summands: each summand appears at one scale and disappears at another, and the multiset of those intervals is the barcode. A long interval is structure of the cloud; a short one is its sampling.

A coloured cloud, and two nested families. Let now $s\colon X \to \{0,1\}$ be a colouring of the points — in our case the class label. The chromatic alpha complex [7] produces, first, one filtration on the *whole* cloud: for every scale a complex L_α whose vertices are all the points of X, with $L_\alpha \subseteq L_\beta$ for $\alpha \le \beta$.

The colouring enters at the second step, and it enters as a *choice*. One of the two colours is singled out, and the simplices all of whose vertices carry that colour form a subcomplex

$$K_\alpha \subseteq L_\alpha,$$

which is again a filtration, and which is an alpha complex of the points of that colour alone. Singling out the other colour would give a different subcomplex and a different answer, so the choice is part of the question being asked. In everything below the choice is fixed once: the chosen colour is **blue**, which in every dataset of this work is class 0, the class that forms the curve; **red** is class 1, the class that fills the region the curve bounds. The chromatic construction exists precisely in order that these monochromatic parts be alpha complexes of their own points, which is what taking the ordinary alpha complex of the union and keeping one colour's simplices would not give.

The maps between them. Fix a homological degree p and coefficients in a field, and let $\alpha_0 < \alpha_1 < \cdots$ be the scales. Two kinds of inclusion act here. Growing the scale includes each complex in the next, which induces

$$\iota_i^K\colon H_p(K_{\alpha_i}) \to H_p(K_{\alpha_{i+1}}), \qquad \iota_i^L\colon H_p(L_{\alpha_i}) \to H_p(L_{\alpha_{i+1}}),$$

and including the subcomplex in the full complex at a fixed scale induces

$$\varphi_i\colon H_p(K_{\alpha_i}) \to H_p(L_{\alpha_i}).$$

All of them come from inclusions of simplicial complexes, which are chain maps, so the diagram

$$\begin{array}{ccccc} H_p(K_{\alpha_{i-1}}) & \xrightarrow{\iota_{i-1}^K} & H_p(K_{\alpha_i}) & \xrightarrow{\iota_i^K} & H_p(K_{\alpha_{i+1}}) \\ \downarrow{\scriptstyle \varphi_{i-1}} & & \downarrow{\scriptstyle \varphi_i} & & \downarrow{\scriptstyle \varphi_{i+1}} \\ H_p(L_{\alpha_{i-1}}) & \xrightarrow{\iota_{i-1}^L} & H_p(L_{\alpha_i}) & \xrightarrow{\iota_i^L} & H_p(L_{\alpha_{i+1}}) \end{array}$$

commutes: both ways around either square are induced by the same inclusion of simplices, already at the level of chain complexes, and homology is a functor. The family $\varphi = (\varphi_i)$ is therefore a morphism of persistence modules, and the object of interest is not the homology of either row but this morphism.

The six modules. Commutativity is what makes the constructions below persistence modules rather than a list of unrelated vector spaces. If $x \in \ker \varphi_i$ then $\varphi_{i+1}(\iota_i^K x) = \iota_i^L(\varphi_i x) = 0$, so ι^K carries kernels into kernels; and $\iota_i^L(\varphi_i x) = \varphi_{i+1}(\iota_i^K x)$, so ι^L carries images into images. Each of the six families below therefore inherits maps of its own, is a persistence module, and has a barcode:

- the **domain** $H_p(K_{\alpha_i})$ and the **codomain** $H_p(L_{\alpha_i})$;
- the **image** $\operatorname{im}\varphi_i \subseteq H_p(L_{\alpha_i})$ and the **kernel** $\ker\varphi_i \subseteq H_p(K_{\alpha_i})$;
- the **cokernel** $H_p(L_{\alpha_i})/\operatorname{im}\varphi_i$;
- the **relative** homology $H_p(L_{\alpha_i}, K_{\alpha_i})$.

Together they are called the *six-pack* of φ.

The six are not independent. The long exact sequence of the pair $(L_{\alpha_i}, K_{\alpha_i})$,

$$\cdots \to H_p(K_{\alpha_i}) \xrightarrow{\varphi_i^p} H_p(L_{\alpha_i}) \to H_p(L_{\alpha_i}, K_{\alpha_i}) \to H_{p-1}(K_{\alpha_i}) \xrightarrow{\varphi_i^{p-1}} \cdots,$$

in which the superscript is the homological degree and the subscript the scale — the sequence lives at one scale and moves in degree. It breaks into short exact sequences

$$0 \to \operatorname{coker}\varphi_i^p \to H_p(L_{\alpha_i}, K_{\alpha_i}) \to \ker\varphi_i^{p-1} \to 0,$$

so the relative barcode in degree p is assembled from the cokernel in degree p and the kernel in degree $p-1$. This is a usable check rather than a remark: in the cloud of Figure 12 every component of K stays a separate component of L, so $\ker\varphi^0 = 0$, and the relative and cokernel barcodes in degree one must agree — which, in the table below, they do.

What each of them says. Read in the language of the classification problem, with blue the chosen class and red the other one:

- the *domain* holds the holes of the blue points on their own, and the *codomain* the holes of the two classes taken together;
- the *image* holds the blue holes that are still holes once the red points are there;
- the *kernel* holds the blue holes that the red points fill in;
- the *cokernel* holds the holes of the whole cloud that blue does not produce on its own — holes of red, or holes that only the two colours together enclose;
- the *relative* module describes the cloud modulo the blue class.

The kernel is the one that carries the question of Section 4.2. A blue loop with red inside it is precisely the configuration in which a classifier has no choice: no continuous map that leaves the blue loop intact can separate the red points from it, so the network must break that cycle in order to solve the task. A blue loop with nothing inside is a shape the network may use or ignore; a blue loop that red fills in is a shape it is *forced* to act on. That distinction is what makes the kernel, and not the topology of either class alone, the computable form of task-relevance — and it is why the pair of datasets in this work differs in the kernel and in nothing else.

A cloud where all of this is visible. Figure 12 shows *one* point cloud, made of three configurations set far enough apart that nothing joins them at the scales that matter. One six-pack is computed, for the whole cloud, with blue the chosen colour as everywhere else; the three configurations are there so that all six modules come out non-trivial at once. They are:

1 eighteen blue points on a circle, with seven red points in a small cluster at its centre;

2 eighteen blue points on a circle, with nothing inside;

3 eighteen red points on a circle, with nothing inside.

All three loops are born at the same scale, $\alpha \approx 0.18$, when neighbouring points on a circle are joined, and a circle of radius one fills in at $\alpha \approx 1.0$. The six barcodes in degree one then read as follows, with the bars rounded to two decimals.

module	bars	what they are
domain $H(K)$	(0.18, 1.00) twice	the loops of 1 and 2
codomain $H(L)$	(0.18, 1.00) twice	the loops of 2 and 3
	(0.18, 0.40)	1, until it is filled
image	(0.18, 1.00)	2, all the way
	(0.18, 0.40)	1, until it is filled
kernel	(0.40, 1.00)	1, from the moment it is filled
cokernel	(0.18, 1.00)	3, absent from K
relative	(0.18, 1.00)	the same class as above

Every entry can be read off the two pictures. Configuration 2 contributes a loop to K that is still a loop in L: it appears in the domain, in the codomain and in the image, and nowhere else. Configuration 1 contributes a loop to K that stops being one in L at $\alpha = 0.40$, when the red cluster is joined to the blue circle: before that scale the class is in the image, after it in the kernel, and the two bars are the two halves of its life. Configuration 3 contributes a loop that exists only in L, since the blue subcomplex contains none of its points: it appears in the codomain and in the cokernel. And the relative barcode reproduces the cokernel, as the exact sequence above requires when $\ker\varphi_0$ vanishes.

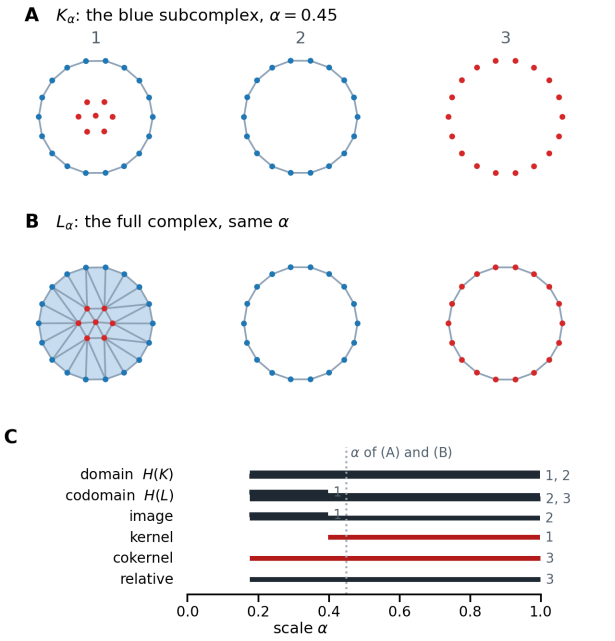


Figure 12: One cloud, in three separated configurations, on which all six modules come out non-trivial. (A) the subcomplex K_α spanned by the blue points at $\alpha = 0.45$ — the loops of configurations 1 and 2, while 3 has no blue points at all. (B) the full complex L_α at the same scale — the loop of 1 is filled by the simplices that mix the two colours, that of 2 is untouched, and that of 3 has appeared. (C) the six barcodes in degree one, computed once for the whole cloud; kernel and cokernel are drawn in red, the number at the right of a bar says which configuration it comes from, and the dotted line is the scale of (A) and (B).

On the matched pair. Figure 13 shows what the six-pack says about the two datasets of Figure 2, with the subcomplex taken to be the class that forms the curve. Both contain the same 500 points on the same circle, and in both the circle's loop is born at the same scale, $\alpha = 0.009$, as soon as neighbouring points on it are joined. From there the two histories differ completely. In the task-irrelevant dataset the loop stays non-trivial in the full complex until $\alpha = 1.50$, where it dies of its own accord — the circle's own disk fills in at the scale of its radius — and the kernel diagram is empty. In the task-relevant dataset

the interior of the opposite class bridges the gap at $\alpha = 0.30$ and fills the loop in; from that scale on the class lives in the kernel, where it stays until $\alpha = 1.50$, giving a single kernel class of persistence 1.20.

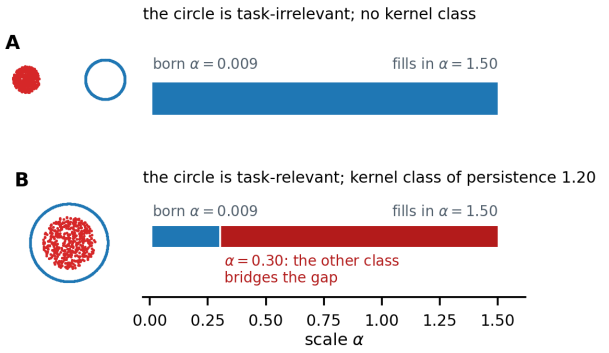


Figure 13: The life of the circle's one-dimensional class in the two datasets of Figure 2, with the subcomplex taken to be the class of the curve. Blue: the class is non-trivial in the full complex. Red: it has been filled in by the opposite class and lives in the kernel. The circle is the same 500 points in (A) and in (B), so the two bars begin together; what differs is whether anything ever fills the loop. The scale α is the one at which points are joined, in the units of the plane.

The informal argument of Section 4 and this computation therefore agree on the pair everything else is calibrated on, which is the one place where both can be checked.

What is used, and how. Of the six modules this work reads exactly one, the **kernel**. Its degree is one only because the features here are closed curves in the plane: the degree follows the feature. The chosen colour is blue, the class that forms the curve, and the computation is run once per dataset; its output is a single statement, that the dataset either carries a kernel class in degree one, with a persistence, or carries none.

That statement is about the data and the task and about no network, which is both its limit and its use: it can be made before anything is trained, and its job here is to certify the design. The matched pair is built so that the two datasets differ in whether the curve's loop is filled by the opposite class and agree in everything else, and the six-pack is what confirms that they do.

5.7 From scores to criteria: thresholds

A score is a number; a *criterion* is a score together with a threshold, and the threshold is a separate decision that has to be made and defended on its own. A neuron is called a *structural witness* for A when

$$\frac{\mathrm{Score}_{\mathrm{str}}(f, A)}{\mathrm{diam}(A)} \leq \tau,$$

and a *homological witness* when the margin of Section 5.4 is positive, that is, when the dominance and the floor are both cleared.

The value $\tau = 0.1$ was fixed by inspection: level-set components scoring below it look like the target up to small perturbations, larger values began to admit components that no one would call a match, and smaller ones began to reject components that plainly were. Figure 14 says what the number means in units anyone can check. A copy of the target displaced by about a sixth of its diameter scores 0.10, and so does a copy concentric with it but with its radius changed by a tenth of the diameter — that is, by a fifth of the radius.

Two things follow, and both shape the rest of the paper. First, since the two deformations in Figure 14 are equally admissible, a verdict says that the curve is close, not that it is close *in a particular way*; a claim about the size or the shape of what a neuron traces needs the descriptor, not the criterion. Second, and more consequentially, τ was fixed on one geometry and does not travel: as soon as the target changes shape, or the box around it changes, the number the same neuron would score

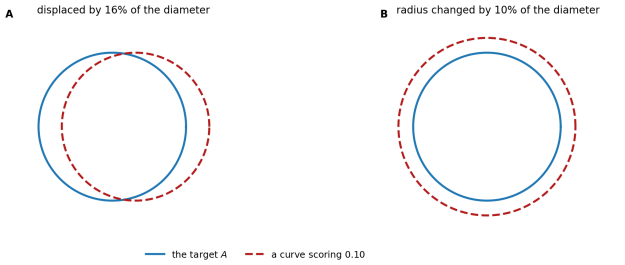


Figure 14: What the threshold $\tau = 0.1$ admits. Both dashed curves score exactly 0.10 against the solid target: in (A) by being displaced, in (B) by being the wrong size. The structural score does not separate these two ways of being wrong — both are a tenth of a diameter away from the target in the only quantity it reports.

changes with it. Section 6.5 replaces the fixed threshold by a rule applied afresh in every configuration, and for the same reason the continuous score, rather than the fraction of neurons above a threshold, is the quantity reported wherever a choice exists.

5.8 Why more than one instrument

None of the four subsumes another, and Table 1 is the short form of what each can and cannot see.

Where two of them agree, a claim is stronger; where they disagree, the disagreement is itself a measurement. Two of those found here carry results: the structural witnesses turn out to be a strict subset of the homological ones and never the reverse (Sections 9 and 10), and in the last layers of a deep network the two move in opposite directions (Section 8).

6 Protocol: how a number becomes a result

Section 5 ends with numbers: a Chamfer distance, the length of a bar, a vector of harmonics. None of them carries a scale of its own. A structural score of 0.04 is small — against what? A cycle was found in the ball around the feature — would one have been found there anyway? Supplying the comparison is what turns a number into a result, and it takes four things: something to read a number against (Sections 6.1 to 6.3); a unit of observation, since a network has many neurons and a run has many networks (Section 6.4); a rule that turns a number into a verdict where a verdict is wanted (Section 6.5); and a population to draw the conclusion about (Section 6.6).

6.1 Two error modes, and the three numbers they leave

A detector can fail in two directions, and the two are not established in the same way.

It can *respond where there is nothing to find*, and each instrument is exposed to this in its own way. The structural score minimises over some two hundred pairs of level and tolerance, and the more it searches the likelier it becomes that some component of some level lands near the target. The homological score does not search, but it asks only whether a cycle is present in a ball around the feature, and a cycle can be present there for reasons that have nothing to do with the feature. In neither case is this a defect to be engineered away — it is the price of what the instrument does — and its size is knowable only by running the instrument where the answer is known to be negative. That is the negative control of Section 6.2, and each instrument takes its own floor from it.

It can also *stay silent where there is something to find*. This one is not settled by a run at all: for any non-degenerate configuration the weights that put a feature into a level set can be written down, so a detectable case is exhibited rather than sampled. Section 6.3 exhibits it, and reads off it the one quantity a

	answers	**cannot see**
structural score	how close, in shape and position	which of the three is wrong
homological score	whether a cycle is there	where the cycle runs
shape descriptor	shape, size and position, apart	whether anything is a witness
chromatic six-pack	whether the task forces the feature	anything about the network

Table 1: What each instrument answers, and what it is blind to.

constructed case does give — the size of the measurement's own overhead.

The two marks a score is read between. Between them the two error modes leave a scale with two marks on it:

- the **floor** — what the instrument returns on a feature that is not there at all, measured on congruent copies of the target placed elsewhere (Section 6.2);
- the **ceiling** — the best it can attain in this configuration at all, which the discretisation fixes (Section 6.3).

On its own a score is a number; against the floor it is a claim; against the ceiling it is a claim with a size. Both marks belong to the configuration they were measured in, and neither is carried from one experiment to another.

What the instrument returns on a feature that is present in the data but not required by the task is *not* a third mark on this scale. It is the quantity Experiment 1 is built to measure, and where it falls is a result rather than a setting.

A floor belongs to a configuration. Sensitivity is not a property of an instrument alone but of the whole configuration — the geometry of the dataset, the architecture, the resolution and the field source, the instrument and its backend. Section 5.4 holds a measured instance: the same neurons, the same field samples and the same decision rule return no homological witness at all on a radius graph and sixteen of twenty-five on a Delaunay complex. A floor is therefore measured in the configuration it will be used in, and never inherited from another one.

6.2 The negative control

The floor is measured on *null features*: copies of the target placed where the task never involved it. A null is the image of the target under a rigid motion — a rotation about its own centroid followed by a translation — so it agrees with the target in shape, in size, in point count and in topology, and differs from it in one respect, which is where it is. Whatever an instrument reports for a null, it reports for an object identical to the target in everything except position.

The placements. They are assembled by hand for each dataset, so that between them they are representative of the problem: they sit in intuitively different parts of it — on the data of the opposite class, in the empty corridor between the two classes, beside the target, and outside the data entirely (Figure 5). How many there are is decided by the configuration rather than by a budget, since a placement only counts as a separate control if the configuration's symmetry does not identify it with another: a round feature inside a concentric round interior looks the same in every direction, and one placement is then the whole control; an elongated interior makes two axes inequivalent; displacing it distinguishes left from right; elongating the feature gives the copy an orientation of its own. The number runs from one to four across the cells of Experiment 4; it is four on the matched pair in Experiment 1 and two in Experiment 2, which keeps the pair of placements that can bind a verdict. Each placement has to sit where the instrument could respond at all — inside the evaluation box, on a part of the field that is not constant to numerical precision — and all of them are fixed in the experiment's specification before the run and stored beside the results, so that what a floor was measured on can be checked afterwards.

Because they are assembled for coverage, the floor they give is a floor *for this problem*, and the comparison it licenses is with the control itself: a target may be reported as no better than its own nulls, and nothing is said about where a uniformly random copy would have landed.

The floor is a distribution. Different placements give different numbers, and the spread between them is not noise to be averaged away: it is the width of the floor. The distribution is therefore what gets reported, and a verdict is read against its *softest* placement — the one that gives the detector the easiest time — rather than against the mean of it.

One control, several instruments. A null is a copy of the target, so it belongs to no instrument in particular: the same placements serve every instrument that takes the target as an argument. The structural score takes a floor from them and so does the homological score, and the two floors are different in kind — one bounds an accidental geometric match, the other an accidental cycle inside the localisation ball.

6.3 What a constructed witness gives

The mirror image of a null would be a neuron that is a witness by construction: weights arranged so that the feature lies in one of its level sets, with the verdict known before the measurement. Such a construction exists here and is used, but not as a control run per experiment. Planting replaces the weights that produce the measured neuron, so its verdict is a deterministic function of the feature, the box, the resolution and the instrument's configuration, and of nothing about the run. A quantity of that kind belongs in the regression gates that hold the instruments against stored values, which is where it lives.

Existence is not in doubt either. For a feature of positive diameter, in a network wide and deep enough to carry the construction, the weights can be written down and the neuron exhibited; where this fails, the configuration is degenerate in a way the experiments do not use, and Proposition 1 says exactly when that happens.

What it does measure: the overhead. The construction's geometric error is 0.0016 and the structural score returns 0.0343 for it, so ninety-five per cent of that number is what the measurement costs: a finite grid, a finite list of levels, a band of finite thickness. It is the decomposition that transfers and not the number. The number is not even a bound: 4 461 of the 30 300 neuron–target pairs of Experiment 2 score below it, the best at 0.0147, and on the homological side the probe's margin of 0.4124 stands against 1.63 attained. Figure 15 shows why. Overhead follows the profile of the field: the planted cap is a gentle dome whose values span 0.76, so the adaptive ε of Section 5.3 gives it a wide band, while a trained neuron that answers on the same circle is a steep wall spanning 1.62 and gets a thin one.

What is attainable at all. The bound comes from Section 5.2 instead. A band read off a grid cannot be located better than half a grid step, and the target is itself a sample with spacing $\delta(A)$, so the scale the two set between them is $\sqrt{(h/2)^2 + (\delta(A)/2)^2}$. In the configuration of Experiments 1 and 2 — $h = 0.068$, $\delta(A) = 0.019$, $\mathrm{diam}(A) = 3.0$ — that is 0.0117 after normalisation. The best trained neuron of Experiment 2's pool, at 0.0147, sits a quarter above that scale; the planted probe, at 0.0343, three times above it. The ceiling is set by the discretisation, and trained neurons come close to it. Where the grid is too coarse for this to hold, the probe says so

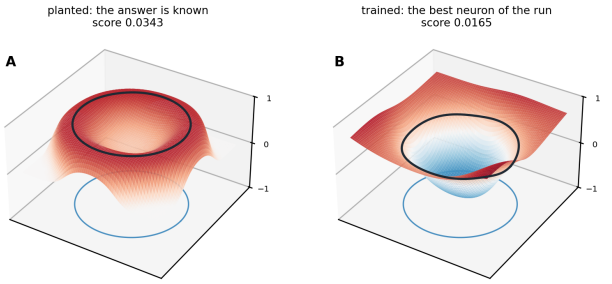


Figure 15: The constructed witness beside a real one, as surfaces over the input plane. **(A)** the planted neuron: the weights on the path into it are overwritten so that its ridge lies on the circle, drawn in black on the surface and in blue on the floor of the box. Its values span 0.76 and the dome is gentle. **(B)** the best-scoring neuron of the fifty task-relevant networks of Section 7, which was trained and not built: a steep wall spanning 1.62, with the same circle on it. The trained neuron scores better, 0.0165 against 0.0343, and the difference is the width of the band each of them gives the measurement.

directly, and Section 5.2 reads its answers off at five resolutions to fix the one used here.

What plays the part of a positive control for a network. What is wanted is a score that fires on a network which solved the task, and a planted field is a different kind of object — nothing in it was trained to solve anything. The control that gives what is wanted is a dataset in which the feature is *required*: the networks measured there are ones that solve the task, and the verdict is not a constant. That control is Experiment 1.

6.4 Units of observation

The three numbers of Section 6.1 say what a score is compared with. They do not say which score. Every run produces one for each pair of a neuron and a target — from a hundred thousand to more than half a million of them in the four runs reported here — and none can be summarised before it is fixed what counts as a single observation.

Three units are used, and they answer different questions.

The neuron. The unit is one neuron against one target. It answers *how much of the network does this*: the fraction of a model's neurons that witness, how that fraction is distributed over the layers, how many neurons a model has that witness at all, and the distribution of the scores themselves over a pool. The same unit carries the false-positive baseline, which in Experiment 1 is one neuron in ten thousand on the nulls against 41.9% of the neurons of the relevant ensemble below the threshold.

The best neuron of a network. The unit is a (dataset, model) pair, valued by the best score any of its neurons attains. It answers *was there any neuron at all*, which is the question a witness rate over models asks. Across the models of a cell such units are summarised by a median, with a bootstrap interval.

One described curve. The unit is a single curve taken off one neuron at one rung of the ladder, and there are some thousands of them in a cell. It answers *where the family as a whole sits*, which is the question the shape descriptor asks, so it is the unit wherever a shape rather than a verdict is reported. These are summarised by a median over all the curves at once, with no averaging per neuron or per model first, and by a median rather than a mean because the tail is populated by curves that barely closed.

Which unit is in force has to be said with every number, because they can move in opposite directions on the same run: in Experiment 2 depth raises the fraction of neurons that witness fivefold without improving the best neuron of the network at all. A statement about the best neuron and a statement about the population of neurons are different statements, and there the difference is the result.

One rule follows from the units and holds everywhere. **A bootstrap resamples models, never neurons.** Neurons of one network are not independent draws: the field of a neuron in the fourth layer is a function of every weight below it, so an interval computed over neurons would be an interval about a single network, not about the population the conclusion is meant to be about.

6.5 The decision rule

Everything so far is continuous; a verdict is not. Section 5.7 fixed one threshold by inspection, $\tau = 0.1$, and it cannot serve the whole work. The score is normalised by $\mathrm{diam}(A)$, so a threshold on it is a tolerance in diameters of the target — and the target is what the later experiments change: the same τ allows a circle of diameter 3 a tolerance of 0.3 units, and a 4:1 ellipse of the same area, diameter 12, one of 1.2.

The floor behaves the same way, and the floor is where a threshold has to come from: measured on each configuration's own nulls it runs 0.14 to 0.38 across the cells of Experiment 3 and 0.038 to 0.131 across those of Experiment 4.

The rule. The threshold is the first percentile of the scores of *that cell's own nulls*, computed afresh in every configuration and never inherited from another one. It is the operational form of the principle already stated for floors: a number is read against the configuration it was produced in.

Two things about it are worth stating plainly. A percentile is the **name of a rule**, not a measured error rate: on null placements that did not take part in setting it, the realised rate comes out two to three times the nominal one, so "first percentile" names how the line is drawn and not how often it is crossed by accident. And the choice of percentile is not delicate. Tightening it from the fifth to the first costs a witness rate of 0.94 against 0.76 in the worst configuration of Experiment 3, with the median over cells unchanged at 1.00, and on the task-irrelevant side it removes almost nothing, 0.08 against 0.06: what fires there is not threshold noise but genuine closed level curves that happen to lie near the target.

The second condition. A score below the threshold is necessary and not sufficient. A network is a composition of layers, so a feature singled out at layer l and singled out by no neuron of any later layer did not go further; whatever that neuron computed, the network's answer does not carry it. The verdict used from Experiment 3 onwards therefore has two parts: the score is below the cell's threshold, *and* the witness survives on every later layer including the head.

On the pool of Experiment 3 the first condition alone admits 296 false positives; adding the second admits none. The layer at which the chain breaks is recorded with each of them: the third for 165, the fourth for 96, the fifth for 35, counting the hidden layers from one. All 1 245 genuine witnesses pass both conditions. The head can be read this way because it is itself a scalar field: one linear output neuron on the plane. Read as one, it witnesses the relevant target in 100 models of 100, and is silent on all 47 irrelevant targets, on all 395 nulls of the relevant datasets, and on 407 of the 408 nulls of the irrelevant ones.

Where a verdict is not needed. A threshold destroys information, and it is applied only where the question is genuinely binary — how many models carry a witness, how many neurons do. Where a comparison can be made on the continuous score, it is made there: the score has a scale of its own once its floor is beside it, and a rate does not.

6.6 Ensembles, the training gate, and trainability

Every unit of Section 6.4 is attached to a model — a neuron belongs to one, a best-of is taken over one, a curve is read off one — so the population a conclusion is about is an ensemble: $N = 50$ independently initialised networks per cell. A single

network says nothing, since initialisation alone moves the best neuron's score by more than several of the effects measured here.

Each of the N has to earn its place. A network enters the ensemble only after it classifies the held-out fifth of its dataset perfectly for five epochs running, and that criterion is what is meant here by the *gate*. It is part of the design and not a formality, in two ways.

It is counted in successes. The training loop makes up to a hundred attempts and stops at the N-th success, so a cell that comes back with 34 models of 50 has not been quietly topped up by relaxing the gate: the ceiling is reported and the cell is left short. That is a statement about the task, which is what it should be.

The ensemble is conditioned on passing it. Success rates differ across a grid — 50 of 50 attempts at one edge, 50 of about 74 at another — so the models being compared are not equally representative of the networks one would train. The rate is therefore reported as a second outcome rather than being left implicit.

Read as an outcome, the rate is a second instrument on the same grid, and it does not always agree with the score. Where the two part company the disagreement is the informative half: trainability measures how hard it is to reach perfect separation, the structural score measures where the boundary that was learned ends up, so a variable that moves one and not the other separates the difficulty of the task from the place of the boundary. Experiment 3 has such a variable.

The gate also acts on comparisons made across architectures, and not neutrally. In Experiment 2 the epoch at which the gate closes varies fourteenfold along the width axis — a median of 124 epochs at width three against 9 at width forty — so a width sweep is also a sweep of training length. Where that confound runs against the conclusion drawn, as it does there, the conclusion survives as a lower bound; where it would run with it, the comparison is not made.

What the four experiments share. The machinery is now complete, and it is the same in all four: an evaluation box and a grid fixed before the run (Section 5.2), a target and its null placements (Section 6.2), a declared unit of observation, a threshold taken from that configuration's own nulls, and an ensemble of $N = 50$ models — fewer in the cells where 50 could not be trained — whose success rate is reported with the rest. What changes from one experiment to the next is a single thing each time, which is what makes the four readable as one argument.

7 Experiment 1: does a neuron see what the task needs?

7.1 The question, and the design that isolates it

Sections 4 to 6 built a question, four instruments and a protocol. The first measurement on trained networks has to answer the existence question — *is there anything here at all* — in a form that geometry alone cannot answer, and the matched pair of Figure 2 is that form. The blue circle is the same 500 points at the same radius in the same place in both datasets. In one the red class lies outside it and a straight line solves the task; in the other the red class fills its inside, so separating the classes *is* separating the inside of that curve from its outside.

On each dataset $N = 50$ networks were trained through the gate of Section 6.6 — tanh, five layers of width five, hence 25 neurons each. Every one of those neurons was scored against five features: the circle, and the four null placements of Figure 5. Both scores were computed for every pair. The unit of observation is the (model, feature) pair valued by its best neuron (Section 6.4).

Two numbers have to be settled before any of that can be read, because the structural score does not define itself without them: how fine the grid is, and how many levels are swept. They are fixed first, in Section 7.2. The experiment then asks six questions, in this order:

1. can the instrument answer where the answer is known in advance?
2. what does it return where there is nothing to find?
3. what does it return on a feature the task does not need?
4. what does it return on one the task does need?
5. do the two differ?
6. did the choice of settings decide any of it?

7.2 Fixing the instrument before measuring with it

The resolution of the grid and the number of levels are free parameters of the structural score (Sections 5.2 and 5.3), and a number produced before they are fixed means nothing. They were therefore chosen first, by two sweeps run over the same ensembles: five resolutions at 101 levels, and six level counts at resolution 200, with the level grids nested exactly one in the next, so that "more levels" means the same levels plus new ones. Together the sweeps make ten distinct configurations, the working point shared, and 200 000 scored rows.

How the resolution was chosen. The rule is to take the smallest setting at which every reported quantity has settled, and two quantities are reported. The first is dimensionless: p_{sup}, the probability that a unit drawn at random from the target beats a unit drawn at random from the floor, where 0.5 would mean the two are indistinguishable. It carries no units and no scale, so a plateau in it is a statement about the verdict rather than about the numbers, and Figure 16A shows it flat across the whole axis, with successive steps inside a tolerance of 0.01 from 100 nodes per axis on. The second is the counted verdict, how many of the fifty models witness; its grain is 1/50 and it settles one step later — two models fail τ at 100 nodes, one at 200 and one at 400.

Resolution 200 is the smallest value at which both have settled, and doubling it again is the expensive half of the axis: 17 320 seconds of evaluation for the two ensembles against 6 834, for p_{sup} unchanged to four decimals, the same count of models, and a floor that moves by 0.009.

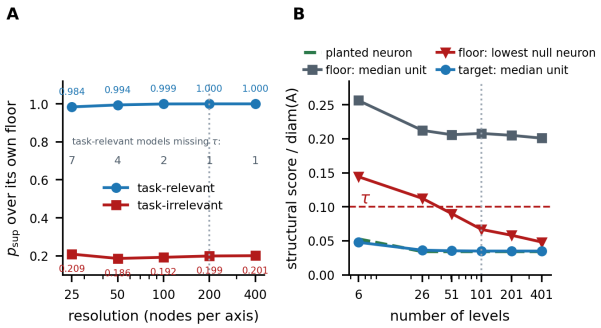


Figure 16: Choosing the configuration. **(A)** p_{sup} of each target over its own floor, against resolution, with the values printed since the steps are small; under them the second quantity the choice waits for, how many of the fifty task-relevant models miss τ, which settles at 200 nodes — the dotted line. **(B)** the same design against the number of levels. The target and the planted neuron stop moving at 26 levels; the median of the floor barely moves; the lowest of the 10 000 null neurons of the design keeps falling and crosses τ, because the score is a minimum over levels and more levels are more attempts.

What never settles, and why it matters later. The low end of the floor does not. The structural score is a minimum over levels, which makes it an extremum statistic: more levels are more attempts at the same field, and an extremum over more attempts can only fall. Figure 16B separates the two behaviours. The target stops at 26 levels and never moves again (0.0349 from

there on), and so does the planted probe of Section 7.3 (0.0343); the median of the floor drifts by under a tenth; but the lowest of the 10 000 null neurons falls from 0.1443 to 0.0482, crossing τ on the way, and the number of false positives goes from one to three. Nothing about the networks changed between those columns.

This is a property of the instrument, not of the experiment, and it is the reason the configuration is frozen once rather than chosen per run. It also has a consequence that Experiment 2 will collect: the number of attempts is exactly what grows when a network is made wider.

How the level count was chosen. By the same rule, with one more condition that the level grid imposes on itself. The adaptive tolerance of Section 5.3 is cut from below at half a level step, so with few levels the tolerance stops being a property of the field and becomes a property of the sweep: at 26 levels that cut-off sets ε for 41% of the target's neurons, at 51 for 6%, at 101 for 0.2%. One hundred and one is the first value at which the target is measured with the tolerance it is supposed to have, and it is also the last at which the middle of the floor is still on its plateau: the step from 51 to 101 is exactly at the tolerance of 0.01, and the two after it are 0.005 and 0.002. Going further buys nothing that is reported — p_{sup} within 0.0002, the same 49 models of 50 — and costs margin: the lowest null neuron falls 0.0667 → 0.0582 → 0.0482 and the false positives go from one to three.

Both settings were chosen against their own interest, which is checkable. A resolution of 400 gives a *larger* gap between target and floor than 200 (0.1632 against 0.1509), and 51 levels a larger one still (0.1604) — the latter because at 51 levels the cut-off sets ε for a quarter to a third of the *null* neurons, which lifts the floor, and a lifted floor flatters the result. What transfers to the later experiments is not the number 200 but the dimensionless ratio it realises, $N_A \approx 40$ grid steps across the feature (Section 5.2); the threshold used below is the fixed $\tau = 0.1$, this being the configuration it was fitted on (Section 5.7).

The ten configurations are used once more at the end, in Section 7.7, to ask whether the choice decided anything.

7.3 Where the answer is known in advance

The planted construction of Section 6.3 is run first in the frozen configuration (Figure 15A). Its true geometric error is 0.0016 and the structural score returns 0.0343: a witness, with a factor of three to spare under τ. The implementation answers on a field that contains the feature.

That is the whole of what it establishes, and in particular it is not an anchor for what follows. The number is a high one by the standards of this ensemble — the best neuron of the fifty task-relevant networks scores 0.0165, half of it, and Figure 15B is that neuron — because the overhead of the measurement follows the profile of the field, and the planted cap is gentle where a trained wall is steep. The scale on which the results below are read therefore has to come from the other end: from what the instrument returns where there is nothing to find.

7.4 Where there is nothing to find

The four null placements of Figure 5 are congruent copies of the circle, one in each intuitively different part of the problem, and they are scored on the same networks as the target. Eight cells — four placements by two ensembles — give medians running from 0.1650 to 0.3675: the floor is a distribution.

Which part of that distribution binds is decided by what a verdict claims. The score is a minimum and the claim is existential — some neuron of this network answers on this feature — so what a target has to beat is the *low end* of the floor: the best that a copy of the circle achieves where the task never involved it. Over the fifty models of an ensemble and its four placements, the two hundred null units reach down to 0.1144 on the task-relevant side and to 0.0667 on the task-irrelevant one, and the fifth percentile of those null units is 0.1378 and 0.1421 respectively.

At $\tau = 0.1$ that gives a false-positive rate over the whole design of one unit in four hundred, 0.25%, and one neuron in ten thousand, 0.01%. The one crossing is the 0.0667 above, and it is looked at in Section 7.6.

7.5 The two targets, and what separates them

Where the task does not need the circle. No model has a witness: 0 of 50, and the median unit scores 0.3923. The comparison that matters is with the low end of its own floor, and the target does not come near it (Figure 17, the two lower rows): not one of the fifty units reaches even the fifth percentile of the two hundred null units (0.1421), and the best of them, 0.1547, is more than twice the lowest null unit and half again above the threshold. The instrument does not answer on this circle, and the failure is not a near miss: over the ensemble $p_{\text{sup}} = 0.1991$, one target unit in five.

So a feature that is present in the data but not required by the task leaves the instrument nothing intermediate to report: on this pair it produces nothing that reaches even the easiest part of the floor.

Where the task needs it. A witness in 49 models of 50, with a median unit of 0.0349 (Figure 17, the two upper rows). Against the low end of the floor the separation is almost complete: all fifty target units lie below the fifth percentile of the floor, and forty-nine of the fifty lie below its *minimum* — the two populations are disjoint but for a single unit, which is the model examined in Section 7.6. In the middle of the distributions the gap is +0.1509, and +0.1300 against the softest placement; $p_{\text{sup}} = 0.9995$. Nor is this one lucky neuron per network: 41.9% of all 1 250 neurons of the ensemble score below τ.

The two together. The same circle, two tasks, and the separation between the two ensembles is complete: $p_{\text{sup}} = 1.0000$, the worst task-relevant model at 0.1304 ahead of the best task-irrelevant one at 0.1547 by 0.0243, and not one of the 2 500 pairs overlapping (Figure 17). This is not a significant difference between distributions but two sets with nothing between them.

The other instrument agrees. The homological score reads the same neurons and asks a different question — whether a cycle is there, not where a curve runs — and it separates the two datasets at least as sharply. On the task-relevant target the median best margin is 1.6273 against −0.0416 on the nulls — the margin is a length, so it is read the opposite way from the structural score and more of it is the stronger detection — and a bar of the required dimension is present in 50 units of 50 against 84% of null units. Two instruments measuring different things return the same verdict on the same networks.

7.6 The two exceptions, looked at

Two units of the whole design go against the counts above: one task-relevant model of the fifty has no witness, and one null unit of the four hundred has one. Each is worth a picture, because each shows something about what the instrument reads (Figure 18).

The model without a witness solves the task perfectly and does it with the wrong shape (Figure 18A). Its decision boundary, and the deep level sets that follow it, form a rounded square inscribed in the circle: a closed curve separating inside from outside, topologically what the task requires and geometrically not the circle. Its best neuron scores 0.1304. The structural score measures geometry, and this is what that means in one network — the failure is a different realised shape, not noise. It is also the first appearance of a distinction the last experiment is built on.

The false positive is a neuron in the second layer of a task-irrelevant network, answering on the null placement above the target with 0.0667 (Figure 18B). Its level set is an ellipse that

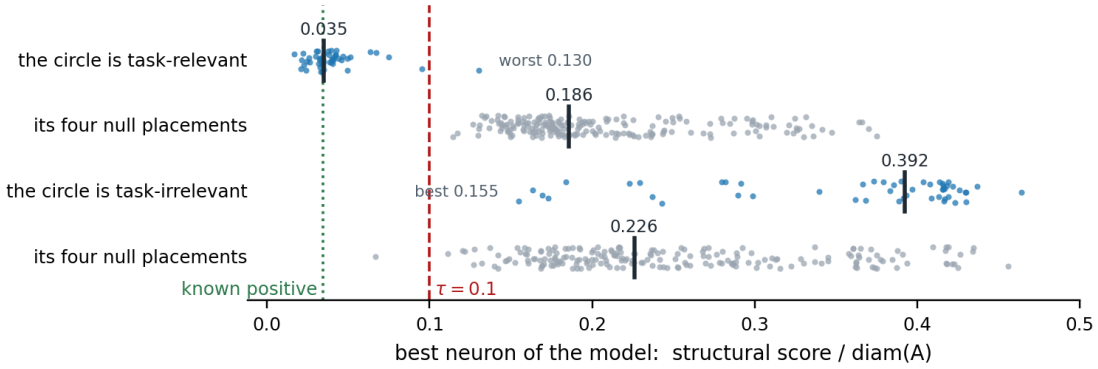


Figure 17: Experiment 1 on one axis. Each dot is one unit — one model valued by its best neuron — and the bar is the median of the row. The circle is the same 500 points in both datasets; only the task differs. The two target rows do not overlap: the worst task-relevant model scores better than the best task-irrelevant one. Read against the low end of the floors, the task-relevant target lies below the lowest null unit in 49 cases of 50, and the task-irrelevant target reaches none of the floor at all.

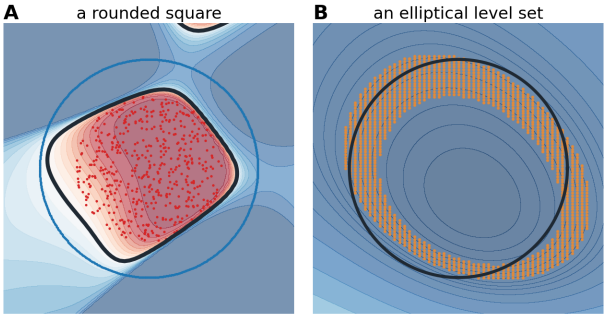


Figure 18: The exceptions. **(A)** the one task-relevant model of 50 without a witness, drawn as the field $\sigma(F(x))$ it computes with its 0.5 contour in black: the same 100% accuracy as the rest, reached with a rounded square inscribed in the circle, and a best neuron scoring 0.1304. **(B)** the one false positive of the design: a second-layer neuron of a task-irrelevant network whose level set (orange) falls around the null placement above the target, scoring 0.0667.

happens to fall around the copy, and nothing in the network carries it any further: in every later layer of the same model the best score on that placement is 0.42 or worse. The second condition that Experiment 3 will add to the verdict — that a witness survive on the layers after it (Section 6.5) — removes this one on its own.

7.7 Did the choice of settings decide any of it?

The two sweeps chose the configuration; the same ten points answer the last question, since every quantity above was computed at each of them. Nothing that is reported moves. No task-irrelevant model has a witness in any configuration, which is 500 (model, configuration) pairs without an exception; the sign of the gap never changes, positive on the relevant dataset and negative on the irrelevant one; p_{sup} on the relevant dataset never falls below 0.98. What does move is the absolute level of the floor, the count of false positives among the nulls (0 to 3 of 200), and the absolute score — none of which is what the experiment reports. Complete separation of the two targets is the one claim that needs a minimum: it holds from $N_A \geq 22$ and 26 levels, and below that the two populations touch.

What Experiment 1 establishes. That there is something to measure, and that the instrument measures it. Where the circle is task-irrelevant, no model of fifty has a witness and the target does not reach even the softest part of its own floor; where the same circle is task-relevant, forty-nine of fifty do, and every one of the fifty units lies below the fifth percentile of the floor. The two ensembles do not overlap at all, and the homological score, reading the same neurons for a different property, separates them as sharply. None of it turns on the settings: the verdict is the same at all ten configurations of the sweep. One reservation belongs to the instrument and returns in Experiment 4: the structural score is a minimum over levels, so what has been established is that a level set repeating the circle *exists* among a neuron's level sets, not that the neuron is built around the circle.

What is carried forward. Not 200 nodes but $N_A \approx 40$ grid steps across the feature; 101 levels; the adaptive ε; the unit of observation and the bootstrap over models (Section 6.4); the floor read as a distribution, with the verdict compared against its softest placement (Section 6.2); and, as a minimum for later designs, two placements rather than four, both away from the decision boundary and in two different directions from the target — the second is not a duplicate but the check that the soft direction was found.

8 Experiment 2: is it an artefact of capacity?

8.1 The question, and what "more network than the task needs" means

Experiment 1 measured one architecture, fixed in advance, and the most economical objection to it needs no new mechanism: twenty-five neurons are more than a circle and a disk require, and a network with units to spare might answer on whatever it is asked about. Two things have to be settled before it can be tested — how much more twenty-five is, and how to read a score that falls with the size of a network for reasons that have nothing to do with what the network learned.

How much is too much. The bottom of the axis is fixed by a theorem rather than by a measurement. A network whose hidden layers are no wider than its input has only unbounded level sets (Section 3.4), and its decision boundary is a level set of its output — so such a network cannot fence off a bounded region of the plane. The task-relevant dataset asks for exactly that fence, since the class inside the circle has to be separated from the class outside it (Figure 2B). Two neurons per layer are therefore never enough, at any depth, and three are: at width three the task is solved at every depth, in 47 initialisations of 50. What depth adds at that width is speed rather than power — at $d = 1$ the gate closes after a median of 2 439 epochs against roughly a hundred at any greater depth — so the practical minimum is six neurons, two layers of three. On the task-irrelevant dataset, where a straight line separates the classes, one neuron suffices.

Against that minimum, the networks of Experiment 1 are eight times the size, and the widest cell below is forty neurons in each of five layers.

The pool. A cell is named by its width and depth — $w = 5$, $d = 5$ for the architecture of Experiment 1 — and $S = wd$ is the

number of hidden neurons it has. Sixteen architectures were trained: a width row at $d = 5$ ($w = 3, 5, 10, 20, 40$, and $w = 1, 2$ as well on the task-irrelevant dataset, the only one where they can be trained at all), a depth row at $w = 5$ ($d = 1, 2, 3, 5, 8, 10$), and a diagonal of five shapes that all have the same $S \approx 24$. On the two datasets together that is 30 cells of $N = 50$ networks and 184 050 scored neuron–feature pairs. The evaluation box is written into the specification rather than derived from each dataset, so the grid is equally fine on both sides ($N_A = 44.2$ steps across the feature) and every score here is comparable with every score of Experiment 1; the floor is measured in each cell separately, from the two null placements that section carried forward. The reference cell repeats Experiment 1 to the sixth decimal.

How the numbers are read. Two of the units of Section 6.4 are in force, and they answer different questions: the share of a network's neurons that witness, and the best neuron of a network. The second falls with width by construction, since a minimum over more attempts is smaller even when the attempts are drawn from the same distribution — which is the reservation Experiment 1 ended on. The rule carried from there is that a score at width w is read only against a floor measured at width w; how much of an improvement multiplicity accounts for is then a question the nulls of the same cells can answer, because multiplicity is all they have behind them.

8.2 Excess capacity does not manufacture a witness

The task-irrelevant dataset carries the same circle as the other one, bit for bit, in a problem that is solved without it (Figure 2A). If spare units produce witnesses, this is where they would appear.

They do not. In fourteen of the sixteen cells no neuron witnesses at all — 0 of 16 050 scored pairs — and the whole dataset produces seven firings: four at $w = 20$ and three at $w = 40$, every one of them on the second or the third hidden layer. Three of the seven have their matched component cut by the edge of the evaluation box, so what their score describes is the box.

The four that remain look like genuine firings — closed level curves running around the circle, off-centre but around it, which is what the criterion is built to find. None of the seven, however, goes anywhere: Section 8.4 flags every one of them as a witness with no witness deeper than itself in the same network.

Two facts about the cells as wholes point the same way. The rate on the target never stands clear of the rate on its own floor — doubling the width leaves four models of fifty, then three, while the nulls of the same cells go from one firing in a hundred units to eight. And the lowest score anywhere on this dataset belongs to a null placement, the neuron already drawn in Figure 18B. The homological score, which asks whether a cycle is present rather than where a curve runs, agrees independently: it fires on fifty-two neurons of thirty-one thousand here, against fifteen thousand of thirty thousand on the task-relevant dataset.

What width does change here. Both curves of Figure 19 fall as the network widens, so neither means anything read alone. Between them the gap does narrow: the chance that a unit of the target beats a unit of the floor rises steadily with width, and the interval on that trend excludes zero. But it starts and ends far from parity — even at $w = 40$ a target unit loses to a null unit about two times in three — and along the depth row there is no such trend at all.

Nor does the narrowing point anywhere. The target's fall steepens over three steps of the row and then flattens again on the last one; had the row stopped at $w = 20$, those three steps would have read as an acceleration towards the threshold, and the fifth cell is what forecloses that reading.

One honesty condition attaches to all of this. The floor is a lower estimate by construction (Section 6.2): of the placements of Experiment 1 the two kept here are the ones that score low, because

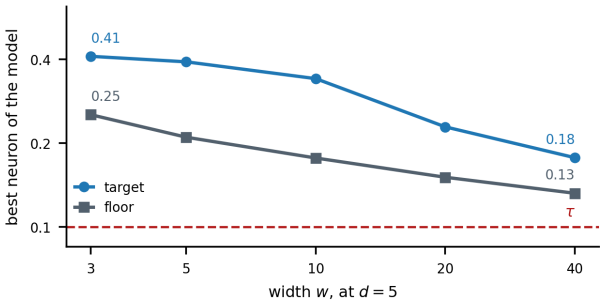


Figure 19: Width against the task-irrelevant circle. Each point is the median over fifty networks of the best neuron of a network; the floor is measured on the same networks, from null placements of the same shape. Both fall as the network widens, and at every width the target stays above its own floor: the instrument comes closer to an answer on a copy of the circle placed where the task never mentioned it than on the circle itself.

a floor is wanted as a bound and not as a typical value. The defensible statement is therefore that the score on a task-irrelevant circle stays above a lower estimate of what an arbitrary circle gets in this problem — not that the two distributions differ, and not that the network suppresses what it does not need.

8.3 A feature the task forces is re-used redundantly

The same architectures on the task-relevant dataset answer a different question: not whether a witness appears, but what happens to one when the network is given room.

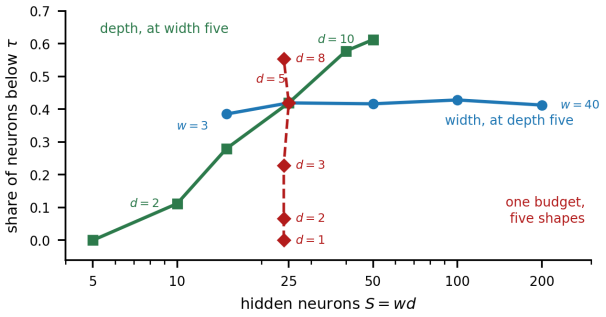


Figure 20: How many neurons of a network witness the task-relevant circle. Width leaves the share where it is and only adds neurons to it; depth raises it fivefold. The dashed column is the control for that comparison: five shapes of the same twenty-four neurons, where the network can only be rearranged, and the share climbs along it just as it does along the depth row.

Width multiplies witnesses without diluting them. At a fixed $d = 5$, the share of a network's neurons that witness stays at about two in five as w goes from 3 to 40 (Figure 20, the flat row); the trend fitted across the row is indistinguishable from flat. Their number therefore grows with the network, from six witnesses at $w = 3$ to eighty-two at $w = 40$: every neuron that is added picks the circle up with about the probability the earlier ones had, and none of the added width stands idle. Wide ensembles are also more uniform — at $w = 3$ a few models miss the threshold altogether and the best score scatters eight times more widely between models than at $w = 40$, where none misses it.

One thing that share is not. It is an average over all the neurons of a network, and the layers of a network differ from one another far more than these cells differ from one another (Section 8.4). What holds constant is the network-wide average; the per-layer curves behind it do move with width, in both directions, and they are read separately below.

Are the neurons better, or only more numerous? A wider network offers the search more level sets to choose from,

and the best of more attempts is better even when no single attempt has improved. How much of the gain is that alone cannot be computed, because how many independent attempts an architecture really offers is unknown. It can be measured instead. The null placements sit in the same cells, with the same neurons available and nothing to find, so whatever their own best score gains from width is multiplicity and nothing else. Sliding the floor's own curve down until it passes through the target at $w = 5$ therefore draws the dotted line of Figure 21A: where the best neuron would sit if width bought attempts and nothing else.

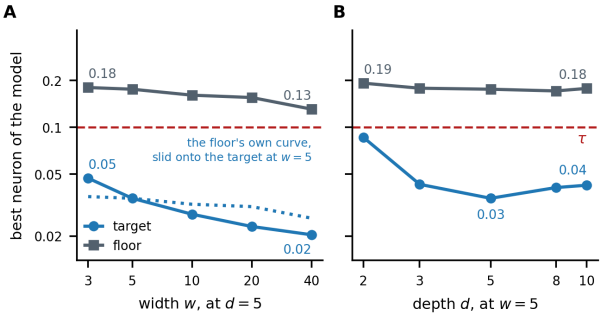


Figure 21: The best neuron and its floor on the task-relevant dataset, along the two axes of the pool and at one scale. **(A)** width, at $d = 5$; the dotted line is the floor's own curve slid down onto the target at $w = 5$, that is, what width would buy if it bought nothing but more attempts. **(B)** depth, at $w = 5$, with $d = 1$ left out since no neuron there can witness. The floor does not move along this axis at all, and the target stops improving after five layers.

The target runs below that line from $w = 10$ on. Across the row it loses more than half of its value where the floor loses a quarter, which on the logarithmic scale of the figure is a slope almost three times as steep, with the two intervals disjoint — so the neurons of a wide network really are closer to the circle, and not merely more numerous. The gain appears only past the reference cell, though: at $w = 3$ the best neuron is a third *worse* than the multiplicity line, and it stops gaining on that line at $w = 40$ — not because the target stalls, which it does not, but because there the floor itself drops sharply. With forty level sets to choose from in a single layer, the search finds a fair component even on a placement that is certainly wrong.

What lowers the floor is the width of a layer, not the number of neurons. Depth adds neurons too, and it does not lower the floor at all: along the depth row the floor is flat from $d = 2$ to $d = 10$ while S grows fivefold (Figure 21B). The two panels of that figure are drawn at one scale, so the comparison can also be made at equal S: at $S = 50$ the floor stands near 0.16 for $w = 10$, $d = 5$ and near 0.18 for $w = 5$, $d = 10$. The wider shape has the lower floor, which is to say the larger search.

The asymmetry is the measurement; the reading attached to it is that neurons of one layer are parallel attempts at the same field, whereas neurons of the next layer are functions of them — a longer chain is not a wider search. This is also what makes the rule of Section 8.1 operational: the multiplicity a cell has to be discounted for is set by its width.

Depth moves the share, and only the share. Over the depth row the share goes from 0.11 at $d = 2$ to 0.61 at $d = 10$ (Figure 20, the rising row), an effect an order of magnitude larger than width's. It survives the obvious deflation, since counting only the neurons that *can* witness barely changes it, and it survives holding the size of the network fixed: along the diagonal of five shapes at $S \approx 24$ the share still climbs from nothing at $d = 1$ to more than half at $d = 8$. What depth does not do is sharpen the best neuron (Figure 21B): that improves up to $d = 5$ and then drifts back, against a floor that stays where it is. Depth buys coverage and not sharpness; Section 8.4 says why.

8.4 How the feature is built across the layers

Everything so far counts over a whole network. Adding one axis — the index of the layer — turns the counts into a *profile*: how a property of witnessing depends on the layer, at a fixed architecture and a fixed feature. Hidden layers are numbered from one, as in Proposition 1, and the head is scored like any other scalar field and stands one place after the last hidden layer.

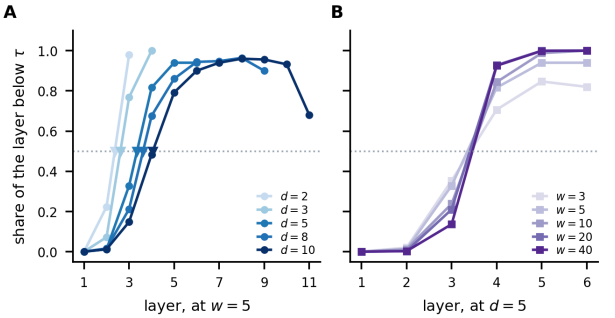


Figure 22: The share of a layer's neurons that witness, along the two axes of the pool; the head is the last point of every curve, so that all of them have common ends. **(A)** depth, at $w = 5$: the curve is flat, then rises over one or two layers, then runs just under one, and the triangles mark where it crosses one half. **(B)** width, at $d = 5$: widening empties the third layer and fills the last one.

The share of a layer does not grow steadily; it jumps. A profile is flat near zero for the first layers, climbs over one or two, and then runs along a shelf just under one (Figure 22A). It is the whole population of a layer that moves and not a tail of it crossing the threshold: the median neuron of a layer falls by an order of magnitude across the same layers, and it passes τ where the share passes one half (Figure 24A). The nulls have no profile anywhere in the pool — their share stays under half a per cent at every layer of every cell — and neither has the task-irrelevant circle, whose only non-zero entries are the seven firings already counted. So the rise is not a general property of deeper fields being curvier.

A network with no depth to spare has to start at once. The second hidden layer witnesses with 0.224 of its neurons when the network has two layers, and with 0.012 of them when it has ten (Figure 22A). Width does the same thing mirrored: on the third layer the share falls by more than half as w goes from 3 to 40, while on the last layer it rises until every neuron of the layer witnesses (Figure 22B). Both readings point one way — a network that has room can postpone the construction, and one that has none cannot — but this is an interpretation and not a measurement: these data do not separate a network that had no need to occupy an early layer from one that could not.

The rise sits $\sqrt{d}$ layers into the network. Measured from the first hidden layer, the five crossings follow $\sqrt{d}$, to within seven per cent at every depth — neither a fixed layer nor a fixed share of the depth (Figure 23).

A transfer rule comes with it, and is used everywhere after this: **a statement about a particular layer is valid only inside the depth it was measured at.** The profile of Experiment 1 is a statement about five layers, not about layer three.

The construction has a length, and a deep network is longer than it. By the construction is meant here the stretch of layers over which the share climbs from nothing to the shelf. In a five-layer network that stretch runs to the last layer; in deeper ones it ends earlier, and what comes after is not more of it — at $d = 10$ the profile turns back down over the last two layers (Figure 22A). That is not a share running out of room, since the quantity with no ceiling does the same: the median neuron of a layer stops improving around the same layer and then gets worse (Figure 24A, the two deepest curves). What stops is the building, not the measuring of it. That refines the depth result

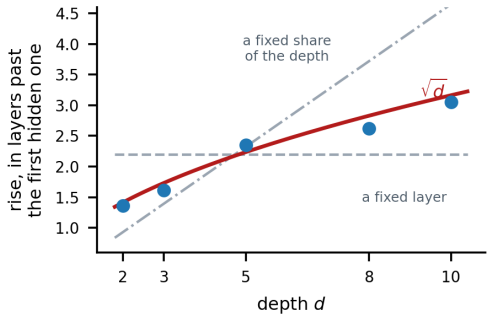


Figure 23: Where the rise sits, against the two laws it is not: a fixed layer (dashed) and a fixed share of the depth (dash-dotted). The five measured crossings follow $\sqrt{d}$ instead.

of Section 8.3, because the share over a whole network is by definition the average of its per-layer shares: a deeper network scores higher on it not because the feature is built better but because more of its layers stand after the building is over. Depth extends the feature through the network; it does not sharpen it, which is exactly what the best neuron of Figure 21B was saying.

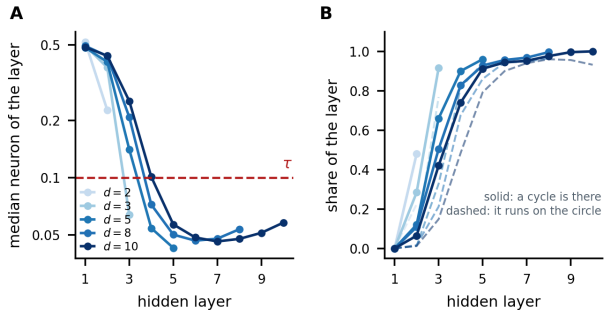


Figure 24: The same profile without a threshold and by the second instrument, at $w = 5$. **(A)** the median neuron of a layer, which needs no threshold and falls by an order of magnitude. **(B)** the share of a layer whose neurons carry a cycle at all (solid) against the share whose level set runs on the circle (dashed, the curves of Figure 22A): the cycle comes first, and at the end of a deep network the two part company.

What the homological score adds. It fires earlier, by about a layer: where a third of a layer traces the circle, two thirds of it already carry a cycle of the right kind (Figure 24B). A closed level curve appears before it becomes this circle, and the gap between the two curves is widest exactly at the rise. At the end of a deep network they part company in the other direction — the structural share slips over the last layers while the homological one climbs to one, and at no depth does the homological profile fall anywhere. Closed level curves do not disappear in the last layers; there are more of them, and their geometry stops resembling the circle. A plausible reading, and no more than that: on the early layers a level curve is unlike the circle because the neuron has not the means to make it so, in the middle it lands on the circle, and on the last layers the network goes on optimising a task it has already solved and arrives at something that solves it no worse while looking less like a circle. What the curves become instead is not measured here, and Experiment 4 is where that question is taken up.

A filter that does not use the threshold. A network is a composition, so a feature singled out at one layer and by no neuron of any later layer has gone nowhere: the head reads the last hidden layer, and whatever that neuron computed, the network's answer does not use it. Flagging such witnesses marks all seven firings on the task-irrelevant circle and four of the 11 073 on the task-relevant one — and those four are two cases, all the neurons of one layer of one minimum-width model and one neuron of one ten-layer model, both of which singled the circle out and then lost it. This is where the second condition of the decision rule (Section 6.5) comes from; it needs no new computation and does not depend on τ.

8.5 Where no neuron can witness, the network still traces the circle

One cell of the equal-budget diagonal is $w = 24$, $d = 1$: twenty-four neurons in a single hidden layer. By Proposition 1(i) none of them can witness anything, and the measurement returns exactly zero. The head of the same networks, read as what it is — one scalar field on the plane — witnesses the circle in 50 models of 50. The zero is thus a statement about neurons and not about networks: these networks do trace the circle, and no single unit of them traces it. At depth one a distributed encoding is not an interpretation of the data but a consequence of the proposition together with the measurement.

Read across the pool the head separates the two datasets as sharply as the neurons do: no head of a task-irrelevant network witnesses, 0 of 250, and the worst task-relevant head is still nearly twice as close to the circle as the best task-irrelevant one. That comparison carries its own caveat, which is why the head is not used as evidence elsewhere: on the task-relevant dataset the decision boundary *is* the target, so a head that witnesses is close to forced by the task. The informative layers are the middle hidden ones.

8.6 A confound the design had not named

The gate of Section 6.6 stops training at the first stable perfect accuracy, and on this task a wider network reaches it sooner: a median of 124 epochs at $w = 3$ against 9 at $w = 40$. The width axis of Section 8.3 is therefore also an axis of training length, running the other way and by a factor of fourteen, and every curve on it can be read as a curve against how long the network trained.

Observation cannot settle it. Within a cell, the correlation between the epoch of the gate and the best score is strongly positive in some architectures and strongly negative in others. Where it is positive it admits two incompatible readings — training longer made the network worse, or the runs that needed eighteen hundred epochs were the ones whose initialisation was poor — and no amount of the same data separates them.

The design. The early and the late measurement come off *one* run, so that the two differ in nothing but training length. Each network trains to three thousand epochs and its state is stored at its own gate, at ten times its own gate, and at the end — multiples of its own gate, because the gates of different cells differ by an order of magnitude and one absolute epoch would fall before the gate in one cell and long after it in another. Five architectures were run this way on each dataset, among them the $w = 24$, $d = 1$ cell of Section 8.5: no witness is possible there, so any movement of the score in that cell is movement of the instrument's baseline rather than of a representation.

What longer training does. It improves the feature the task needs, in every capable cell, and no interval on the paired difference covers zero (Figure 25B). Since the comparison is made inside a run, the variance over initialisations is gone from it entirely — and that settles the question of sign: the observed correlation was selection. Slow runs are worse because their initialisation was poor, not because they trained for a long time; made to train longer, the same run improves.

Two consequences follow. The confound runs *against* the conclusion of Section 8.3, since wide networks were measured after less training and less training is worse, so the advantage of width measured there is a lower bound. And the smallest network in the study turns out to be short of time rather than of capacity: at $w = 3$, $d = 2$ — six neurons in all — a witness is present in 41% of models at the gate and in every model after training on, where the median best neuron ends up level with the reference cell.

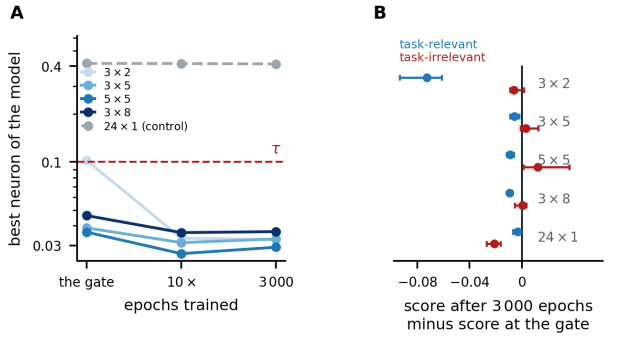


Figure 25: Training length as a variable of its own, measured inside single runs. **(A)** the median best neuron at the gate, at ten times the gate, and at three thousand epochs, on the task-relevant dataset; the dashed line is the cell where no neuron can witness, so its movement is the movement of the baseline. **(B)** the difference between the last rung and the gate, paired within each run, with bootstrap intervals over models: negative in every capable cell of the task-relevant dataset, and covering zero or positive on the task-irrelevant one.

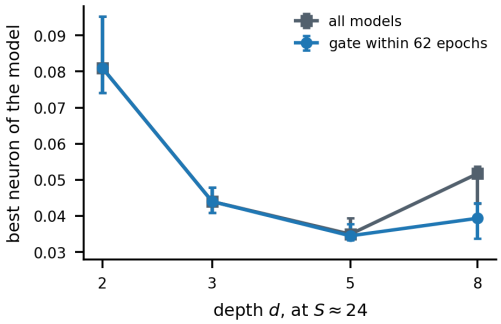


Figure 26: The equal-budget row, all models against those whose gate closed within 62 epochs. The two agree wherever training length agrees; at $d = 8$, the one cell that trains three times longer than the rest, the dip in the best neuron belongs to the extra training and not to the depth.

Past about ten gates, more training makes the structural score worse. The optimum is the middle rung in three cells of four, and by three thousand epochs a little of the gain is given back. The homological share meanwhile rises monotonically to the end: the closed curves go on multiplying while they stop resembling the circle — the same divergence as in the last layers of a deep network, and the same reading, a network that goes on optimising a task it has already solved.

On the task-irrelevant dataset nothing appears at any rung. Fifteen pairs of a cell and a rung, and fifteen zeros; the homological share rises in none of them. Three thousand epochs is up to two hundred gates of extra training, and it creates nothing. The scores there cannot be read at all, and the control cell is what says so: it drifts further than any target on that dataset moves. What is readable is the count, and the count is zero.

Is the effect the instrument's? The score sweeps a hundred and one levels laid across the observed range of a neuron's field, and three thousand epochs change that range out of recognition: in the control cell the weight norm grows sixfold and three quarters of its units end up saturated. A range that moves could therefore move the score on its own, with nothing about the level curves having changed. What rules that out is which cell moves. The control, whose field moved most, is where the score moved least — twenty-three times less than at $w = 3$, $d = 2$, whose field moved far less. A drifting scale would have shown itself where it drifted most.

The same confound, seen in the pool itself. Along the equal-budget diagonal one cell is out of line in training length — $w = 3$, $d = 8$, with a median of 125 epochs against 38 to 59 — and it is also the one cell whose best neuron is worse than the trend. Restrict every cell to the models that passed the gate within 62 epochs and the two curves of Figure 26 lie on top of each other everywhere except there, where the dip disappears: what looked like depth spoiling the best neuron was training length. A matched subsample is not an intervention — selecting on the epoch of the gate also selects on the initialisation — which is why the paired design above is what the conclusion rests on, and why this check is read only as a bound.

8.7 The verdict does not depend on the activation

Every network so far has been a tanh network, and the smoothness that makes level sets curves is an assumption of the whole setting (Section 3.2). The reference cell was therefore retrained with two other activations, with deliberately different roles. The logistic sigmoid is a control on the instrument: $\sigma(z) = (\tanh(z/2) + 1)/2$, so a sigmoid network and a tanh network run through the same class of functions and the answer is obliged to agree — a disagreement would be evidence about the score and not about networks. ReLU is the real test of generality: its fields are piecewise linear, so the regularity Section 3.4 works under is simply absent.

The verdict transfers whole. On the task-relevant dataset all three detect the circle — a witness in at least 96% of the models that trained, and p_{sup} against each activation's own floor above 0.995 — and on the task-irrelevant dataset none of them detects anything, on the target or on the nulls.

The numbers do not transfer. Each activation carries its own floor, and reading a score against it changes the order the raw scores give: a piecewise-linear field lands near an arbitrary circle less often, so landing near the right one is worth more. The floor is the scale — the rule of Section 8.3 along one more axis, and the reason no raw score is carried from one activation to another.

Two particulars belong to the reading of the numbers rather than to the verdict. Sigmoid witnesses with 0.741 of its neurons against tanh's 0.419 and yet its best neuron is worse: its level curves are more numerous — the homological score says so independently — and they lie around the circle more loosely. A share and a best neuron answer different questions, here as along the depth row. And a quarter of the ReLU units are constant on the feature, the third class of Proposition 1, so they cannot witness for the same kind of reason as a first-layer unit; counted over the units that can, ReLU witnesses more often than tanh does. Its homological score, on the other hand, fires ten times more often on the task-irrelevant circle — closed polygonal curves appear where a smooth field has none — so a homological threshold fitted on tanh would have to be recalibrated for it.

What Experiment 2 establishes. Capacity is not the explanation. No amount of it produces a witness of the task-irrelevant circle — not more neurons, not a layer eight times wider than the reference, not two hundred times longer training — and the few firings that do occur are indistinguishable from what the controls produce in the same cells. On the task-relevant circle capacity does something else entirely: a fixed share of whatever neurons are available carries the feature, so their number grows with the network, and width makes each of them better than multiplicity alone would. That correction is measurable only because the floor of the same cell measures multiplicity, and it is the width of a layer that sets it, not the number of neurons. The feature itself rises about $\sqrt{d}$ layers in, takes some four layers more to build, and afterwards is only carried, so depth buys coverage and not sharpness. Where no neuron can witness by Proposition 1, the network traces the circle anyway, which is a distributed encoding demonstrated rather than assumed. And none of it is an artefact of tanh, though no raw score may be carried from one activation to another.

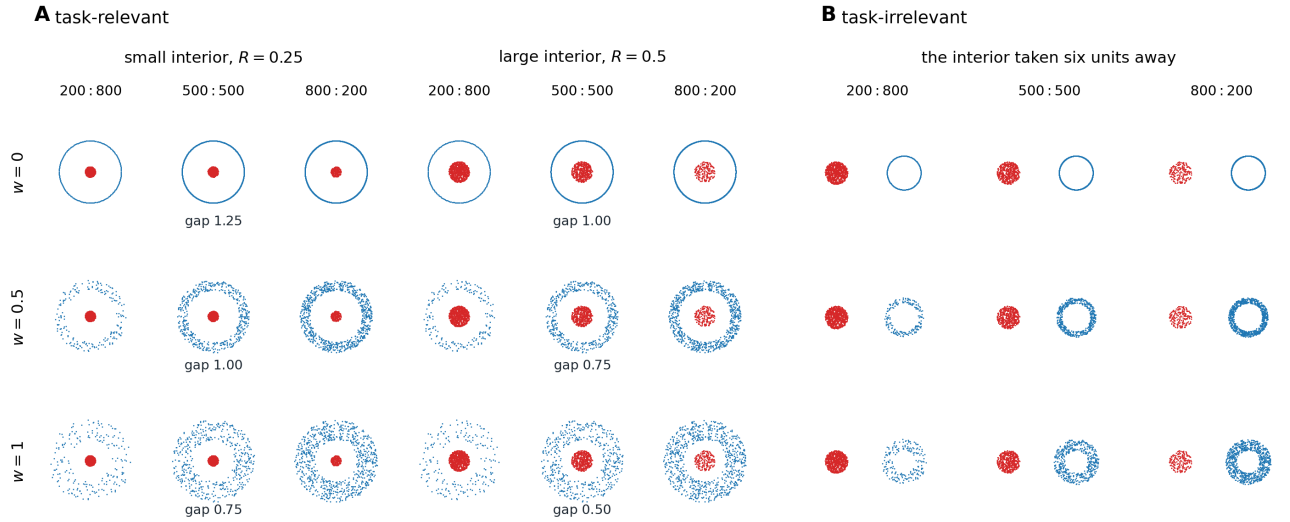


Figure 27: The sweep. Each panel is one configuration of one dataset, drawn to scale: the feature in blue, the interior of the opposite class in red, and the column heading giving how the thousand points are divided between them. Down a column the same points are spread over an ever wider ring — the distance between neighbours on the feature goes 0.019, 0.051, 0.071 down the three rows, against a grid step of 0.068, so the sweep carries the sample across the resolution of the instrument — and the ring's inner edge travels inwards as it widens, which narrows the gap drawn on the balanced columns. Across the six left-hand columns the interior grows and the budget moves between the two objects. The three right-hand panels of each row are the task-irrelevant twins, drawn from further back so that both objects fit: the same feature with the interior taken six units away, where a straight line solves the task. The remaining twelve cells of the sweep repeat twelve of these geometries with five times the points, and the cell that reproduces Experiment 1 exactly is the dataset of Figure 2B.

9 Experiment 3: is it density?

9.1 A sweep that moves the density and leaves the rest standing

One objection survives both experiments, and it is about the data rather than about the network. In the matched pair the circle that gets witnessed is also the one with the opposite class packed against it from inside, and a boundary has to run between two classes whether or not anything topological is involved. The programme carries this as a named confound — level sets stick to where the points are — and testing it means moving the density of the sample on its own, which turns out to be harder than it sounds.

Why the points cannot simply be counted out. The direct way to make the feature denser is to give it more points, and it cannot be used here, because the feature and the interior carry different labels: every point added to the curve is a point taken from the other class, so the class balance moves with the density and no measurement separates them afterwards. The density is therefore varied by *geometry at a constant point count*. The circle becomes a *ring* of width w, its points uniform over the area of the band $[1.5 - w/2,\ 1.5 + w/2]$, and the same five hundred points that lay on a curve are spread through a strip (Figure 27). Three things follow, and all three are the reason for the construction: the class balance becomes an axis of its own, $\rho = n_{\text{ring}}/n_{\text{disk}}$, free to be set independently; the support never degenerates; and $w = 0$ is not a special point but exactly the circle of Experiments 1 and 2, so that their cells sit inside this sweep rather than beside it.

The four axes. A cell is fixed by a task and four numbers, and each of the four is an axis of the sweep with a question of its own.

- **The width of the ring** $w \in \{0,\ 0.5,\ 1.0\}$ is the density axis: at a constant point count, spreading the feature over a wider band is the only way to make it sparser.
- **The radius of the interior** $R \in \{0.25,\ 0.5\}$ is the geometry of the object on the other side.
- **The balance** $\rho = n_{\text{ring}}/n_{\text{disk}} \in \{0.25,\ 1,\ 4\}$ divides the same budget of points between the two objects — two hundred on the feature against eight hundred inside it, five hundred each, or the reverse.
- **The total count** $T \in \{1\,000,\ 5\,000\}$ raises both densities at once, leaving the geometry and the balance where they are.

The first three are crossed in full on the task-relevant side; on the task-irrelevant side the interior is fixed six units away and the width and balance axes are swept exactly as they are here, which is what makes the two sides comparable. The fourth is a slice across the other three rather than a full crossing. With one further cell reproducing the geometry of Experiment 1, that is 40 cells of $N = 50$ networks five neurons wide and five layers deep, and 551 250 scored neuron–feature pairs. Figure 27 draws every configuration that carries the axes.

What else the width axis moves, named before it is used. Widening a symmetric ring pushes its inner edge inwards, $1.5 \to 1.25 \to 1.0$, and with it narrows the *gap* between the feature and the interior, $1.5 - w/2 - R$ (Figure 27, where it is drawn on the balanced columns). This was a constraint of the design rather than a variable of it: the gap is held at half a unit or more in every cell, so that no cell asks the network to change value sharply over a short distance. It correlates with the manipulated variable, and the grid was built with that in mind — two pairs of cells reach the same gap through different geometry, which is what Section 9.5 uses.

The cells are task-relevant before any network is trained. That the disk inside the ring makes the feature necessary is a statement about the data, and it is checked rather than assumed. The kernel of the chromatic filtration (Section 5.6) — a cycle of the feature's own points that dies when the points of the other class are added — is present in all 25 task-relevant configurations and in none of the 15 task-irrelevant ones, with no intermediate zone. Every relevant cell is therefore a positive example, so a cell where the detector says nothing is a miss of the instrument and not a property of the task; and every irrelevant cell is a negative, which turns the first question below into a measurement instead of an assumption.

9.2 Which circle the score is read against

Until now the target was the feature itself. A ring does not name one: the target can be the ring, or any circle inside it, and the choice fixes every number that follows. Four candidates

are registered in every cell and scored on the same neurons — the ring, and the circles along its inner edge, its midline and its outer edge. Each carries its own nulls, which are rigid copies of that same shape, so each carries its own floor and its own threshold.

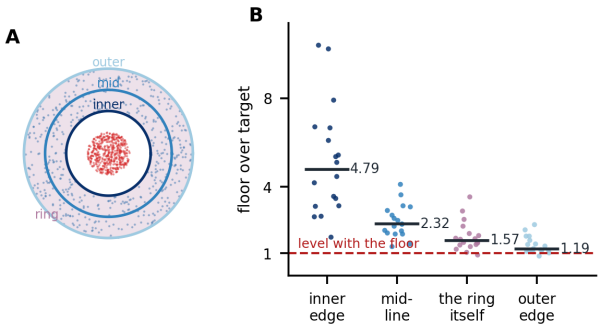


Figure 28: (A) The four candidate targets in one cell. Three of them are circles that no point of the dataset lies on. (B) How far each separates the target from its own floor, one point per configuration and a bar at the median. The quantity is dimensionless, so the four are comparable despite their different diameters, and it uses no threshold.

The ring is not usable as a target. Its thousand points have to cover an area rather than a curve, so they lie 0.097 apart — 1.43 grid steps, against the condition $\delta(A) < h$ of Section 5.2. Part of any score measured against it would be its own sampling, and that part cannot be told from the network's contribution. This is the one argument available before a single number is computed.

Among the circles, the inner edge separates best. The four candidates are drawn on one cell in Figure 28A, and the measure that chooses between them is the median floor over the median target in the same cell: dimensionless, so the diameters cancel, and free of any threshold. It gives 4.79 at the inner edge — twice what the midline gets and four times what the outer edge does — and the ordering of the four holds in each of the 18 configurations separately rather than on average (Figure 28B). The outer circle in places scores no better than its own nulls. A second reading, this one through the threshold, agrees: counted against the chromatic ground truth, where a miss is an error, the median cell has a witness in every model at the inner edge, in 0.83 of them at the midline, and in 0.18 and 0.16 of them at the ring and the outer edge.

The verdict is carried by the inner edge. It also has the only motive stated before the run: if what makes a feature relevant is the pair it forms with the interior of the opposite class, then the place to look is where the two meet. At $w = 0$ the ring degenerates and its inner edge is the circle itself, so the target is defined in all forty cells.

The midline circle is kept for one thing. It is the only target that stays where it is as the ring widens. Along the width axis the two therefore ask different questions — one whether the picture at a *fixed* place decays when the points there thin out, the other whether the network's structure follows the edge of the feature — and Section 9.4 needs both. One consequence has to be carried with the moving target: its diameter shrinks $3.0 \to 2.5 \to 2.0$, so a normalised score is not comparable along that axis and the dimensionless ratio is read beside it. The direction of the bias is known in advance: the ceiling of the instrument is $h/2$ in plain units, so normalised it *rises*, $0.0113 \to 0.0136 \to 0.0170$, and normalisation works against an improvement rather than for it.

Thresholds. Each target's threshold is the first percentile of that target's own nulls in that cell (Section 6.5), taken over two placements on the task-relevant side and three on the task-irrelevant one, where a corridor between the classes exists. On the verdict target they run from 0.14 to 0.38 across the forty cells. This is also the experiment on which the second condition of the verdict was first applied.

9.3 Density does not manufacture a witness

The decisive cells are the fifteen task-irrelevant ones (Figure 27B): the same feature, with the opposite class taken six units away, where a straight line solves the task and the circle is of no use for it. If level sets stick to where the points are, then making the feature dense is enough and a witness has to appear.

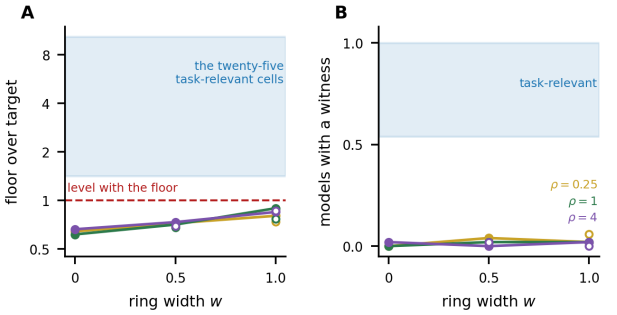


Figure 29: The task-irrelevant cells along the width and balance axes, on the verdict target: filled markers at $T = 1\,000$, open ones at five times the sample. (A) the target against the floor of the same cell, which is a lower estimate by construction (Section 6.2): nothing on either axis carries the target up to it. (B) the share of models with a witness, against the band spanned by the twenty-five task-relevant cells measured the same way.

None appears on any of the three axes that exist there — the interior is fixed six units away, so its radius is not one of them. The share of models with a witness runs from 0.00 to 0.06 with a median of 0.02, at a threshold built to admit one null in a hundred, and neither widening the ring nor moving points between the objects nor raising the whole sample fivefold shifts it (Figure 29B). Nor does any of them bring the target nearer to the floor of its own cell (Figure 29A), which is where an effect too small for the threshold would have shown. The hypothesis is refused in the most direct form available: the thing it predicts was attempted and did not happen.

The few firings that do occur are not threshold noise — they survive even the strictest percentile, because the neuron really has found a closed level curve near the target — and they are what the second condition of Section 6.5 was built for. Not one of them reaches the head, and of the genuine witnesses not one fails to.

None of this is the instrument failing to respond. The same fifty models per cell, the same criterion and the same thresholds give a witness in between 0.54 and 1.00 of the models on the task-relevant side, which is the band drawn in Figure 29. The zero is a property of the dataset.

9.4 Density does not improve a witness that exists

Density was not the cause; it could still be the modulator, and if it were, every comparison between experiments run at different sample sizes would need correcting for it. The width axis answers, read once with each target.

At a fixed place, thinning the points does nothing. On the target held at radius 1.5 all six lines are flat: the median moves by between 0.003 and 0.023 within a line, with no consistent sign, against 0.121 between the lines (Figure 30A). Over that axis the points of the feature come to lie 3.8 times looser. Whatever separates one line from another — and the lines are well separated — it is not how tightly the points are packed.

On the moving target everything improves, and density has the sign wrong. Against the inner edge the median improves in all six lines (Figure 30B), and the dimensionless ratio

rises from between 1.4 and 3.3 at $w = 0$ to between 3.2 and 7.9 at $w = 1.0$. The density hypothesis predicts the opposite: the feature comes to lie almost four times looser along this axis and the witness becomes better. Normalisation does not produce it either, since it pushes the other way — measured against the instrument's own ceiling, which worsens by half along the axis, the improvement is by a factor of 1.7 to 2.4. And the same geometric move on the task-irrelevant side, where there is nothing to follow, gives the arithmetic of a shrinking target and nothing else: the normalised score there *worsens*, 0.635 to 0.81, and the ratio stays below chance at every width (Figure 29A).

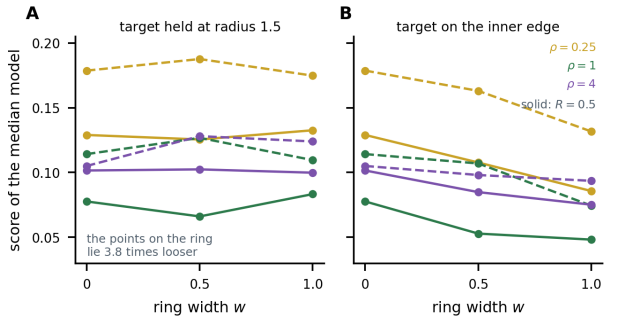


Figure 30: The same six task-relevant lines along the width axis, scored against a target that stays at radius 1.5 (A) and against the moving inner edge (B). At $w = 0$ the two targets coincide, so each line starts from the same point in both panels. Colour is the class balance, and a dashed line is the small disk.

Nor does the absolute level of density. That is the total-count axis, the one place where both densities rise together. Five times the sample, at unchanged geometry and unchanged balance, moves the median by 0.008 to 0.036 — consistently in sign, unlike anything density does, but on a verdict that has nowhere left to go: the share of models with a witness was 1.00 and becomes 0.98 to 1.00. What the points are needed for is to make each object resolvable; past that, adding more is worth little.

9.5 What does move it: the budget of each object, and the gap

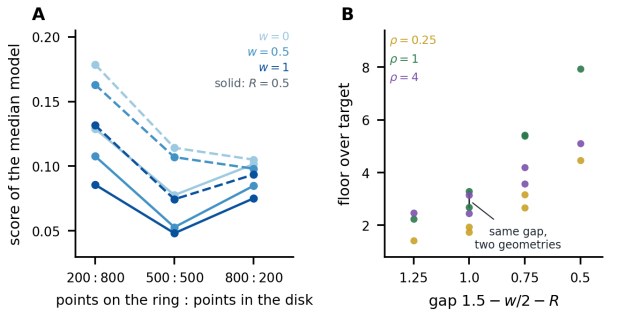


Figure 31: (A) The balance axis: the same thousand points divided differently between the feature and the interior, six lines through the grid. (B) Every task-relevant cell against the gap between the feature and the interior, with the axis running the way the gap narrows; a segment joins the two cells that reach the same gap through different geometry.

Each object needs the points its own size requires. Along the balance axis, starving the feature — two hundred points on the ring against eight hundred in the disk — is the worst point of all six lines, on the score and on the dimensionless ratio alike (Figure 31A). The optimum sits at equality in four lines of six; the two exceptions are both the small disk, and they are not noise but the same rule seen from the other end. A disk of radius 0.25 is already resolved by two hundred points — they lie 0.031 apart, closer than the 0.040 of the large disk at five hundred — so it has nothing to gain from more, and what is taken from it goes to the feature with profit. Packing does not explain any of this: along the balance axis the spacing on the feature changes as much as it does along the width axis, where nothing happened at a fixed place.

The interior is a working variable. Along its own axis, doubling the radius of the disk improves the score in all nine combinations of the other three, and the dimensionless ratio in eight of them, the ninth being a tie in the second decimal; the median gain is 0.037, about a third of the score's own size. The floor stays where it is — the threshold of a cell shifts by 0.016 between the two radii, half the size of the effect, and upwards in only four pairs of nine. The interior does not change how easy the circle is to hit by accident. It changes how well the trained network hits it.

A third outcome, measured on the same grid and knowing nothing about targets or level sets, says the same about it. The share of initialisations that reach the gate (Section 6.6) follows the point budget exactly as the score does — best at equality in all six lines, where the score had it in four, and worst wherever the feature is starved — and does not follow the radius of the disk at all, improving in five pairs of nine against the score's nine of nine. A larger interior does not make the task harder to solve; it moves where the boundary ends up.

Two of the four axes turn out to be one axis. Widening the ring brings the feature's inner edge towards the centre; growing the disk brings the interior outwards to meet it. Both narrow the gap, and both improve the picture at the inner edge, which is why the whole task-relevant grid orders itself along the gap (Figure 31B). The zone in which two objects of opposite classes interact lies in that gap, and the narrower it is, the more tightly that zone is pressed against the inner edge — which is where the target is.

The gap is the mechanism and not the only variable. Two pairs of cells reach the same gap through different geometry, and in both the cell with the larger interior is at least as good (Figure 31B, the joined points): ahead by a fifth on the dimensionless ratio in one pair, and level on it in the other, where in plain units — the only unit in which targets of different diameter may be compared — it is still ahead, 0.132 against 0.149. The size of the interior enters on its own account as well.

What this settles. A statement that has been carried since Section 3.3 as an interpretation is now measured: the relevant feature is a property of the *pair*, not of the curve. Were it a property of the curve alone, the radius of the disk could do nothing at all; instead the two objects move the score from opposite sides, six lines of six and nine pairs of nine, and what they have in common is the region between them.

9.6 Structure sits at the feature's inner edge

Having four targets in every cell answers something they were not registered for. Three of them are concentric circles at different radii, scored on the same neurons of the same models, so together they ask at what distance from the centre the level sets of a network lie best. In this role a circle is not a correct target but a *probe*, and the structural score by itself cannot ask the question, since it minimises over all levels and constrains no radius.

The answer is the same in every unit and in every configuration. The floor barely depends on the radius of the probe — it moves by four per cent across the three — so the scale is honest, while the distance from the target grows nearly fivefold from the inner edge to the outer one at $w = 1.0$ (Figure 32A). The dimensionless ratio falls with it, from 5.38 at the inner edge to 1.14 at the outer, where the target is level with its own floor; both orderings hold in all eighteen configurations separately (Figure 32B). Normalisation, once more, favours the outer probe, which is the largest target — and it is still the worst. The wider the ring, the further apart the probes and the more the three separate.

The measurement is solid and its reading is not. Two explanations predict exactly this. The structure may sit where the

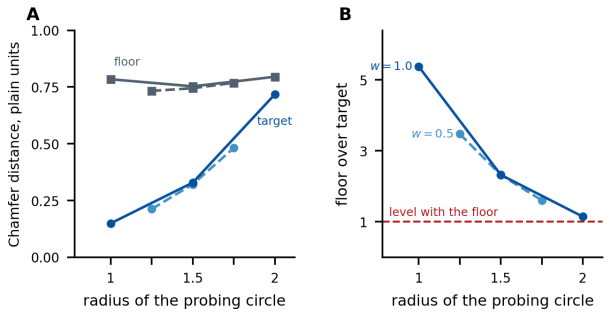


Figure 32: Three concentric probes read by the same neurons, at two ring widths. (A) In plain units, because the probes differ in diameter: the floor hardly depends on the radius of the probe, and the target's distance grows steeply with it. (B) The dimensionless ratio of the two.

feature meets the interior, in which case its position means something; or a neuron may simply form a cap with an extremum somewhere near the centre, and the level curves of a cap are roundest near the extremum and lose their roundness with distance, in which case the monotonicity is a property of the shape of a field and the network could not have done otherwise. Here the two cannot be told apart, because the ring and the disk are concentric in all forty cells, so "follows the inner edge" and "is round near the centre" describe the same curve. Separating them requires moving the two objects with respect to each other, which is Experiment 4.

9.7 The other instrument gives the same verdict, and gives it harder

Everything so far is one instrument. The homological score asks a different question of the same neurons — whether a closed loop is present near the target at all, rather than how well a curve lies on it — and it has no threshold of ours in it.

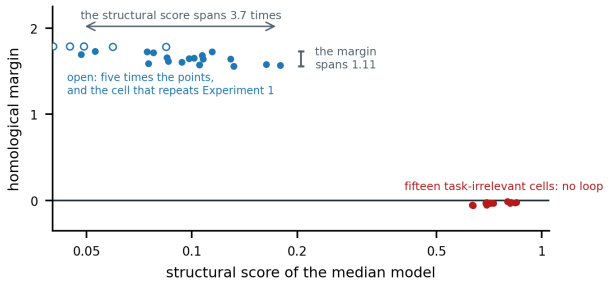


Figure 33: Every cell read by both instruments on the verdict target: the structural score of the median model against the median of its best homological margin. A margin above zero is a dominant loop. The three geometric axes stretch the cells across the horizontal scale and leave them where they are on the vertical one.

On the density question it leaves no loophole. A dominant loop is found for 1 247 models of 1 250 on task-relevant targets and for exactly one of 750 on task-irrelevant ones, and along both axes of the task-irrelevant grid the rate is zero everywhere except a single cell at 0.02 (Figure 33). The structural score could in principle have stayed silent because the loop runs somewhere other than where we look; this one looks at the whole neighbourhood and finds nothing there. No density builds a closed structure around a feature the task does not use.

On the geometric axes it does not move at all. Over the eighteen task-relevant configurations the structural score spans a factor of 3.7 and the margin a factor of 1.11, with a dominant loop in every model of sixteen cells of eighteen. This is not a scale running out of room: the unsaturated reading, how many of a model's twenty eligible neurons carry the loop, gives 13 to 16 and is equally flat. Neither the width of the ring nor the radius of the disk reaches it.

One instrument sees a subset of what the other does. Of the 1 250 task-relevant pairs, 1 161 pass both criteria and 86 pass only the homological one; *none* passes only the structural one, and at the level of individual neurons the exceptions number 14 in 14 295. If a level curve lies on the target then a loop is there of course; the converse fails, and it fails in one corner of the grid — 77 of the 86 are at the small disk, which is exactly where the structural score is at its worst. There the network does build a closed structure and does not run it along the inner edge.

Read together: a closed structure is present whenever the task forces one and absent whenever it does not, and what the geometry of the pair moves is *where* that structure runs. Margins themselves are not comparable between targets of different radius, since the neighbourhood the score looks in grows with the target; everything above is measured on the verdict target.

What Experiment 3 establishes. Density does not make a witness and does not improve one. It does not manufacture one where the task has no use for the feature — at no width, no balance and no sample size, and nothing on those axes brings the target as far as the floor of its own cell — and where the task does use the feature, thinning the points at a fixed place moves nothing. The confound is closed, and the conclusions of Experiments 1 and 2 need no correction for how the sample was packed. Two things do move the score, and neither is density: how many points each object is given, which is a question of whether an object of that size is resolved at all, and the gap left between the feature and the interior of the opposite class. The second is measured from two independent sides — the ring growing inwards and the disk growing outwards — and it is the first direct evidence that what is being measured belongs to the pair rather than to the curve. Within the gap, the structure lies against the feature's inner edge. The homological score agrees about density in a stronger form and stays flat across the geometry that moves the structural one, which separates the two questions cleanly: a closed structure exists exactly when the task forces it, and geometry decides where it runs.

10 Experiment 4: whose shape does the separating curve carry?

10.1 The blind spot of experiments 1 and 2

Experiment 3 ended with a statement it had not been built to make: what moves the score is the gap between the feature and the interior of the opposite class, measured from both of its sides, so the thing being detected belongs to the *pair* and not to the curve. That statement was obtained where the pair is degenerate.

In all forty cells of that sweep the ring and the disk are concentric and both are round. Two objects like that have the same shape, the same centre and no direction between them, so "the level curve follows the feature" and "the level curve wraps the interior" are not two hypotheses but one picture (Figure 35A). Nothing measured in Sections 7 to 9 distinguishes them, and this is not an oversight that a sharper instrument would repair: in that configuration the two predictions coincide exactly. The same degeneracy is what left the localisation of Section 9.6 with two readings — structure sitting where the two objects meet, or a field with a cap whose level curves are roundest near its extremum — and no arrangement of round concentric objects can separate them.

A claim about a pair is tested by taking the pair apart. Three moves do it, and each changes one object while the other stays bitwise where it was (Figure 35B–D): elongate the two independently, so that they stop having the same shape; turn one across the other, so that they stop having the same direction; move one off centre, so that they stop having the same place. If what the earlier experiments read was the feature, none of the three should matter much. If it was the pair, all three should.

What has to be read is therefore not whether a curve coincided with a target. It is whose shape that curve has, which way it

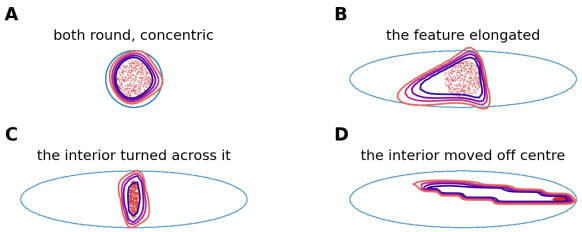


Figure 35: One neuron's ladder of level curves (colour runs from the rung nearest the interior to the rung nearest the feature, Section 5.5) in four configurations of the same kind of pair, all four in one window. (A) both objects round and concentric, as in Experiments 1 to 3: a curve that follows the feature and a curve that wraps the interior are the same curve. (B–D) the three ways the pair is taken apart. In each of them the two readings predict different pictures, and the picture is not the one the feature predicts.

points and where it sits — three questions that a single distance compresses into one number. The instrument for them is the descriptor of Section 5.5; the two scores return at the end of the section, as a check and as a result of their own.

10.2 Four blocks, and what is held fixed across them

Both objects become ellipses. The feature is an elliptical curve of class 0 and the interior an elliptical region of class 1 inside it, and each can be elongated, turned and displaced on its own. A cell is one such pair; there are 56 of them, each carrying $N = 50$ networks five neurons wide and five layers deep, for 320 000 scored neuron–feature pairs.

- **Block A**, 16 cells: the aspect of the feature crossed with the aspect of the interior over 1, 1.5, 2.5 and 4, the objects concentric and their long axes aligned (Figure 34). It asks whose *shape* the curve carries.
- **Block B**, 9 new cells: the same aspects with the interior turned across the feature. Read together with cells of block A it forms a 4×4 grid whose edges measure each object alone — a round interior leaves only the feature with an axis, a round feature only the interior — and whose interior measures what happens when the two axes *collide* (Figure 38). It asks about *direction*.
- **Block C**, 8 new cells: four of these geometries kept bitwise as they are while the thousand points are split differently between the two objects, 200, 500 or 800 of them inside (Figure 43). It asks whether shape is a function of geometry at all.
- **Block D**, 23 cells: the interior halved in size — otherwise there is nowhere to move it — and displaced along the feature's long axis and across it (Figure 40). It asks about *position*.

The feature holds its minor semi-axis, the interior its major. Elongating an ellipse means fixing one semi-axis and stretching the other, and the two objects fix different ones. The feature holds its *minor* semi-axis, so elongating it pushes its ends outwards and multiplies its area by four from 1:1 to 4:1; the interior holds its *major* one, so elongating it makes it thinner. An elongated interior is therefore also a smaller interior, and an elongated feature is also a roomier one. The axis "aspect of the feature" is consequently not a pure axis of shape, and where that matters it is said again.

What is held fixed. The point count is 1 000 in every cell and the class balance is a separate axis, so that the objection Experiment 3 closed cannot reappear here as a confound. The evaluation grid is not fixed: its step is chosen per cell so that the feature's diameter always spans $N_A = 54.5$ steps. This is the correction Experiment 3 could not make. There the target changed diameter along an axis and the normalised score stopped being comparable along it (Section 9.2); here the diameter changes fourfold between cells and the resolution changes with it, so the discretisation limit $h/2\,\mathrm{diam}(A)$ is the same number in all 56 cells and the scores are comparable across the whole design.

The threshold, and why it is taken cell by cell. The score is normalised by the diameter of the target, so any threshold on it is a tolerance measured in diameters — and the diameter is what this design changes, from 3 at a circle to 12 at a 4:1 ellipse of the same area. The floor moves with it: measured on each cell's own null placements it runs from 0.038 to 0.131, a factor of three and a half, and the two ends are different cells rather than different samples of one. Every rate below is therefore read against the first percentile of that cell's own nulls (Section 6.5); at the fifth the floor runs 0.052 to 0.150 and no conclusion of this section changes.

Which curve is described, and how the curves are counted. The description is taken on all five rungs of the ladder in values (Section 5.5), which knows only that there are two objects. Two units of observation are in use and each figure says which (Section 6.4): where the question is whether *some* neuron does something — every rate of witnesses, every score — a model is valued by its best neuron and the fifty models are the sample; where the question is where the family as a whole comes to rest, the unit is one described curve, a pair (neuron, rung), of which a cell holds about 2 400, and the summary is their median with no averaging inside a model first.

A round curve separates the classes in fifty cells of fifty-six. In 50 of the 56 cells a round level curve centred at the feature's centre separates the two classes: the interior fits inside a circle inscribed in the feature, so the network is never obliged to draw anything else. What is measured below is therefore

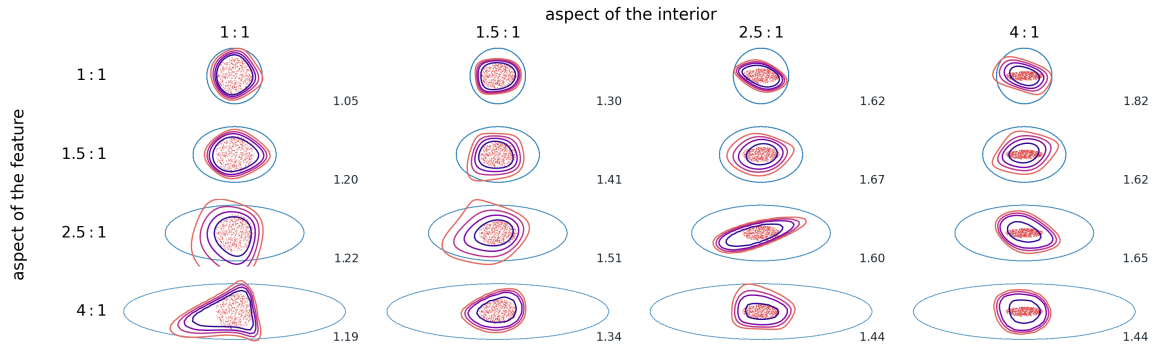


Figure 34: Block A: the aspect of the feature down the page against the aspect of the interior across it, the objects concentric and aligned. Each panel carries one neuron of that cell's median model — the middle neuron by score among those that close a curve, not the best one — with the four inner rungs of its ladder, coloured from the rung nearest the interior to the rung nearest the feature. All sixteen panels are drawn in one and the same window: the feature really is four times longer at the bottom of the page than at the top, and the curves really do stay the size they are. The number in each panel is that cell's median elongation over all of its described curves.

what a network does when it has a choice. The six exceptions are all in block D and are taken up in Section 10.7.

10.3 Shape: the interior moves the curve, the feature does not much

Block A crosses the two shapes, and the answer is visible before it is computed. Figure 34 draws all sixteen cells with one ordinary neuron's ladder over each pair. Down the left column the feature stretches fourfold while the interior stays round, and the curves stay compact and nearly round, leaving the ends of the feature empty over several radii. Across the top row the feature is a circle and does not change at all, and the curves nevertheless elongate, following the interior.

Measured over every curve, the two effects differ by a factor of three. The elongation of a curve is read from its second harmonic as the axis ratio of the ellipse with the same harmonic, which puts curve and objects on one scale. Averaging each axis over the other, the aspect of the interior carries the median from 1.17 to 1.63, monotonically (Figure 36A); the aspect of the feature carries it from 1.45 to 1.35 through a maximum at 2.5:1, a span of 0.14 with no order in it (Figure 36B). Over the whole grid the median elongation of a cell stays between 1.05 and 1.82 while the objects run from 1 to 4.

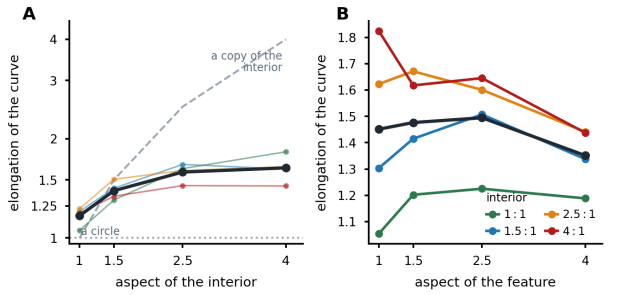


Figure 36: The elongation of the level curves of block A, one point per cell, median over all of its described curves; the heavy line is the mean over the axis not drawn. (A) against the aspect of the interior, on a log scale that fits both references: a curve copying the interior would lie on the dashed diagonal, a round one on the dotted line. (B) against the aspect of the feature, magnified. The effect of the feature changes sign with the interior — upwards when the interior is round, downwards when it is elongated — which no interpolation between the two shapes can do.

The feature is not silent; it is outvoted. Reading the same numbers where the interior is round and cannot drown it, elongating the feature does move the curve: from 1.05 in the cell with two circles to between 1.19 and 1.23 at the three elongations, the lowest line of Figure 36B. The correct statement is that the influence of the feature is the weaker of the two and loses whenever the two meet directly, not that the feature does nothing. Part of even this may not be shape: elongating the feature also quadruples its area, and the two readings are not separable on this grid.

The median is the larger of two modes. Where the interior is most elongated, a third of the curves do not lie near the median at all but above an elongation of 2 (Figure 37C), and that minority is a population rather than noise. It belongs to neurons rather than to rungs: a neuron that draws one such curve draws them along its whole family (Figure 37A, B) — over block A, 0.110 of neurons are elongated at every rung they have, against the 0.0003 that independent rungs would give. It belongs to networks rather than to the cell: half the networks of this cell carry none of them and a third carry almost nothing else, where scattering the same curves at random would put nearly every network near the middle (Figure 37D). And it answers to the pair as the median does — these curves lie along the objects in 0.96 of cases against 0.87 for the moderate ones, and the minority grows with the elongation of the interior.

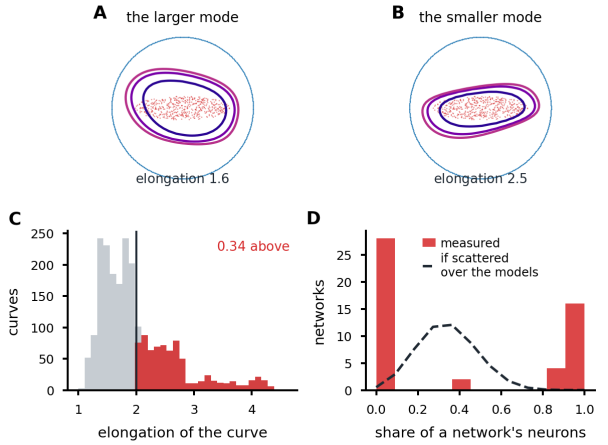


Figure 37: The two modes, in the cell where the smaller one is heaviest: a round feature about an interior of aspect 4:1. (A, B) two neurons of one and the same network, three inner rungs each — the one is nearly round along its whole family, the other elongated along its whole family. (C) the elongation of every described curve of the cell; the part above 2 is the smaller mode. (D) how those curves are distributed over the fifty networks, against what scattering them at random would give.

10.4 Direction: both objects turn the curve, and the interior wins

Elongation is one coordinate of shape and the coarser one. Direction is the finer, because a curve elongated by chance points anywhere while a curve responding to the pair points along it, and because a mistake about direction is a right angle and cannot be argued away as a matter of calibration. The direction of a curve's long axis is the phase of its second harmonic (Section 5.5), and it is read where there is an axis to read: the curves that take part are those elongated by 1.22 or more, that is with $a_2 \geq 0.10$, and every panel below reports what share of its curves that is.

In block A both objects lie along the same axis, so direction cannot separate them there. Block B turns the interior across a feature that is bitwise the same object (Figure 38), and the 4×4 grid it completes has three regimes in it (Figure 39).

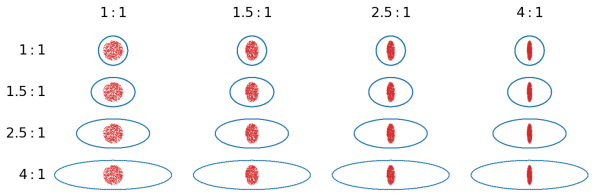


Figure 38: The sixteen configurations in which the two axes are crossed: the aspect of the feature down the page, the aspect of the interior across it, the interior turned. Nine of them are trained here; the first column, where the interior is round, and the first row, where the feature is, are cells of Figure 34 — a round object is unchanged by the turn. The window is the one of Figure 34.

Alone, each object turns the curve to itself. Where the interior is round and only the feature has an axis, between 0.74 and 0.93 of the curves lie within 30° of it (Figure 39A). Where the feature is round and only the interior has an axis, between 0.93 and 0.99 lie along the interior (Figure 39B). Neither object is neutral.

Together, the interior wins every cell. In the nine cells where the two axes are perpendicular, between 0.57 and 0.96 of the curves lie along the interior against between 0.02 and 0.34 along the feature, and the interior is ahead in nine of nine (Figure 39C). The margin is narrowest where the interior is least elongated and widest where it is most.

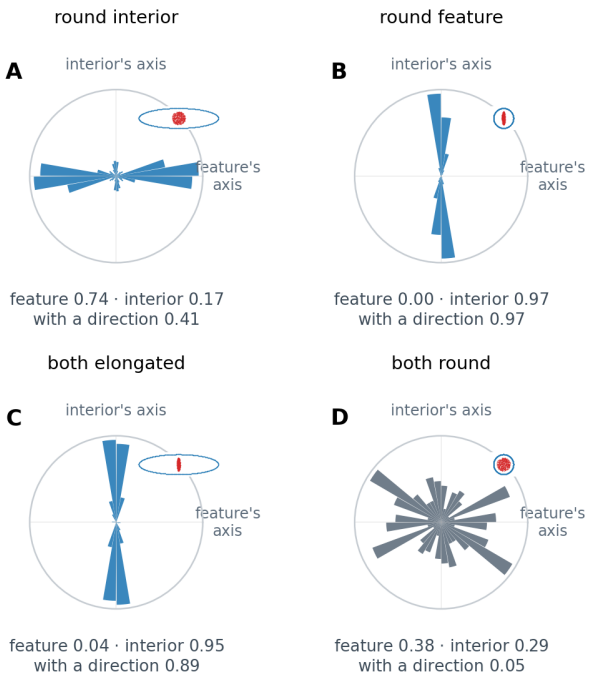


Figure 39: The direction of the long axis of every curve of a cell, as a circular histogram; a direction is an axis and not an arrow, so each measurement is entered twice and the rose is symmetric. The panels are drawn in a frame where the feature's long axis is horizontal and the interior's vertical; the cell of panel B is borrowed from the aligned grid and turned with its dataset, which is exact because the object that is round there is unchanged by the turn. Under each rose: the share of curves within 30° of each object's axis, and the share that had an elongation worth a direction at all.

The instrument does not invent a direction. In the cell with two circles only 0.05 of the curves have an elongation worth an angle at all, and their rose is isotropic (Figure 39D). The share that does have an angle is itself informative: about half of the curves where the interior is round, and most of them wherever it is elongated. What makes a curve elongated in the first place is mostly the interior.

10.5 Position: the curve follows the interior as far as the feature allows

The last thing the pair can stop sharing is a place. Block D moves the interior off centre (Figure 40) and asks where the curve goes. The quantity is the centroid of the curve and its two distances — to the centre of the interior and to the centre of the feature — which need no normalising and are defined at zero displacement too. Displacement is measured as a share of the largest the pair admits, since the absolute room differs sevenfold between the two directions: along the long axis of an elongated feature the interior can go 5.25 units, across it only 1.12.

The curve goes with the interior, and stops when the feature stops it. Displaced along the long axis of an elongated feature, the curve stays with the interior while its distance to the feature's centre grows with the displacement, in all three such families (Figure 41A). It does not keep this up to the end: at half the admissible displacement the centres are 2.5 apart and the curve is 0.4 from the interior, at the full displacement they are 5.1 apart and it is 2.1 behind. The interior is then pressed against the end of the feature and there is no room left to follow it into.

Across the feature it does not go at all. There the two distances exchange roles: the curve holds its distance to the feature's centre at about 0.2 while the interior walks away from it (Figure 41B). The same happens along the axis wherever the feature is round, so that there is little room in either direction. What decides is not the direction but whether the feature leaves

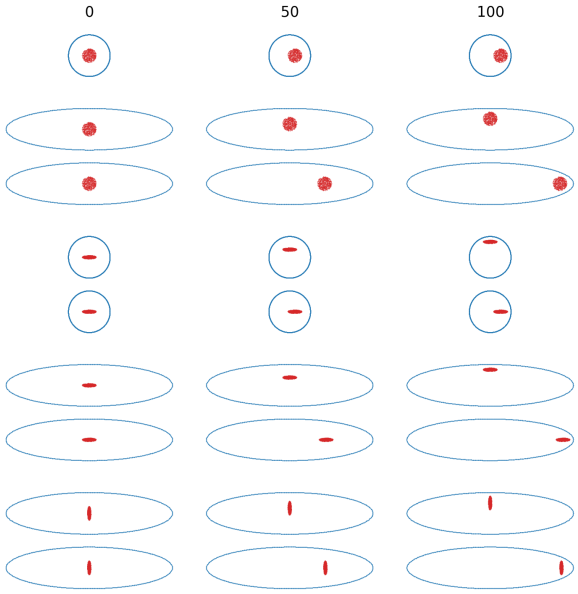


Figure 40: The twenty-three configurations of the displacement block, in the window of Figure 34; the columns are the three displacements, in per cent of the largest the pair admits. Each pair of objects takes two rows — displaced along the feature's long axis, then across it — with the undisplaced cell repeated at the head of both; the round pair has no two directions to distinguish and takes one row. The interior is half the size it has in the other blocks, and the same per cent is a different absolute distance in every row.

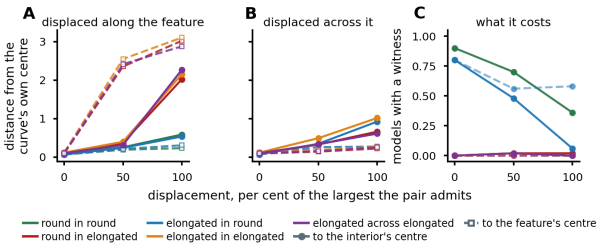


Figure 41: Block D in plain units, every point a median over all the described curves of a cell. (A, B) how far the centre of the curve stands from the centre of the interior (solid, filled) and from the centre of the feature (dashed, open), as the interior is displaced along the feature's long axis and across it. (C) the share of models with a witness, solid for displacement along the long axis and dashed for across it.

room in it — which is the gap of Experiment 3, now read as a distance rather than as a width.

Following costs the witness. Where a witness existed before the displacement it is taken away by displacement alone, the feature untouched: from 0.90 to 0.36 in the pair of two circles and from 0.80 to 0.06 for an elongated interior in a round feature (Figure 41C). Across the feature, where the curve does not follow, the same pair keeps its witness almost intact, 0.80 to 0.58. The asymmetry of the first two panels runs through the third.

Halving the interior is itself an intervention. At zero displacement it already turns the direction over in two families of five and raises the share of curves that have a direction at all several fold. The reference point in this block is therefore its own zero-displacement cell and not the cell of the same name in block A.

10.6 Witnessing is destroyed with the feature held fixed

Everything so far has been about the shape of a curve. The verdict of Sections 7 to 9 is a different object — a score below a threshold, and a witness that survives to the head (Section 6.5)

— and it is what the two moves that leave the feature untouched do to *it* that decides how those sections are to be read.

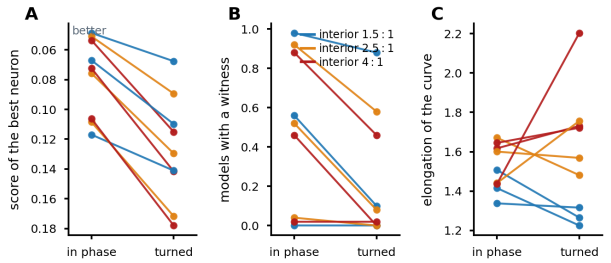


Figure 42: The nine cells in which the interior is turned across the feature, each joined to the cell with the same two objects in phase; the feature is the same point set in both. (A) the median over fifty models of the best neuron's score, the axis inverted so that up is better. (B) the share of models with a witness, read against each cell's own floor. (C) the median elongation of the curves, which does not move systematically: the turn does not remake the curve, it turns it.

Turning the interior. The score of the best neuron worsens in nine pairs of nine, often by half again (Figure 42A), and the share of models with a witness falls in seven of nine, from 0.92 to 0.58 in one pair and from 0.56 to 0.10 in another (Figure 42B); in the two remaining pairs it was already zero in phase. The feature in each pair is the same set of points in the same place, so there is nothing for the difference to attach to except the interior. And the elongation of the curves does not move with any consistent sign — down in five pairs, up in four (Figure 42C) — which says what the mechanism is: the curve is not remade, it is rotated off the feature.

Displacing it. The other move does the same, and Section 10.5 has already shown it: displacement alone takes the witness away wherever the curve follows the interior, 0.90 → 0.36 in the pair of two circles, and leaves it almost intact where the feature gives the curve nowhere to follow.

Witnessing is a property of the pair. One and the same circle can be made witnessed or not witnessed without touching it, by turning or displacing the set of the opposite class inside it. Witnessing is a property of the pair. This is the statement Section 9.5 could only reach from one side, and it is now shown twice by two independent moves.

10.7 The curve is rounder than what it follows

Following the interior is not copying it. Taking the second harmonic with its sign, so that a curve along the interior and a curve across it do not cancel, the share of the interior's elongation that reaches the curve has a median of 0.41 over the 27 concentric cells whose interior has a shape at all, with a spread of 0.10 to 0.78. The measured shape lies between a circle and a copy of the interior in 27 cells of 27: never outside the bracket, and systematically nearer its round end (Figure 36A, where a copy would lie on the diagonal).

The third reference makes the same point from the other side. Comparing each curve with three shapes — the feature, the interior, and a circle — the circle is the only one whose share does not fall to zero anywhere in block A. In the hardest cell, where both objects are 4:1, a neuron whose curve has the shape of the feature is found in 0.02 of the models and one with the shape of the interior in 0.02 as well, while a neuron whose curve has the shape of a circle is found in 0.66.

So between "the interior chooses" and "the curve is the interior" there stands something that rounds, and 0.41 is a measured number rather than a derived one. What that something might be is the subject of Section 10.9.

A reading of where in the bracket the curve sits. One account of the two facts of this section is worth stating as a hypothesis, with what supports it and what does not. A level curve of itself tends towards a circle; the shape of the interior reaches it not directly but through how much room — how much *freedom* — the curve has; and the feature acts chiefly as a ceiling rather than as a source of shape. Chiefly and not wholly: where the interior is round and leaves its voice audible, the feature's own elongation does reach the curve, weakly (Section 10.3). Where there is room the curve takes the round shape; where there is little it takes the path of least resistance and copies the interior, and even then only in part. While the ceiling does not obstruct, its own shape explains almost nothing — over the fifty cells where a round curve separates, the midline of the gap, which follows both objects faithfully, correlates with what is measured at +0.11. In the six where a concentric circle stops separating, the promise of Section 10.2, the ceiling does obstruct and at once decides: there the elongation is 3.05 to 7.06, a median of 3.54 against 1.60 over the other fifty, and the two ranges do not meet.

Its middle link is the one that is not measured. Every dimensionless measure of freedom — the ratio of the two sizes, the gap relative to the interior, the area of one relative to the other — correlates with the share of shape that passes, in the expected direction and between −0.40 and −0.50. But on this grid freedom is tied to the elongation of the interior at +0.68 to +0.79, since elongating it at a fixed major semi-axis also makes it smaller; hold the elongation fixed and the correlation goes (−0.10 to −0.22). The hypothesis is therefore unrefuted rather than supported, and testing it needs an axis this design has not got: one interior, of one shape, at several sizes inside an unchanged feature.

10.8 Geometry alone does not determine the curve

Everything so far has been a statement about geometry, and block C asks whether geometry is enough. Four of the pairs are kept bitwise as they are — the same ellipses, the same semi-axes, the same gap — while the thousand points are split differently between them: 200 inside and 800 on the feature, five hundred each, or the reverse (Figure 43). Any description of the curve computed from the geometry alone is obliged to return the same number three times in a row.

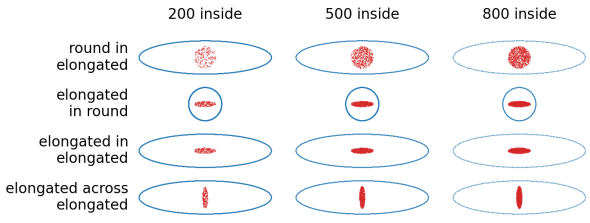


Figure 43: The twelve configurations of the density block, in the window of Figure 34. Along a row the two objects are the same point sets in the same places; only the thousand points are divided between them differently, which is visible as the feature thinning out from left to right.

It does not. In the most strained of the four — both objects 4:1 with their axes perpendicular — moving points from the feature into the interior carries the median elongation from 1.59 to 2.97 (Figure 44A). The shift is populational rather than the work of a few networks: 0.92 of the fifty models at the far end lie above the median of the models at the near end (Figure 44B). It is also a shift of elongation and not of direction — the share of curves lying across the feature stays between 0.87 and 0.95 along the axis — so what the points move is how elongated the curve is, not which way it points. The effect is not uniform: it is strong in one geometry of the four, half as strong in two, and absent in the one where the feature is round.

The right form of the next question is therefore not "what function of the geometry is the shape of the curve" but "what function of the geometry *and* the data". This section measures the first argument with the second held fixed, which is what the constant point count of every other block is for.

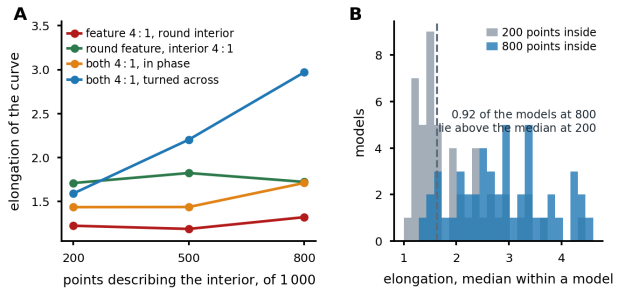


Figure 44: Block C: the same four geometries described by a different split of the same thousand points. (A) the median elongation of the curves of a cell. (B) the most strained of the four geometries, one point per model: the median elongation inside each of the fifty networks, at the two ends of the axis.

10.9 What the curve is instead, and a family that can be excluded outright

If the curve is neither object, the question is what it is. Five geometric constructions were computed for every cell from the pair alone and compared on equal terms (Table 2): one constant each, fitted on the *size* of the curve, with the shape predicted rather than fitted. What each predicts is a population number and not a neuron: the median elongation over all the curves of a cell. Three quantities set the scale it is read on, and all three are measurements of this section rather than choices of ours. The number itself is known to about ±0.09, the half-width of a bootstrap over the fifty networks of a cell. Across the grid it ranges over 0.77. And the two dependences the grid was built to separate are worth 0.46 and 0.14 (Section 10.3), so a construction in error by more than that cannot resolve them.

Table 2: Five constructions on the sixteen cells of block A, each computed from the pair alone with one constant fitted on size. The error is the mean absolute error in elongation, to be read against the ±0.09 to which the measured numbers themselves are known and the 0.46 and 0.14 of the two dependences they carry.

construction	error	rank corr.
the midline of the gap	0.79	+0.11
the curve of maximal margin	0.40	+0.38
the average admissible curve	0.65	+0.24
dilation of the interior	0.24	+0.88
the equilibrium potential of the pair	**0.12**	+0.78

The equilibrium potential of the pair. Treat the two objects as the plates of a capacitor and solve $\Delta u = 0$ in the gap with $u = 0$ on the interior and $u = 1$ on the feature; the candidate for the neuron's level curve is a level curve of u. Nothing about shape is put in: u is determined by the pair. Fixing one level, $u = 0.40$, for the whole grid, the mean error is 0.12 (Table 2) — little more than the uncertainty of the numbers it predicts, a quarter of the interior's effect, though still of the size of the feature's. Two of its properties are exactly the two the measurements demand. Both boundaries enter, so the effect of the feature is free to change sign with the interior, which the measured dependence does (Section 10.3) and no interpolation between two shapes can. And it rounds at the right rate: the multipole of a conductor's field that carries elongation decays as r^{-2} while the one that carries size decays logarithmically, so shape is lost faster than size with distance, which is what 0.41 of the elongation arriving describes.

Dilation of the interior. The set of points at a fixed distance from the boundary of the interior orders the cells better than anything else, at a rank correlation of 0.88, and it explains in one sentence why the curve follows the interior and why it is rounder: adding the same length in every direction takes $(a + \delta)/(b + \delta)$ towards 1. As a prediction of the number it is twice as far out, and it is bound to be: the dilation of a circle is a circle, so wherever the interior is round the construction must return exactly 1.000 against a measured 1.19 to 1.23.

Three of the five are excluded together. The midline of the gap, the curve of maximal margin between the two objects and the average over all the curves that fit in the gap fail for one and the same reason: each is monotone in the shape of each boundary, so none of them can give an effect of the feature that changes sign with the interior, which is what Section 10.3 measures. Direction refutes them a second time and harder: where the two objects lie across one another the midline points along the feature and the measured curves point along the interior, a miss by a right angle in all nine cells. Their errors are 0.79, 0.40 and 0.65. The third is the most tempting of the three — if a curve has to lie in the gap, perhaps the observed one is just a typical admissible curve and no mechanism need be looked for — and it fails for a second reason that needs no fitting: under any measure on those curves that is symmetric about the middle of the gap the average curve *is* the middle of the gap exactly, and at a fixed size no such measure can come out rounder than it either. The networks draw something far rounder. In the cell of Figure 45A the midline is an ellipse of aspect 3.99 where the measured median is 1.44.

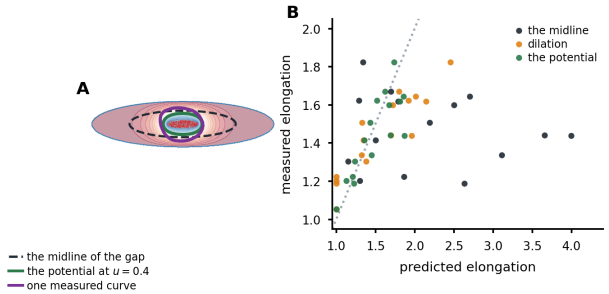


Figure 45: (A) the cell in which both objects are 4:1: the equilibrium potential of the pair as a background, the midline of the gap, the level curve of the potential at $u = 0.40$ and one measured curve. The midline is an ellipse of aspect 3.99; the measured curves of this cell have a median of 1.44. (B) predicted against measured elongation over the sixteen cells of block A, for three of the five constructions.

What no construction reproduces. Both halves of the feature's effect are beyond all five: the rise at a round interior is given only by the potential, the fall at an elongated one only by dilation. And in block D, where the objects are no longer concentric, all of them fail together and fail badly: in the most extreme cell the measured elongation is 7.06 and the best of them predicts 1.55. What is described is the configurations in which the two objects share an axis, and not those in which they do not. As everywhere in this work these are statements about agreement between two measured quantities; nothing here says that a network computes a potential.

10.10 What the other instruments say

The descriptor has answered the question the section was built around. The two scores of Section 5 read the same neurons and answer a different one — whether the curve coincided with the target, and whether a loop is there at all — and what they do here is a result in its own right.

The structural score breaks where the descriptor does. Along the rows of block A its witness rate runs 0.89, 0.95, 0.49, 0.02 as the feature elongates (Figure 46A), with the break between 1.5:1 and 2.5:1 — the same place at which the share of models possessing a neuron of the feature's *shape* collapses. Two instruments measuring different things agree about where the transition is, which is the consistency check this section owed the earlier ones.

At the best neuron the homological score does not move at all. In every one of the sixteen cells a dominant loop is found in every one of the fifty models, and the median of the models' best margins stays between 1.70 and 1.75 across the whole grid. Over all 56 cells there are 1 950 models in which a loop is present and no level curve lies on the feature, and *none* in

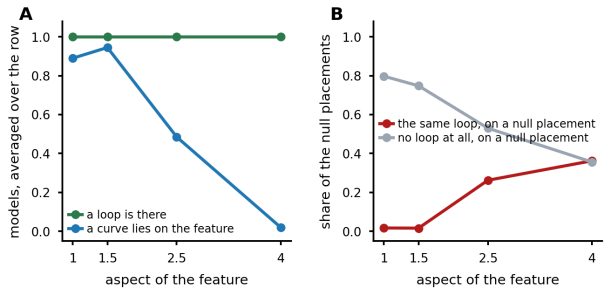


Figure 46: The sixteen cells of block A, averaged over the four interiors of each row. (A) the share of models in which some neuron carries a loop of the right dimension near the target, against the share in which some neuron's level curve lies on the target at that cell's own threshold. (B) what the first of the two costs: how often the same loop is found at a null placement, and how often no loop is found there at all.

which the reverse holds — at the level of individual neurons the exceptions number five among the 39 325 that one instrument or the other admits. Elongating the feature therefore does not destroy closed structure. It takes the structure off the feature, so the collapse just described turns into a positive statement: a persistent loop is there regardless, and what the geometry of the pair decides is where it runs.

As the feature elongates it starts firing on the nulls too. The neighbourhood in which the homological score looks is a ball of one and a half times the feature's own radius (Section 5.4), and the feature's radius grows with its longest axis, so the ball grows fourfold across this grid. It shows: the share of null placements at which the same loop is found rises from 0.02 to 0.36 along the rows, and the share at which no loop is found at all falls from 0.80 to 0.36 (Figure 46B). What weakens is specificity rather than sensitivity, and two consequences follow for reading any of these numbers. A rate on a target is not a verdict unless the rate on that same cell's nulls is beside it, and a margin is not comparable between cells whose features differ in aspect. Neither touches the containment above, which is a statement inside a cell, on the same models and the same neighbourhood.

What Experiment 4 establishes. The curve a network draws between the two objects is chosen by the interior within limits set by the feature. Its shape follows the interior three times more strongly than the feature and monotonically, where the feature's own effect is real but weaker and changes sign; its direction is turned by either object alone and by the interior in all nine cells where the two compete; its position follows the interior as far as the feature leaves room, staying with it where there is room and with the feature where there is none. The consequence for the verdict is direct and is the result the experiment was built for: one and the same circle can be made witnessed or not witnessed without being touched, by turning or by displacing the set of the opposite class inside it. What the structural score answers about is therefore the pair and not the feature. And the curve it looks for is neither of the two objects: wherever the two share an axis it lies between a circle and a copy of the interior, nearer the circle, carrying about 0.41 of the interior's elongation. In those cells that shape is reproduced to 0.12 in elongation, about the uncertainty of the measurement itself, by the equilibrium potential of the pair, a construction with no shape parameter in it; which is agreement between two measured quantities and not a mechanism. Two limits come with this. Shape is not a function of the geometry alone: at bitwise identical geometry, redistributing the same points between the two objects moves the median elongation by nearly a factor of two. And the homological score, reading the same neurons, stays flat across every geometry that moves the structural one, so the closed structure survives what the geometry destroys — which is where the two instruments must be read apart.

11 Synthesis: what the scores are measuring

11.1 What each experiment contributes

Experiment 1: whether a feature is relevant to the task is visible in single neurons. Two datasets carry the same 500 points in the same place and differ only in whether the region the circle encloses is occupied by the opposite class — a difference that is purely topological, since it survives any deformation that keeps the circle closed. Where the circle is relevant to the task, 49 networks of 50 have a neuron whose level set follows it; where it is irrelevant, none of 50 has, and the two ensembles do not overlap at all. The circle is shared bit for bit, so no property of the circle itself can account for that. The same experiment fixed the scale every later number is read on: what a congruent copy of the circle scores where the task never involved it, which turns out to be a distribution and not a number.

Experiment 2: capacity creates nothing, and the feature is carried redundantly. No network is large enough to find the circle where the task has no use for it — not a layer eight times wider than the reference, not ten layers, not two hundred gates of extra training. Where the task does need the circle, capacity does something else: it spreads the feature rather than concentrating it. A constant share of a network's neurons picks the circle up however wide the network is — though width does make each of them a little better than multiplicity alone would — and depth extends the feature through more layers without making the best of them any better. What capacity buys is therefore redundancy, and its clearest form is a network of depth one, where Proposition 1 forbids *any* neuron to witness and the network traces the circle all the same.

Experiment 3: what matters is not the density of the points but the room between the objects. Spreading the same points of the feature over a band nearly four times looser changes nothing at a fixed place; no redistribution of the points between the two objects, and no fivefold increase of their number, produces a witness where the task has no use for the feature. Points are there to resolve an object, and once an object is sampled finely enough to be seen at all, adding more to it changes nothing. What does move the score is the room left between the feature and the interior of the opposite class, and it moves it from either side: growing the feature inwards and growing the interior outwards do the same thing. That was the first sign that the object measured is not the feature alone, and it was obtained where the two objects are concentric and both round, so it could not say which of them the neuron's level set was following.

Experiment 4: the object measured is the pair, not the feature. Elongating the two objects independently, turning one across the other and moving one off centre break that degeneracy. The level curve then follows the interior: in its shape three times more strongly than the feature, in its direction in every cell where the two compete, and in its position as far as the feature leaves room. The verdict obeys the same rule, and this is the result the experiment was built for: one and the same circle can be made witnessed or not witnessed without being touched, by turning or displacing the set of the opposite class inside it. What the level curve takes from the interior is not a copy either; in every concentric cell it lies between a circle and a copy of the interior, nearer the circle.

11.2 The three effects, separated

One result runs through all four, and it comes before any explanation of it. Where the task has no use for the feature nothing is ever found — in no architecture, at no activation, after no length of training and under no sampling of the data — and nothing on any of those axes brings the target even as far as the floor measured in the same cell. Where the task does need the feature, something is always found. Section 1 named three properties of the data that could be responsible and that a single picture

cannot tell apart; the four experiments separate them, and the three answers are different in kind.

The topological effect is confirmed, and it decides whether there is anything to find. Whether a task forces a closed boundary is a property of the data, settled before any network is trained: the chromatic filtration finds a cycle of the feature's own points that the opposite class fills in, in all 25 task-relevant configurations of Experiment 3 and in none of the 15 task-irrelevant ones, with no intermediate zone (Section 9.1). Where it is present a closed structure is found and stays found — even where the structural score has stopped firing. Over the sixteen cells of block A the homological score, which reads topology and not geometry, holds at 1.00 while the structural rate falls from 0.89 to 0.02. That robustness is bought rather than free: the neighbourhood the homological score looks in grows with the feature, so the same loop begins to be found at the null placements as well (Section 10.10).

The density effect does not make a witness, and it does reach the shape. This is the confound the programme carried by name from the start — level sets stick to where the points are — and on the verdict it does nothing. It is not zero, though. At bitwise identical geometry, the same ellipses with the same semi-axes and the same room between them, moving points from one object to the other carries the median elongation of the level curves by nearly a factor of two (Section 10.8). The sample reaches the shape of what is drawn while leaving the verdict alone, so the question this work leaves is not what function of the geometry the shape is, but what function of the geometry *and* the sample.

The geometric effect is real, and it decides where the level curve runs. What the topological requirement leaves free — the forms of the two objects, their relative size, where they stand with respect to each other — sets the shape, the direction and the position of the level curve that gets drawn. The canonical form of this worry is exactly the configuration Experiments 1 to 3 measured in: two round concentric objects share a centre, so a level curve that wraps the interior lies on the feature for nothing, and "the neuron found the circle" cannot be told from "the neuron wrapped the region". Experiment 4 separates the two and finds the second to be the better description.

11.3 What a task-relevant topological feature turns out to be

What a task forces is not the feature. It is a closed boundary separating the inside of the feature from the outside, and any such boundary stands in the annulus between the two classes — the *gap* of Section 3.3. Everything above says that the level sets of neurons answer to that annulus: both of its boundaries move them, the room between the boundaries bounds them, and the densities on the boundaries reach them too.

The structural score does not ask about the annulus. It compares a component of a level set with one chosen point set, the feature, and the interior does not enter its definition at all (Section 5.3). Where the two objects are round and concentric this costs nothing, since a level curve that wraps the annulus is a level curve that lies on the feature; everywhere else the two part company. The score then returns "no witness" for a neuron that has drawn exactly the closed boundary the task requires, for the one reason that the boundary does not run along the point set we chose to compare it with. That is what the two containments count: over the 56 cells of Experiment 4 there are 1 950 models carrying a loop of the required dimension with no level curve on the feature and *none* the other way round, and on the unrelated grid of Experiment 3 the same holds with 86 models against none.

Which properties of the gap decide is open. The forms of the two boundaries leave a trace and so do the angle and the distance between them; the relative size of the two objects leaves one too, and this design cannot isolate it, every dimensionless measure of the room left being tied to elongation (Section 10.7). What is established is that the feature alone is not enough.

What this leaves for the second step. The aim stated in Section 2.2 is the reverse of what was done here: to look into the neurons of a network whose data nobody designed, and read off the structure it has learned. This work is the controlled half of that, and what it says about the general case is a warning about the question to ask. A criterion pinned to one chosen point set answers about agreement with that set, while the object a task forces is a region with two boundaries; where the two are not interchangeable — which is the general case, the concentric one being the exception — such a criterion will miss what the network has built.

11.4 Scope

All four experiments intervene in the data and none of them in the network. What is measured throughout is an agreement between two quantities, a level set of a neuron and a point set chosen by us, and an agreement is not a use: that a network would be worse without the neuron is a statement only ablation can make, and none is made here (Section 4). The setting is narrow as well. Every dataset here is two objects in the plane, one of them a closed curve, and every network belongs to one family of architectures with three activations. A procedure transfers out of a setting like that and numbers do not: a floor is measured on the cell's own nulls and a threshold is taken from it, so the same reading can be made anywhere, while a score, a rate and a margin belong to the configuration that produced them.

12 Conclusion

Conclusion

Section 1 named three properties of a dataset that act at once on the level sets of a trained network's neurons and cannot be told apart in a single picture. This work separates them by experiment, and they turn out to be different in kind.

The topological effect is real, and it is the one that decides whether there is anything to find. Where a task forces a closed boundary, nearly every network has a neuron whose level set follows the feature; where the same feature, sampled by the same points and standing in the same place, is of no use to the task, no network does. Neither competing explanation survives. No amount of capacity manufactures a witness where the task has no use for the feature, and no packing of the sample does either. What the sample decides is different in kind: how well each of the two objects is resolved, the picture being clearest where neither is starved, and, at geometry held fixed, how elongated the curve that gets drawn comes out. The geometric effect is real as well, and it answers a different question: what the topological requirement leaves free, the forms of the two objects and their relative size and position, is what sets the shape, the direction and the place of the level curve that gets drawn, and it can take a witness away from a feature that has not been touched.

That is what makes the object of the whole question the pair, and not a feature on its own. What a task forces a network to build is a closed boundary in the region between the two classes, so a criterion that compares a level curve with the feature alone agrees with the task only where the two objects are round and concentric; elsewhere it returns "no witness" for a neuron that has drawn the boundary the task requires, because that boundary does not run along the feature. A witness is thus a narrower finding than the word suggests. It says that among the many level sets of a neuron there is one whose component agrees, in shape and in place, with a point set we fixed in advance — and what decides the shape and the place of the level sets a network actually draws is the pair of objects and not the feature on its own.

All of this is an agreement between two measured quantities. That a network *uses* what its neurons agree with is a stronger statement, and nothing here is evidence for it.

References

[1] Gregory Naitzat, Andrey Zhitnikov, and Lek-Heng Lim. *Topology of Deep Neural Networks*. Journal of Machine Learning Research, 21(184):1–40, 2020.

[2] Binxu Wang and Carlos R. Ponce. *On the Level Sets and Invariance of Neural Tuning Landscapes*. Proceedings of Machine Learning Research, 197:278–300, 2023.

[3] Jesse Johnson. *Deep, Skinny Neural Networks are not Universal Approximators*. International Conference on Learning Representations, 2019.

[4] Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. *Finding Neurons in a Haystack: Case Studies with Sparse Probing*. Transactions on Machine Learning Research, 2023. arXiv:2305.01610.

[5] Nelson Elhage et al. *Toy Models of Superposition*. Transformer Circuits Thread, 2022. arXiv:2209.10652.

[6] Victor Lecomte, Kushal Thaman, Rylan Schaeffer, Naomi Bashkansky, Trevor Chow, and Sanmi Koyejo. *What Causes Polysemanticity? An Alternative Origin Story of Mixed Selectivity from Incidental Causes*. arXiv:2312.03096, 2023.

[7] Sebastiano Cultrera di Montesano, Ondřej Draganov, Herbert Edelsbrunner, and Morteza Saghafian. *Chromatic Alpha Complexes*. arXiv:2212.03128, 2022.

[8] Bastian Rieck, Matteo Togninalli, Christian Bock, Michael Moor, Max Horn, Thomas Gumbsch, and Karsten Borgwardt. *Neural Persistence: A Complexity Measure for Deep Neural Networks Using Algebraic Topology*. International Conference on Learning Representations, 2019.

[9] Karthikeyan Natesan Ramamurthy, Kush R. Varshney, and Krishnan Mody. *Topological Data Analysis of Decision Boundaries with Application to Model Selection*. International Conference on Machine Learning, 2019.